Optimization and Statistical Foundations for Management Analytics

Lecture Notes

Master of Management Analytics (MMA) Program

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Contents

1 | Course Structure

This chapter covers how the course is organized, and what is expected of students.

1.1 Course Format and Schedule

MGSC 660 is a foundation course in statistics and optimization for the Master of Management Analytics (MMA) Program.

The first three weeks use a hybrid structure to ease students into the program: Students are expected to use the prerecorded lessons, practice questions, and activities on the course site.

Beginning in the fourth week, the course moves to fully in-person teaching. Each week includes two meetings, with the last one devoted to final-exam review.

1.2 Assessment and Assignments

Assessment includes five assignments. Assignments 1 and 5 are group assignments, while Assignments 2, 3, and 4 are individual.

Teams should ideally contain four students, although smaller teams are acceptable. Students may form teams across different sections, since the sections are treated as one combined class. One team member must complete the final submission before the deadline; merely placing a file in the assignment area may not count as a completed submission. Deadlines are strict and generally fall on Saturdays at 11:30 p.m. Eastern Time.

The first group assignment uses the Merton Truck Company case, and the final group assignment uses the Freemark Abbey Winery case. These are the two mandatory readings; the suggested textbooks are optional references.

The five assignments create a rhythm for the six-week term. Assignment 1 is not due during the first weekend, but almost every later weekend contains a deliverable, so group formation should happen immediately. Students who cannot find a team should contact the teaching assistant and can be assigned to one. Teams are free to remain together for Assignment 5 or reorganize.

1.3 Participation

Participation can occur in two equivalent ways: contributing during live class discussions, or contributing online through the module discussion boards. For each module, the expected online alternative is at least two meaningful posts — questions, answers, clarifications, or relevant professional experiences.

At the end of each module, students complete a short participation survey indicating how they contributed. Online posts can make up for missed live participation, but doing neither results in no participation credit.

Three kinds of online discussion space are distinguished: module boards for conceptual questions and experiences, assignment boards for questions about assignment requirements, and a separate final-exam area. This organization creates a searchable repository for the whole class; credit is based mainly on meaningful contribution rather than on whether an answer is selected as uniquely best.

1.4 Software: Python and Excel

The course is programming-language agnostic at the conceptual level. That being said, python is the default implementation language and is strongly encouraged, since it will matter throughout the program. Excel is also acceptable, especially for students transitioning into Python; Excel users should activate Solver and the Analysis ToolPak. The key message is that software is a means of implementing optimization, not the subject of the course itself.

Students are given a shared online Python workspace containing examples organized by lesson. They should fork the workspace to create an editable personal copy, experiment with the code, and practice. A local Python setup (for example, an editor such as Visual Studio Code) is another option.

1.5 The Final Exam

The final exam is a three-hour, in-class, conceptual exam. Computers are not allowed, and students will not be asked to reproduce Python, Excel, or other programming syntax. Calculators are allowed, as is one double-sided page of notes, and any calculations should be relatively simple. A practice exam from a previous year illustrates the expected style of questions.

1.6 Success

The overall takeaway from the logistics is that success depends on steady engagement: complete the asynchronous lessons before the next live meeting, practice with the examples, participate in each module, and read the two cases in time for their respective group assignments. The course rewards conceptual understanding — being able to explain *why* a model is structured a particular way, not merely obtaining an answer from Solver or a Python package.

Part I

Optimization Foundations (Module 1)

2 | Optimization in Analytics and Machine Learning

2.1 Training as Optimization

Optimization is a core mechanism behind modern analytics, machine learning, and artificial intelligence. Every machine-learning model must be trained: training means adjusting model parameters so that the model's output moves closer to a known target, often called the ground truth or label. Behind the scenes, that adjustment is an optimization problem. The model begins with imperfect predictions, measures the error, and repeatedly changes its parameters to reduce that error.

The training loop can be stated in business terms: a model produces an output, the output is compared with what actually happened, and the difference is converted into a numerical loss. The optimizer then changes the model parameters and tries again, and this sequence may be repeated thousands or millions of times. Training is therefore not a mysterious feature added to a model; it is a repeated decision process governed by an objective.

2.2 Classification and Regression

Two broad machine-learning tasks illustrate this idea. *Classification* predicts a discrete category, such as whether an image is a cat or a dog, or which token should come next in a language model, and commonly uses a cross-entropy objective. *Regression* predicts a continuous quantity, such as a future price, and commonly uses squared error.

For classification, the output is one of a finite set of possibilities. A language model's next-token choice feels open-ended, but it is still a classification over a discrete vocabulary; cross-entropy translates the quality of the predicted probability distribution into a loss, penalizing a correct label that receives low predicted probability.

For regression, the target lies on a continuous scale (a future stock price, for example). Squared error makes both positive and negative residuals cost something and gives greater weight to large misses. In both cases, training searches for parameter values that improve an objective.

2.3 Optimization Beyond Machine Learning

Optimization is much broader than machine learning. In supply chains, a company such as Amazon must choose which fulfillment center should serve an order, whether to reserve inventory for future demand, and which transportation mode to use. A nearby warehouse may be cheaper for the present order but strategically valuable for another customer; trucks and aircraft have different costs and capacity limits. These competing considerations turn fulfillment into an optimization problem.

Retailers optimize prices and promotions: when to offer a discount, how deep it should be, and whether to customize an offer using a customer's history. Manufacturers decide what to produce, in what quantities, and how much inventory to hold. The business domain changes, but the mathematical structure is similar.

2.4 The Three Components of an Optimization Model

Every optimization model has three essential components.

Decision variables are the quantities the model is allowed to choose. In a pricing model, the decision variable may be price; in an AI model, the variables are the trainable parameters, potentially numbering in the billions. More variables do not automatically create a better model — a model must balance expressive power, training cost, deployment feasibility, energy use, and reliability.

The objective function evaluates how good a candidate solution is (minimizing prediction error, minimizing fulfillment cost, maximizing profit). A useful objective should be simple enough to calculate and optimize, yet rich enough to capture the central trade-off in the real problem.

Constraints translate business rules and physical limits into mathematical conditions: a transportation plan cannot exceed vehicle capacity, a price may need to stay within an approved range, certain promotions may not be offered together, model parameters may need to be nonnegative or bounded. Constraints define the feasible space — the set of solutions that are allowed and implementable.

2.5 Formulation Requires Judgment

An optimization system can solve only the model it is given. If the decision variables omit a crucial choice, if the objective rewards the wrong outcome, or if a business rule is missing, a mathematically optimal answer may be operationally poor. This is why students are asked to compare models generated by AI tools and evaluate their realism rather than accepting them uncritically.

The fulfillment and pricing examples both illustrate why an apparently obvious choice may not be optimal: serving a customer from the nearest warehouse reduces distance, but that warehouse's inventory may be more valuable for another order, and a lower price may increase demand while reducing margin. These examples prepare students to look for trade-offs rather than search for a rule that says one action is always best.

3 | Variables, Objectives, Constraints, and Convexity

3.1 Convexity of Feasible Regions

A feasible region is *convex* when the straight line joining any two feasible points stays entirely inside the region. Convexity matters because optimization algorithms move through a sequence of candidate solutions, and in a convex region movement between feasible solutions behaves predictably. Non-convex regions can contain gaps, isolated areas, or multiple hills and valleys, making them harder and more expensive to search. Linear constraints are especially useful because they generate convex feasible regions and lead to scalable solution methods.

The key modeling discipline is: identify what can be chosen, define exactly what success means, express every important limitation, and then check whether the resulting model is computationally manageable and usable in the real world. Optimization is not simply pressing a solve button — it is the careful translation of a decision problem into variables, an objective, and constraints.

3.2 Cross-Entropy Loss for Classification

For supervised classification with observations indexed by i and classes indexed by c, a common cross-entropy loss is

$$L(\theta) = -\sum_{i}\sum_{c} y_{i,c} \log p_{i,c}(\theta).$$

The parameter vector θ contains the model weights. The indicator $y_{i,c}$ equals one for the correct class and zero otherwise, and the predicted probability $p_{i,c}(\theta)$ depends on θ. Minimizing L assigns high probability to correct labels. A language model uses the same basic structure at the token level: the correct next token supplies the label, while the model distributes probability over its vocabulary.

3.3 Mean Squared Error for Regression

For regression, a standard mean-squared-error objective is

$$\mathrm{MSE}(\theta) = \frac{1}{n}\sum_{i=1}^{n}\Big[y_i - \hat{y}_i(\theta)\Big]^2.$$

The residual is actual minus predicted. Squaring makes positive and negative residuals contribute positively and penalizes large deviations more heavily. Training seeks $\theta^\star \in \arg\min L(\theta)$ (or $\arg\min \mathrm{MSE}(\theta)$). Regularization can be added — for example $\lambda\|\theta\|^2$ — when the analyst wants to discourage overly large parameter values.

3.4 The Gradient Descent Update Rule

A differentiable unconstrained minimization method uses gradient descent:

$$\theta_{k+1} = \theta_k - \alpha_k \nabla L(\theta_k).$$

The gradient is the vector of partial derivatives. The minus sign is used because the goal is minimization; for maximization, the update uses a plus sign. The learning rate α_k is the step size and may be fixed, scheduled, or adaptive.

3.5 Formal Definition of a Convex Set and a Convex Function

A set X is convex if, for every $x, z \in X$ and every $\lambda \in [0, 1]$, the weighted average remains feasible:

$$\lambda x + (1 - \lambda)z \in X \quad \text{for all } \lambda \in [0, 1].$$

A function f is convex when the graph lies below every chord joining two points on the graph:

$$f\Big(\lambda x + (1 - \lambda)z\Big) \leq \lambda f(x) + (1 - \lambda)f(z).$$

For twice-differentiable f, a sufficient and (on an open convex domain) equivalent test is that the Hessian matrix is positive semidefinite everywhere: $v^\top H(x)\, v \geq 0$ for every vector v, where $H(x)$ contains all second partial derivatives. For maximizing a concave function, the Hessian is negative semidefinite.

3.6 Linear Inequalities and the Convexity of Linear Programs

A linear inequality $a^\top x \leq b$ defines a half-space, which is convex. Intersections of convex sets remain convex, so the feasible region of a linear program is convex. If the objective is also linear, any local optimum is globally optimal, although multiple globally optimal solutions can exist along an edge or face.

3.7 First-Order Optimality and the KKT Conditions

First-order optimality for a differentiable unconstrained convex minimization problem is $\nabla f(x^\star) = 0$. With constraints, the correct conditions involve feasible directions or the Karush–Kuhn–Tucker (KKT) conditions. In simplified form, KKT combines primal feasibility, dual feasibility, stationarity, and complementary slackness — this distinction explains why an optimum on a boundary can have a nonzero ordinary gradient.

3.8 Decision Variables versus Inputs

Decision variables must be controllable choices, not merely facts in the environment. A retailer can choose a price or an allocation quantity, but cannot directly choose customer demand — demand is predicted as a consequence of price. Similarly, an AI model's parameters are decision variables during training, while the labeled examples are inputs. Clearly separating choices from inputs is one of the first tests of a sound formulation.

3.9 Coefficients, Right-Hand Sides, and the Feasible Set as an Intersection

Objective coefficients express how a unit of a decision contributes to the performance measure; constraint coefficients express how a unit consumes a resource or affects a business rule; right-hand-side values usually represent limits, requirements, or available capacities.

Constraints can be simple bounds (for example, requiring a price to fall between ten and forty) or complex nonlinear relationships. A collection of constraints acts jointly: the feasible set is the intersection of the solutions allowed by every rule, and satisfying one rule is not enough — an implementable solution must satisfy them all at once.

3.10 Convexity in Practice

The geometric definition of convexity does not depend on the number of variables: choose any two feasible solutions, form a weighted average, and if every such average is feasible, the set is convex. In one dimension, a single uninterrupted interval is convex, while two separated intervals are not; in two dimensions, a filled circle or ellipse is convex, while a crescent or a region with a hole is not.

Calculus can test convexity through second derivatives and semidefinite curvature conditions, but explicitly forming a large second-derivative matrix may be computationally unrealistic. Practitioners often build models from known convex components — linear inequalities are a fundamental example — and combine functions using rules that preserve convexity, rather than attempting a brute-force proof for a model with billions of variables.

Convex objectives and feasible sets are attractive because local improvement methods have stronger guarantees. Modern deep neural networks are generally non-convex, which is one reason their training behavior is more complicated; this intuition should be kept distinct from a universal mathematical claim.

4 | The Pricing Model and Gradient Calculations

4.1 A One-Variable Pricing Problem

The lecture develops the basic ideas through a one-variable pricing problem. A sportswear retailer wants to choose the price of a sweatshirt. Because there is only one decision variable, price, this is called *scalar* optimization; a scalar is a single number, while a vector contains several decision variables.

The retailer's estimated demand is $20 - 0.5p$, so demand falls as price rises; the coefficient 0.5 represents price sensitivity. Each unit sold earns a margin equal to the selling price minus a unit cost of \$2, so expected profit is demand multiplied by unit margin: $(20 - 0.5p)(p - 2)$.

The price cannot be below \$10, and the demand formula should not predict negative demand — demand reaches zero at $p = 40$, so price should not exceed \$40. The feasible set is the interval $[10, 40]$, which is convex, since every point between any two feasible prices is also feasible.

4.2 Gradient Search: The Core Idea

To solve the problem, the lecture introduces gradient search, the basic family of methods behind much of modern model training. The algorithm begins with an initial solution (a price of 30), then calculates an update with two parts: a direction supplied by the gradient and a distance controlled by the step size. The new solution equals the current solution plus that update.

In one dimension, the gradient is the derivative of the objective function: it measures how quickly the objective changes when the decision variable is changed by a very small amount. A positive derivative means moving toward a higher value of the decision variable locally increases the objective; a negative derivative means the opposite. At $p = 30$ the derivative is negative, so profit can be improved by reducing the price.

With several decision variables, the gradient becomes a vector of partial derivatives, and each component measures the local rate of change in one direction. This is why gradients scale to models with enormous numbers of parameters: each iteration uses local rate-of-change information rather than evaluating every possible solution.

4.3 Why "Local" Matters

A gradient describes what happens in a small neighborhood around the current point — it does not promise that moving a long distance in the same direction remains beneficial. The step size limits how far the algorithm travels before recalculating the gradient. A very small step is safe but may require millions of iterations; a large step can overshoot the best region or cause repeated zigzagging. Practical methods seek a useful balance.

4.4 Numerical Illustration

In the spreadsheet illustration, the initial price is 30 and a fixed step size of 0.1 is used. The negative gradient produces a negative update, so the price moves downward; the calculation is repeated (recompute the derivative, multiply by the step size, update the price, and continue), and the sequence approaches a price of approximately $21. At the optimum, the gradient is zero, or close enough to zero that further improvement is negligible.

Expanding the pricing objective, profit $= (20 - 0.5p)(p - 2)$, is a quadratic expression. At low prices, margin is small even if demand is high; at very high prices, margin per unit is high but demand approaches zero. The best price balances these effects, which is why neither the lowest nor the highest allowed price is automatically optimal.

4.5 The Product Rule and the Sign of the Update

The derivative is calculated using the product rule: differentiate the demand term, keep the margin term, then add the demand term multiplied by the derivative of margin. At $p = 30$, the result is negative, indicating that reducing price should improve profit near 30. After the price changes, the derivative must be recalculated, because the objective is curved and its slope changes with location.

The update rule is recursive: if the current price is p_0, the next price is

$$p_1 = p_0 + \alpha \cdot (\text{gradient at } p_0).$$

With a fixed step size $\alpha = 0.1$ and a gradient of -9 at $p_0 = 30$, the first update is -0.9, moving the price from 30 to 29.1. Repeating the calculation generates a sequence that approaches 21.

4.6 Direction of Ascent versus Descent

For a maximization problem, moving in the gradient direction produces a local increase. For a minimization problem, the sign is reversed and the algorithm moves against the gradient — this is usually called *gradient descent*. The broader phrase *gradient search* is used because the same local-derivative idea supports either maximization or minimization; students should pay attention to the stated objective before interpreting the update direction.

The formal stopping condition at an unconstrained interior optimum is a zero gradient. Computers normally use a tolerance (for example, stopping when the gradient magnitude is below 0.01), since exact zero may be unnecessary or impossible with finite numerical precision. A constrained optimum may occur at a boundary, where the unconstrained gradient need not be zero; constrained solvers account for the allowed directions.

The pricing problem could be solved directly by setting its derivative to zero, and a student raises exactly this point. The instructor agrees, but explains that direct algebra is easy for one or a few variables and does not scale to a model with millions or billions of decisions — the toy problem makes the logic of scalable optimization visible. In real software, users normally specify a convergence tolerance instead of waiting for the derivative to become exactly zero; Excel Solver's generalized reduced gradient method performs this iterative work automatically and can respect lower and upper bounds.

Modern training often uses adaptive step sizes. The Adam family of methods changes the effective step size using information about recent gradients and momentum; the details, along with backpropagation (an efficient way to calculate gradients through neural networks), are deferred to later courses. PyTorch is identified as a commonly used Python package that makes these methods convenient.

5 | Solver Methods, Curvature, and Local Optima

5.1 Solving the Pricing Problem Exactly

The pricing example can be solved completely. Let p be price in dollars, demand in thousands of units be $D(p)$, and unit cost be \$2:

$$D(p) = 20 - 0.5p, \qquad 10 \le p \le 40.$$

$$f(p) = D(p)(p-2) = (20 - 0.5p)(p-2).$$

Expanding gives a concave quadratic:

$$f(p) = -0.5p^2 + 21p - 40.$$

Its first and second derivatives are

$$f'(p) = 21 - p, \qquad f''(p) = -1.$$

Because the second derivative is negative, f is strictly concave. The stationary point satisfies $21 - p = 0$, so $p^\star = 21$. This point lies inside the feasible interval, making it the unique global maximum. Demand is 9.5 thousand units, margin is \$19 per unit, and maximum profit is \$180.5 thousand under the model. At $p = 30$, profit is \$140 thousand and the derivative is -9, confirming that price should move downward.

5.2 Convergence of Gradient Ascent

With gradient ascent and fixed step α, the update becomes

$$p_{k+1} = p_k + \alpha(21 - p_k) = (1-\alpha)p_k + 21\alpha.$$

Subtracting the optimum from both sides gives $p_{k+1} - 21 = (1-\alpha)(p_k - 21)$. Convergence occurs when $|1-\alpha| < 1$, equivalently $0 < \alpha < 2$. With $\alpha = 0.1$, the error shrinks by 10% each iteration; from $p_0 = 30$, the first value is 29.1, matching the spreadsheet demonstration.

5.3 Projected Gradient Updates

If an update crosses a bound, a projected method uses

$$p_{k+1} = \mathrm{Proj}_{[10,40]}\big(\text{unconstrained update}\big).$$

Excel's generalized reduced gradient method handles constraints more generally by working with feasible or reduced directions.

5.4 Gradients and Directional Derivatives in Higher Dimensions

For a multivariable objective $f(x)$, the gradient is

$$\nabla f(x) = \left[\frac{\partial f}{\partial x_1}, \ldots, \frac{\partial f}{\partial x_n}\right]^\top .$$

The directional derivative along a unit vector d is $\nabla f(x)^\top d$. By the Cauchy–Schwarz inequality, this quantity is largest when d points in the gradient direction — the mathematical reason the gradient gives the direction of steepest local ascent under ordinary Euclidean distance.

5.5 The Hessian and Newton's Method

The Hessian describes local curvature:

$$H(x) \text{ has entry } H_{i,j} = \frac{\partial^2 f}{\partial x_i \, \partial x_j}.$$

A Newton step for minimization solves

$$H(x_k)\, d_k = -\nabla f(x_k), \qquad x_{k+1} = x_k + d_k,$$

possibly with a damped step. This rescales the gradient using curvature, which can reduce zigzagging, but storing and solving with an $n \times n$ Hessian is costly when n is large.

5.6 Two-Phase Feasibility Methods

A generic two-phase feasibility method defines a nonnegative violation function $V(x)$. For inequalities $g_i(x) \le 0$, one option is

$$V(x) = \sum_i \max(0, g_i(x))^2 + \sum_j h_j(x)^2.$$

Phase one minimizes V until it is approximately zero; phase two then optimizes the true objective while maintaining feasibility. The $\max(0, g_i(x))$ term is needed so that only a positive value — representing violation of a less-than-or-equal constraint — is penalized.

5.7 Multi-Start, Smoothing, and Local Optima

At a local minimum, the gradient may be zero even when another, lower basin exists. Multi-start samples several initial vectors and compares the resulting local solutions, keeping the best. Smoothing or continuation methods replace the original objective with an easier family f_τ and gradually change the temperature parameter τ. Both strategies spend additional computation to reduce dependence on one local starting condition.

Gradient search is powerful, but its reliance on local information creates two weaknesses. The first is *zigzagging*: on a narrow or sharply curved objective surface (visualized using isoprofit curves, whose tangent is perpendicular to the gradient), the steepest local direction can point across the valley rather than along it, so the algorithm bounces from side to side and makes slow progress. A second-order method such as Newton's method uses curvature information to choose a more informed direction, but calculating and using second-order information can be too expensive for very large models.

The second weakness is *local optimality*: in a non-convex landscape, the algorithm can stop at a local minimum or maximum even when a better global solution exists elsewhere, because at the local optimum every nearby direction looks worse and the gradient offers no escape route. Multi-start and smoothing/temperature-based approaches both improve robustness without eliminating this computational trade-off.

6 | Linear Programming, Duality, and Sensitivity Analysis

6.1 Testing for Linearity

These difficulties motivate linear optimization. A linear function is a sum of terms in which each term contains at most one decision variable, appearing only to the first power; a linear constraint has a linear expression on each side of an equality or inequality. The pricing profit function from the previous chapters is *not* linear, because it multiplies two expressions containing price, ultimately creating a squared term.

To test linearity, separate an expression into terms: constants and constant multiples of a single first-power variable are allowed, while products of decision variables, squares, roots, logarithms, and other nonlinear transformations are not. An equality or inequality is linear only when both sides meet this rule — the connector itself (less than, greater than, or equal to) does not make a constraint nonlinear.

Linear models are deliberately restrictive, but the simplicity is valuable: their behavior is predictable, their feasible regions are convex, and they avoid the local-optimum problem that complicates general nonlinear models. A linear program has a linear objective function and a collection of linear constraints.

6.2 The Wine Cooler and Beer Example

A Montreal company produces wine cooler and beer. Let W be tons of wine cooler and B be tons of beer, with profit measured in thousands of dollars. Wine cooler earns \$3,000 per ton and beer earns \$2,000 per ton. Production consumes two limited raw materials, hops and malt: the company has six tons of hops and eight tons of malt, where each ton of wine cooler uses one unit of hops and two units of malt, and each ton of beer uses two units of hops and one unit of malt.

The manufacturing model can be written as a linear program:

$$\text{Maximize } z = 3W + 2B$$

$$\text{subject to} \quad W + 2B \leq 6, \qquad 2W + B \leq 8, \qquad W \geq 0, \qquad B \geq 0.$$

The first inequality is the hops capacity and the second is the malt capacity. In matrix form,

$$\text{maximize } c^\top x \ \text{ subject to } Ax \leq b, \ x \geq 0,$$

where $x = [W, B]^\top$, $c = [3, 2]^\top$, A has rows $[1, 2]$ and $[2, 1]$, and $b = [6, 8]^\top$.

6.3 Solving the Linear Program

The feasible vertices are $(0,0)$, $(0,3)$, $(4,0)$, and the intersection of the two resource boundaries. Solving $W+2B=6$ together with $2W+B=8$ gives $W=10/3$ and $B=4/3$. The corresponding objective values (in thousands) are 0, 6, 12, and 38/3, respectively. Therefore

$$W^\star = \tfrac{10}{3}, \qquad B^\star = \tfrac{4}{3}, \qquad z^\star = \tfrac{38}{3} \approx 12.667 \text{ thousand dollars.}$$

Both resource constraints bind at this solution.

6.4 The Dual Problem

The associated dual problem is

$$\text{Minimize } 6y_1 + 8y_2 \quad \text{subject to} \quad y_1 + 2y_2 \geq 3, \quad 2y_1 + y_2 \geq 2, \quad y_1, y_2 \geq 0.$$

Complementary slackness and the positive production quantities imply that both dual product constraints bind. Solving gives $y_1 = 1/3$ and $y_2 = 4/3$, in thousands of dollars per additional ton of capacity. The dual objective is also $38/3$, illustrating strong duality. Locally, one extra ton of hops is worth about \$0.333 thousand, while one extra ton of malt is worth about \$1.333 thousand, provided the current basis remains optimal.

6.5 The Simplex Method

Simplex moves among basic feasible solutions, which correspond geometrically to vertices. At each iteration it selects an entering variable that can improve the objective and a leaving variable determined by feasibility, terminating when no reduced cost indicates further improvement. For a linear program, an optimal solution can always be found at an extreme point (a vertex) of the feasible region: instead of moving through the interior in tiny gradient steps, simplex moves from vertex to vertex, improving the objective until no adjacent vertex is better.

6.6 Graphing the Feasible Region

Each resource boundary can be graphed using intercepts. For the hops equality $W+2B=6$, setting $W=0$ gives $B=3$, and setting $B=0$ gives $W=6$; the malt boundary is constructed the same way. Because usage must be no more than capacity, the feasible side includes the origin, and the feasible polygon is where both resource limits and non-negativity overlap.

The objective's gradient is the constant vector $(3000, 2000)$; only the ratio matters for direction, so it can be viewed as three units toward wine cooler for every two toward beer, and this direction never changes (unlike the quadratic price objective from the previous chapters). Without capacity constraints, profit could increase without bound — the constraints create a finite opportunity set and force the firm to choose a mix.

A linear objective cannot have a uniquely superior point strictly inside a polygon unless the same value extends toward a boundary, so at least one optimum occurs at a vertex. In two dimensions, students can evaluate the objective at each vertex directly; the simplex algorithm generalizes the idea to many dimensions by moving systematically among adjacent basic feasible solutions instead of enumerating every possible point.

6.7 Sensitivity Analysis and Shadow Prices

Sensitivity analysis (what-if analysis) turns the optimized model into a management tool. An objective coefficient may change if a product's profit margin changes; a constraint coefficient may change if production becomes more or less resource efficient; a right-hand side may change if the company acquires more hops or malt. The analyst asks whether the same production plan remains optimal, how much profit changes, and which resource is most valuable at the margin.

A *shadow price* is the marginal improvement in the optimal objective associated with one additional unit of a binding resource, within a relevant range; a zero shadow price often indicates that the resource is not currently scarce. Dual variables formalize these marginal values — managers are often more interested in the value of extra capacity, or the risk of changing assumptions, than in a single static optimum.

6.8 Key Takeaways from Module 1

Across the session, four ideas carry the most weight: optimization is the decision engine behind both operational planning and machine-learning training; good formulation requires decision variables, an objective function, and constraints that faithfully represent the real situation; gradient search is scalable because it repeatedly uses inexpensive local information, but step size, curvature, feasibility, and local optima all matter; and linear programming trades modeling flexibility for exceptional structure, reliability, and interpretability.

- Optimization underlies operational decisions and the training of machine-learning models.
- Every optimization model needs decision variables, an objective function, and constraints.
- Convexity makes optimization more predictable; non-convex problems can contain difficult local optima.
- Gradient search uses a local slope for direction and a step size for distance.
- Small steps can be slow; large steps can overshoot or zigzag. Adaptive methods adjust the step size.
- Linear programs use a linear objective and linear constraints, creating a structured convex feasible region.
- The simplex method searches extreme points of a linear program rather than taking tiny interior steps.

- Sensitivity analysis studies how the optimal solution changes when coefficients or capacities change.
- The course emphasizes conceptual modeling and interpretation, with Python or Excel used for implementation.

For study purposes, students should be able to identify the three components of an optimization problem, distinguish scalar and vector decisions, explain convexity in plain language, describe how gradient and step size play different roles, recognize why gradient search can zigzag or get stuck, test whether a function is linear, formulate the manufacturing objective and capacity constraints, and explain why simplex focuses on extreme points.

Part II

Probability, Bayes' Rule, Decision Trees, and the Value of Information (Module 2)

7 | Probability Models, Sample Spaces, Events, and Counting

This chapter covers probability models, sample spaces, events, counting, and how these ideas connect to language models and generative AI.

7.1 Defining a Probability

Probability is defined as a normalized number measuring how likely an event is. A probability of zero means impossibility within the model, one means certainty, and intermediate values quantify uncertainty.

$$0 \leq P(A) \leq 1.$$

7.2 Sample Spaces and Events

A probability model has three conceptual ingredients: an experiment, a sample space, and probabilities assigned to outcomes. The sample space, usually written Ω, contains every elementary outcome that the model recognizes. An event A is a subset of Ω and may contain one or many elementary outcomes.

$$A \subseteq \Omega, \qquad \sum_{\omega \in \Omega} P(\omega) = 1.$$

The second axiom is the normalization rule: all elementary-outcome probabilities must sum to one. Increasing the probability of one outcome therefore requires decreasing probability elsewhere. A softmax layer in a neural network is mentioned as a standard way to convert arbitrary scores into positive numbers that sum to one.

7.3 Large Language Models as a Motivating Example

Large language models provide the motivating example. A token is treated as an elementary outcome, the vocabulary is the sample space for the next-token decision, and the model estimates a probability distribution over that vocabulary. A marginal probability such as $P(\text{school})$ ignores context. A language model instead estimates a conditional probability such as the probability that the next token is "school" given the preceding prompt.

7.4 Simple Experiments and Structured Sample Spaces

Simple experiments make the same structure visible. One die has six elementary outcomes. Two dice have thirty-six ordered outcomes. A quality test may have success and failure. A digital image has an enormous structured state space determined by pixel positions and red, green, and blue intensities. Generative modeling is difficult largely because real sample spaces are huge and their probabilities are not uniform.

7.5 Counting With and Without Replacement

For two sequential draws from three colors with replacement, the ordered sample space has three times three, or nine, outcomes. If all colors and draws are symmetric, each outcome has probability one ninth. Without replacement, only six ordered color pairs remain when each color is initially represented once. The lecture uses this distinction to show that counting depends on the experimental protocol.

7.6 Combining Events: Union and Intersection

An event can combine elementary outcomes. The event of drawing the same color twice contains red-red, green-green, and yellow-yellow.

$$P(\text{same color}) = \frac{1}{9} + \frac{1}{9} + \frac{1}{9} = \frac{1}{3}.$$

This calculation uses additivity for mutually exclusive outcomes. More generally, the union of two events needs a correction when they overlap.

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

In the marble example, let A mean at least one red and B mean two marbles of the same color. $P(A)$ is five ninths, $P(B)$ is three ninths, and their intersection is red-red with probability one ninth. Therefore the union has probability seven ninths. Simply adding five ninths and three ninths would double-count red-red.

7.7 The Complement Rule

The complement rule is another consequence of normalization.

$$P(A^c) = 1 - P(A).$$

The practical discipline is to define the experiment before calculating. Specify whether order matters, whether sampling uses replacement, whether outcomes are equally likely, and exactly which elementary outcomes constitute the event. Most probability errors begin with an ambiguous sample space rather than difficult arithmetic.

7.8 Elementary Outcomes versus Events

The distinction between an elementary outcome and an event matters computationally. An elementary outcome is indivisible at the model's chosen resolution. An event is any collection of elementary outcomes. The same real-world description can produce different sample spaces depending on the question. If the experiment is drawing one marble and recording color, the outcomes are red, green, and yellow. If the question is the number of red marbles in two draws, the outcomes are zero, one, and two, even though the underlying ordered color pairs are more detailed.

7.9 Counting Rules Under Uniformity

Counting rules supply probabilities only when symmetry assumptions are justified. If all elementary outcomes are equally likely, the probability of event A is its number of favorable outcomes divided by the number of outcomes in Ω.

$$P(A) = \frac{|A|}{|\Omega|} \qquad \text{(under uniformity).}$$

With replacement, two draws from three colors create nine ordered pairs because the first and second positions each have three possibilities. Without replacement, the count depends on how many physical marbles of each color exist. The simplified three-times-two count assumes one marble per color and ordered observations. This illustrates why analysts must not use a counting formula before specifying inventory, replacement, and order.

7.10 Joint-Probability Tables

A joint-probability table is a useful representation for two categorical variables. Rows encode the first draw and columns encode the second. Each cell is an elementary joint outcome. Row sums and column sums are marginal probabilities, and selected blocks or diagonals represent events. The table makes overlap visible and helps prevent double counting.

7.11 Countable Additivity and Inclusion-Exclusion

For disjoint events A_i, countable additivity gives:

$$P\left(\bigcup_i A_i\right) = \sum_i P(A_i), \qquad \text{when the } A_i \text{ are pairwise disjoint.}$$

The general two-event addition rule follows by splitting $A \cup B$ into three disjoint regions: only A, only B, and the intersection. The subtraction term removes the duplicate intersection. For three or more overlapping events, inclusion-exclusion alternates sums and subtractions of intersections.

7.12 Empirical Estimation from Data

Uniform distributions are useful initial models, but observed systems are rarely uniform. A biased die, unequal marble counts, language frequencies, and image structure all produce unequal probabilities. Empirical estimation uses relative frequencies from data.

$$\hat{P}(A) = \frac{\text{number of observed cases in } A}{\text{total observations}}.$$

As data accumulate, the estimate can be updated. A generative model performs a far more structured version of this task, sharing statistical strength across contexts rather than independently counting every possible sentence or image.

7.13 Softmax as a Normalization Mechanism

Softmax is the normalization mechanism briefly referenced in the lecture. Given model scores z_1 through z_K, it defines:

$$\text{softmax}(z_i) = \frac{e^{z_i}}{\sum_{j=1}^{K} e^{z_j}}.$$

Every output is positive and the outputs sum to one. Softmax does not guarantee that probabilities are well calibrated or correct; it guarantees only the mathematical form of a categorical probability distribution. Training data and the loss function determine whether those probabilities become useful.

8 | Conditional Probability, Bayes' Rule, and Independence

8.1 Defining Conditional Probability

Conditional probability restricts attention to cases in which another event is already known to have occurred. It is written $P(A \mid B)$. The event B becomes the new reference population, so the intersection of A and B must be normalized by $P(B)$.

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \qquad P(B) > 0.$$

For a fair die, let A be a result less than four, so $A = \{1, 2, 3\}$. Let B be an odd result, so $B = \{1, 3, 5\}$. The intersection is $\{1, 3\}$. Therefore:

$$P(A \mid B) = \frac{2/6}{3/6} = \frac{2}{3}.$$

The complement within the same condition is five, so $P(A^c \mid B)$ equals one third. The two conditional probabilities add to one. This illustrates why the denominator is required: it renormalizes probabilities after the sample space is restricted to B.

8.2 The Mask-and-COVID Example

The mask-and-COVID table provides a practical example. Fifteen percent of the population both wears masks and catches COVID. Forty-five percent wears masks without catching COVID. Another fifteen percent catches COVID without wearing masks, leaving twenty-five percent in neither group. The marginal COVID probability is thirty percent and the mask probability is sixty percent.

$$P(\text{COVID}) = 0.15 + 0.15 = 0.30.$$

$$P(\text{COVID} \mid \text{mask}) = \frac{0.15}{0.60} = 0.25.$$

Within this illustrative table, observing mask use changes the estimated COVID probability from thirty percent to twenty-five percent, a five-percentage-point reduction. The thirty percent value is described as a prior or marginal probability; twenty-five percent is the posterior after conditioning on mask use. The table is a teaching example, not evidence for a real-world causal estimate.

8.3 The Multiplication Rule and Bayes' Rule

Rearranging conditional probability yields the multiplication rule.

$$P(A \cap B) = P(B)P(A \mid B) = P(A)P(B \mid A).$$

Equating the two versions and solving for $P(A \mid B)$ produces Bayes' rule.

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)}.$$

Bayes' rule has two interpretations. It reverses the direction of conditioning, allowing $P(A \mid B)$ to be derived from $P(B \mid A)$. It also updates a prior belief $P(A)$ to a posterior belief $P(A \mid B)$ using the likelihood $P(B \mid A)$ and the evidence $P(B)$. The denominator ensures that posterior probabilities across competing hypotheses sum to one.

For the mask example, $P(\text{mask} \mid \text{COVID})$ is one half because half of the COVID group wears masks. Bayes' rule gives:

$$P(\text{COVID} \mid \text{mask}) = \frac{0.50 \times 0.30}{0.60} = 0.25.$$

8.4 Independence

Independence means that observing B does not alter the probability of A.

$$A \text{ and } B \text{ are independent if } P(A \mid B) = P(A).$$

An equivalent multiplication test is:

$$P(A \cap B) = P(A)P(B).$$

Independence is symmetric. If A is independent of B, then B is independent of A. It should not be confused with mutual exclusivity. Two mutually exclusive events with positive probability cannot be independent because occurrence of one makes the other impossible.

8.5 Connections to Generative AI

The lecture connects these formulas to generative AI. Each additional prompt token changes the conditioning information, and the model recomputes a distribution over possible next tokens. Search autocomplete is a simpler example. The mathematical core is repeated conditional-probability estimation under a very large and structured sample space.

Conditional probability can be visualized as filtering and rescaling. First discard outcomes outside B. Then retain the portion of B that also belongs to A. Finally divide by the total mass of B. This procedure explains both the numerator intersection and denominator normalization without relying on memorization.

8.6 The Law of Total Probability

The law of total probability reconstructs a marginal probability from conditional pieces. If B_1 through B_k form a mutually exclusive and exhaustive partition, then:

$$P(A) = \sum_{i=1}^{k} P(A \mid B_i)P(B_i).$$

This formula supplies the denominator in many Bayes calculations. In a diagnostic setting, A might be a positive test and the B_i might be disease states. In the economic-signal example later in the lecture, A is a positive forecast and the partition is good versus bad economy.

8.7 Base Rates and Posterior Odds

Bayes' rule is most informative when base rates matter. A high $P(B \mid A)$ does not automatically imply a high $P(A \mid B)$. The posterior combines the likelihood with the prior prevalence and divides by the overall evidence. Ignoring $P(A)$ is the base-rate fallacy.

Posterior odds provide an equivalent technical form.

$$\text{Posterior odds of } A \text{ vs. } A^c = \text{prior odds} \times \text{likelihood ratio}.$$

$$\text{Likelihood ratio} = \frac{P(B \mid A)}{P(B \mid A^c)}.$$

This form separates prior belief from the evidential strength of B. A likelihood ratio greater than one favors A; below one favors A^c; equal to one carries no information about A.

In the mask table, the posterior comparison is descriptive. Conditioning identifies association in the table, not causation. A causal conclusion would require stronger design assumptions about confounders, selection, measurement, and how mask use was assigned. The lecture's mathematical point is how subgroup information changes probability.

8.8 Conditional Independence

Independence can also be checked through complements and factorization. If A and B are independent, A is independent of B^c, A^c is independent of B, and their joint table factors into row and column marginals. Zero correlation is generally weaker than independence for numeric variables, although the two coincide under special distributional assumptions.

Conditional independence is central to larger models. Events A and B may be dependent marginally but independent after conditioning on C.

$$A \perp B \mid C \quad \iff \quad P(A \mid B, C) = P(A \mid C).$$

Graphical models, naive Bayes, hidden Markov models, and many sequence models exploit conditional-independence structure to avoid estimating an impossibly large unrestricted joint distribution.

8.9 The Chain Rule for Language Generation

For language generation, the chain rule decomposes a sentence probability into next-token conditionals.

$$P(w_1, \ldots, w_T) = \prod_{t=1}^{T} P(w_t \mid w_1, \ldots, w_{t-1}).$$

A language model estimates these factors. Generating text then samples or selects from each next-token distribution and appends the chosen token to the conditioning context. The probability module therefore connects directly to the computational behavior of generative systems.

9 | Decision Trees, Expected Value, and Backward Induction

9.1 From Probability to Sequential Decisions

The second half of this part applies probability to sequential decisions. A decision tree separates controllable choices from uncontrollable states. A square represents a decision node. A circle represents a chance or event node. Branches from a decision node are available actions. Branches from a chance node are possible states with probabilities. Terminal nodes carry final payoffs.

9.2 The Real-Estate Example

The real-estate example has three initial choices: build an apartment, office, or warehouse. The economy is then good with probability 0.6 or bad with probability 0.4. Payoffs are in millions of dollars. Apartment pays fifty in a good economy and thirty in a bad economy. Office pays one hundred in good and minus forty in bad. Warehouse pays thirty in good and ten in bad.

9.3 Expected Monetary Value

Expected monetary value is the probability-weighted payoff.

$$\mathrm{EMV}(a) = \sum_s P(s \mid a)\,\mathrm{payoff}(a, s).$$

With common state probabilities, the three calculations are:

$$\mathrm{EMV}(\mathrm{apartment}) = 0.6(50) + 0.4(30) = 42.$$

$$\mathrm{EMV}(\mathrm{office}) = 0.6(100) + 0.4(-40) = 44.$$

$$\mathrm{EMV}(\mathrm{warehouse}) = 0.6(30) + 0.4(10) = 22.$$

A risk-neutral decision maker chooses office because forty-four is the largest EMV, despite the negative payoff in a bad economy. This criterion evaluates average monetary performance over repeated comparable decisions. A risk-averse decision maker could prefer a different alternative if utility, downside constraints, or risk measures replaced raw monetary value.

9.4 Backward Induction (Rollback)

Decision trees are solved by rollback, also called folding back or backward induction. Begin at the nodes closest to the terminal payoffs and work toward the root. At a chance node, calculate the expected value of its children. At a maximizing decision node, retain the child with the largest value. For cost-minimization trees, retain the smallest value instead.

$$V(\text{chance node}) = \sum_i p_i V_i.$$

$$V(\text{decision node}) = \max_{i \text{ feasible}} V_i \qquad \text{(payoff maximization)}.$$

Suboptimal branches are pruned. A branch marked with selection probability one identifies the chosen action; branches marked zero are not selected under the optimal deterministic policy. If two actions tie exactly, either can be selected without changing value, or a randomized policy can mix between them.

9.5 Software Support and Its Limits

The lecture demonstrates SilverDecisions, an open-source browser tool. A user creates square and circular nodes, attaches probabilities and payoffs, and lets the software roll the tree back automatically. A special placeholder can calculate the missing probability on a chance node so that outgoing branch probabilities sum to one. If they do not sum to one, the tool warns the user.

For nested chance nodes, rollback still works one layer at a time. First evaluate the deepest chance node. Treat its calculated expected value as the payoff passed to its parent. Continue until the root is reached. This recursive structure underlies dynamic programming, reinforcement learning, game-playing systems, and other multistage decision models.

9.6 Timing and the Bellman Recursion

A decision tree encodes timing as well as outcomes. Moving a chance node before a decision node means the information becomes available before the decision. This change in order is the basis for calculating the value of information.

A formal decision tree alternates information states and actions. Let $V(n)$ denote the value of node n. At a terminal node, V equals the stated payoff. At a chance node, V is a probability-weighted average of child values. At a decision node, V is the best available child value under the chosen objective. This recursion is the Bellman principle in a simple finite tree.

$$V(n) = \begin{cases} \text{payoff}(n) & \text{at a terminal node,} \\ \sum_i p_i\, V(\text{child}_i) & \text{at a chance node,} \\ \max_i V(\text{child}_i) & \text{at a decision node.} \end{cases}$$

The rollback order matters because a decision can depend only on information observed before that node. Solving forward by choosing the branch with the largest immediate payoff can fail when later risks or rewards differ. Backward induction summarizes all downstream consequences before the earlier choice is evaluated.

9.7 Risk Aversion and Expected Utility

The real-estate example assumes risk neutrality. Expected monetary value is linear in money, so a gain of ten offsets an equally likely loss of ten. Under expected utility theory, a risk-averse decision maker applies a concave utility function u before averaging.

$$\mathbb{E}[u(a)] = \sum_s P(s)\, u(\text{payoff}(a, s)).$$

The optimal action maximizes expected utility, not necessarily expected money. Decision trees can also include constraints, risk measures, or multi-attribute scores when monetary expectation alone is inappropriate.

9.8 A Nested Chance Node Example

The nested-chance-node classroom question illustrates recursion. Suppose a deeper chance node has value forty-two. Its parent chance node has branches paying fifty with probability 0.6, the nested value forty-two with probability 0.2, and thirty with probability 0.2.

$$\text{Parent value} = 0.6(50) + 0.2(42) + 0.2(30) = 44.4.$$

The inner uncertainty is solved once and replaced by its expected continuation value. This compression makes large trees manageable conceptually, though their number of states may still grow exponentially.

9.9 Pruning and Ties

Pruning records the policy. At a decision node, rejected actions can be visually crossed out because they will not be used under the assumed model. At a chance node, branches are never pruned merely because their payoff is unattractive; chance outcomes remain possible and must be averaged.

When two actions tie, there are multiple optimal deterministic policies. Any mixture between tied actions has the same expected value in the basic linear model. Randomization becomes strategically important in adversarial games because predictable actions can be exploited. The lecture briefly connects this idea to game theory and generative adversarial networks, though those topics lie beyond the module.

9.10 Auditing a Decision Tree

SilverDecisions supports the mechanics but does not validate the model's assumptions. Users must still verify timing, branch labels, probability normalization, payoff units, and whether a node is genuinely a decision or chance. Copying a subtree is efficient only when its structure is truly shared; branch-specific probabilities and payoffs must be edited afterward.

A reliable audit proceeds from right to left: confirm every terminal payoff, confirm that outgoing chance probabilities sum to one, recompute each chance-node expected value, verify each decision-node maximum or minimum, and compare the selected path with the business narrative.

10 | Perfect and Imperfect Information

10.1 The Uninformed Benchmark

Without additional information, the real-estate decision has value forty-four and the optimal action is office. Perfect information means the true economic state is revealed before the building decision. The tree therefore starts with the good-or-bad chance node, followed by a decision node under each realized state.

10.2 Expected Value of Perfect Information

If the economy is known to be good, the best payoff is one hundred from office. If it is known to be bad, the best payoff is thirty from apartment. The expected value with perfect information is:

$$\text{EV}_{\text{perfect}} = 0.6(100) + 0.4(30) = 72.$$

The expected value of perfect information is the improvement over the best uninformed decision.

$$\text{EVPI} = 72 - 44 = 28.$$

Twenty-eight million dollars is an upper bound on what a decision maker should pay for any information about the economy. Real forecasts are imperfect, so their value cannot exceed EVPI when evaluated under the same assumptions.

10.3 Setting Up the Imperfect-Information Example

The imperfect-information example uses a positive or negative economic signal. The prior probabilities are $P(\text{good}) = 0.6$ and $P(\text{bad}) = 0.4$. The signal accuracy is expressed conditionally: $P(\text{positive} \mid \text{good}) = 0.8$, and $P(\text{positive} \mid \text{bad}) = 0.1$. Therefore $P(\text{negative} \mid \text{good})$ is 0.2 and $P(\text{negative} \mid \text{bad})$ is 0.9.

The marginal probability of a positive signal follows the law of total probability.

$$P(\text{positive}) = 0.8(0.6) + 0.1(0.4) = 0.52.$$

$$P(\text{negative}) = 1 - 0.52 = 0.48.$$

10.4 Updating Beliefs with Bayes' Rule

Bayes' rule updates the economic-state probabilities after each signal.

$$P(\text{good} \mid \text{positive}) = \frac{0.8(0.6)}{0.52} = \frac{12}{13} \approx 0.9231.$$

$$P(\text{bad} \mid \text{positive}) = \frac{1}{13} \approx 0.0769.$$

$$P(\text{good} \mid \text{negative}) = \frac{0.2(0.6)}{0.48} = 0.25.$$

$$P(\text{bad} \mid \text{negative}) = 0.75.$$

10.5 Optimal Decisions After Each Signal

After a positive signal, expected payoffs are approximately forty-eight point four six for apartment, eighty-one point two three for office, and twenty-eight point four six for warehouse. Office remains optimal. After a negative signal, expected payoffs are thirty-five for apartment, minus five for office, and fifteen for warehouse. Apartment becomes optimal.

$$V(\text{positive}) = \max\{48.46,\ 89.22,\ 28.46\} = 81.23.$$

$$V(\text{negative}) = \max\{35,\ -5,\ 15\} = 35.$$

10.6 Expected Value of Imperfect Information

Weighting these conditional optimal values by signal probabilities gives:

$$\text{EV}_{\text{imperfect}} = 0.52(89.22) + 0.48(35) = 63.1944.$$

$$\text{EVSI} = 63.1944 - 44 = 19.1944.$$

10.7 The Decision-Analysis Workflow

The complete workflow is: construct priors; specify signal likelihoods; calculate signal marginals; update posteriors with Bayes' rule; optimize separately after each signal; average the conditional optimal values; and subtract the uninformed benchmark. This combines probability modeling and optimization into a single decision-analysis procedure.

Value of information is a decision concept, not merely a prediction-accuracy concept. Information has value only when it can arrive before a decision and can sometimes change the chosen action. A forecast may be statistically informative yet have zero economic value if the same action remains optimal under every signal.

10.8 Comparing the Three Information Regimes

Three models should be compared on the same payoff scale. The no-information model chooses one action using prior state probabilities. The perfect-information model observes the true state and then chooses. The sample- or imperfect-information model observes a noisy signal, updates beliefs, and chooses a signal-specific action.

$$\text{EVPI} = \mathbb{E}_{\text{state}}\Big[\max_a \ \text{payoff}(a, \text{state})\Big] \ - \ \max_a \ \mathbb{E}[\text{payoff}(a, \text{state})].$$

The maximum is inside the expectation under perfect information because the action can depend on the realized state. Without information, the maximum is outside because a single action must be selected before the state is known. This noncommutativity is the source of information value.

For imperfect information Y, the value is:

$$\text{EVSI} = \sum_y P(y) \ \max_a \ \mathbb{E}[\text{payoff}(a, S) \mid Y = y] \ - \ \text{baseline EMV}.$$

The Bayes calculations and decision calculations must be kept separate. First calculate $P(\text{signal})$ using total probability. Second calculate posterior state probabilities for that signal. Third evaluate all actions under those posterior probabilities. Fourth select the best action. Only then average across signals.

In the stated example, a positive signal occurs with probability 0.52 and leads to office. A negative signal occurs with probability 0.48 and leads to apartment. The policy can be written: choose office after positive; choose apartment after negative. That policy is the operational output of the information analysis.

10.9 Bounds and Sensitivity

The imperfect-information value must satisfy a bound when the analysis is internally consistent.

$$0 \leq \text{EVSI} \leq \text{EVPI}.$$

The computed values, 19.1944 and 28, satisfy this ordering.

The value of perfect information is a ceiling on the price of any forecast, not an automatic purchase recommendation. The net value of a real information service is EVSI minus its acquisition and implementation costs. Delays, model maintenance, and the ability to act on the signal also matter.

Signal quality can be studied through sensitivity analysis. If $P(\text{positive} \mid \text{good})$ and $P(\text{negative} \mid \text{bad})$ both move toward one, posteriors become more extreme and EVSI approaches EVPI. If signal likelihoods become identical across states, the likelihood ratio approaches one, posteriors equal priors, the policy stops changing, and EVSI approaches zero.

Perfect information reverses node order by revealing the state before choice. Imperfect information inserts a signal node before choice but leaves the true state uncertain afterward. This distinction is critical: a noisy forecast does not replace the economic-state chance node; it changes the probabilities used at that later node.

The broader workflow supports market research, medical testing, quality inspection, forecasting, and machine-learning-assisted decisions. A model should report not only predictive performance but also the decisions it changes, the expected payoff improvement, and the maximum rational price of the information.

10.10 Rapid Technical Review

- A probability model assigns nonnegative probabilities that sum to one.
- Addition subtracts overlap; conditioning renormalizes within observed information.
- Bayes' rule reverses conditioning and updates priors to posteriors.
- Independence means conditioning does not change probability.
- Chance nodes take expected values; decision nodes optimize across actions.
- Rollback solves a tree from terminal nodes toward the root.
- EVPI compares perfect foresight with the uninformed optimum.
- Imperfect-information value requires Bayesian posteriors and signal-contingent decisions.

Part III

Random Variables, Statistical Distributions, and Statistical Estimation (Module 3)

11 | Random Variables, Probability Mass, Density, and Sampling

This chapter moves from events to random variables, sampling, and statistical inference.

11.1 From Events to Numerical Random Variables

An event is a subset of a sample space. A random variable goes one step further: it assigns a numerical value to every elementary outcome. Formally, if Ω is the sample space, a random variable X is a measurable mapping from Ω into the real numbers,

$$X : \Omega \to \mathbb{R}.$$

The mapping sends outcomes in the sample space into real numbers. The word *random* describes the unknown outcome before the experiment; the mapping itself is fixed. For a coin toss, one may define $X(\text{heads}) = 1$ and $X(\text{tails}) = 0$. The same experiment could support another valid encoding, but once an encoding is chosen it must be applied consistently.

A discrete random variable takes values in a finite or countably infinite set. Examples include the number of children in a family, a zero-one indicator, or the integer identifier assigned to a language token. A continuous random variable takes values across an interval or another uncountable set. Examples include time, temperature, a financial return, or an idealized pixel intensity.

$$\text{Discrete support: } S_X = \{x_1, x_2, \dots\}, \qquad \text{Continuous support: } S_X \subset \mathbb{R}.$$

Generative artificial intelligence provides useful motivation. Text models convert text into tokens, associate tokens with discrete numerical identifiers, and generate a conditional distribution for the next token. Image, audio, and video systems often represent their data as high-dimensional numerical arrays. Although stored pixels are quantized on computers, mathematical models frequently treat latent variables or normalized intensities as continuous.

For an RGB image with height H and width W, a simple representation contains three values per pixel. A 256×256 image therefore contains $256 \times 256 \times 3 = 196{,}608$ scalar components. The joint random vector is enormously higher-dimensional than a single coin toss, which helps explain why training and inference are computationally expensive.

$$\text{Dimension of an } H \times W \text{ RGB image} = 3HW;$$
$$\text{for } H = W = 256, \ \text{dimension} = 196{,}608.$$

11.2 Probability Mass Functions and Probability Density Functions

Every probability model must be nonnegative and normalized. For a discrete variable, the *probability mass function* (PMF) assigns a probability to each attainable value. The masses must sum to one.

$$p_X(x) = P(X = x), \qquad p_X(x) \geq 0, \qquad \sum_{x \in S_X} p_X(x) = 1.$$

For a continuous variable, the *probability density function* (PDF) is denoted f_X. A density is also nonnegative, but normalization is expressed by an integral rather than a sum.

$$f_X(x) \geq 0 \qquad \text{and} \qquad \int_{-\infty}^{\infty} f_X(x)\,dx = 1.$$

A central technical distinction is that density is not probability. A density may exceed one when its support is narrow. Probabilities are areas under the density curve and must lie between zero and one. For a genuinely continuous variable, the probability of any exact point is zero, even when the density at that point is positive.

$$P(a \leq X \leq b) = \int_a^b f_X(x)\,dx; \qquad \text{for continuous } X,\ P(X = x_0) = 0.$$

The word *likelihood* is sometimes used loosely for the height of a density, but density and likelihood should be distinguished. A density $f(x \mid \theta)$ is a function of possible data x when θ is fixed. A likelihood $L(\theta \mid x)$ treats the same mathematical expression as a function of the unknown parameter θ. Density describes curve height; likelihood is reserved for parameter inference.

$$\text{Density: } f(x \mid \theta) \text{ as a function of } x.$$
$$\text{Likelihood: } L(\theta \mid x) \propto f(x \mid \theta) \text{ as a function of } \theta.$$

11.3 Uniform Distributions and Probabilities as Areas

The first continuous example is the uniform distribution. If X is uniform on the interval from a to b, the density is constant inside that interval and zero outside it. The rectangle must have area one, so its height is the reciprocal of the interval length.

$$\text{If } X \sim \text{Uniform}(a,b),\ \text{then } f_X(x) = \frac{1}{b-a} \text{ for } a \leq x \leq b,\ \text{and 0 otherwise.}$$

For $\text{Uniform}(1,10)$, the support has length nine, so the density is one ninth. A proposed constant height of five is positive but invalid because its area is five times nine, or forty-five, rather than one.

$$\int_1^{10} \frac{1}{9}\,dx = \frac{10-1}{9} = 1.$$

For the interval from seven to seven point one, the probability is the area of a thin rectangle: width 0.1 multiplied by height one ninth.

$$P(7 \leq X \leq 7.1) = \frac{7.1-7}{10-1} = \frac{0.1}{9} = \frac{1}{90} \approx 0.0111.$$

The narrow uniform distribution on the interval from one to one point one has width 0.1 and density ten. This density is greater than one but still valid because ten times 0.1 equals one. The probability between 1.05 and 1.06 is ten percent.

$$\text{If } X \sim \text{Uniform}(1, 1.1), \text{ then } P(1.05 \leq X \leq 1.06) = 10(0.01) = 0.10.$$

11.4 Sampling, Empirical Frequency, and Reproducibility

Repeated draws from a specified distribution can be generated in Excel or Python. A sample is a realized value from the random variable; a dataset of size n contains n such draws. If the draws are independent and identically distributed, the empirical proportion of observations falling in an event A estimates its model probability.

$$\widehat{P}_n(A) = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}\{X_i \in A\}.$$

In one set of one hundred simulated draws from Uniform$(1, 1.1)$, seven fell between 1.05 and 1.06, giving an empirical proportion of 0.07. This does not contradict the theoretical value 0.10. Sampling error is expected in a finite dataset. Under the law of large numbers, the empirical proportion converges to the true probability as n grows.

$$\widehat{P}_n(A) \to P(A) \text{ as } n \to \infty, \text{ under standard independent sampling assumptions.}$$

A pseudorandom seed initializes a deterministic number generator. Reusing the same seed with the same generator and software implementation reproduces the same sequence, which is valuable for validation and debugging. Cross-platform results may differ when implementations differ. A seed does not make observations more random; it makes a pseudorandom experiment reproducible.

This process is closely connected to AI inference. A generative model produces outputs by sampling from learned conditional distributions. The analogy should not be taken too literally: modern language models do not merely transform one scalar uniform distribution into English. They learn a high-dimensional parameterized network, compute conditional token probabilities with a softmax layer, and apply a decoding rule such as greedy selection, temperature sampling, top-k sampling, or nucleus sampling.

$$P(\text{next token} = j \mid \text{context}) = \frac{\exp(z_j/T)}{\sum_k \exp(z_k/T)}.$$

Here z_j is the model score for token j and T is temperature. Lower temperature concentrates probability on high-scoring tokens; higher temperature flattens the distribution and increases output diversity. This is the operational connection between probability distributions, sampling, and why the same prompt can produce different responses.

12 | Distribution Transformations, Quantiles, and Expected Value

12.1 Affine Transformations of Random Variables

A transformation constructs a new random variable from an existing one. If $Y = g(X)$, then each realized X value is converted by the deterministic function g. Shifting, squeezing, and stretching are the most common cases; the most important basic case is an affine transformation,

$$Y = aX + b.$$

The coefficient a changes scale. If $|a|$ exceeds one, the distribution is stretched; if it lies between zero and one, the distribution is compressed. A negative a also reverses orientation. The constant b shifts every value by the same amount. If X has support from lower bound L to upper bound U and $a > 0$, then Y has support from $aL + b$ to $aU + b$.

$$\text{If } a > 0 \text{ and } X \in [L, U], \text{ then } Y \in [aL + b,\ aU + b].$$

Consider mapping $\mathrm{Uniform}(1, 10)$ to $\mathrm{Uniform}(4, 6)$. One structured derivation uses three steps. First subtract one, moving the support from zero to nine. Second multiply by two ninths, compressing its length from nine to two. Third add four, shifting the support to four through six.

$$Y = 4 + \frac{2}{9}(X - 1) = \frac{2}{9}X + \frac{34}{9}.$$

Checking endpoints verifies the mapping. When $X = 1$, $Y = 4$. When $X = 10$, $Y = 6$. Because the input is uniform and the transformation is linear and increasing, the output is also uniform. Its density is one divided by two, or one half.

$$f_Y(y) = \frac{1}{2} \quad \text{for } 4 \le y \le 6.$$

For a differentiable one-to-one transformation $Y = g(X)$, density changes according to the Jacobian rule. This rule preserves total probability while horizontal distances are stretched or compressed.

$$f_Y(y) = f_X\Big(g^{-1}(y)\Big) \left| \frac{d}{dy} g^{-1}(y) \right|.$$

A valid random-variable transformation does not always have to be one-to-one. For example, $Y = X^2$ is valid. One-to-one behavior is required only for the simplest single-inverse change-of-variables formula. If several X values map to the same Y, their density contributions must be summed over all inverse branches.

$$f_Y(y) = \sum_{x:\, g(x) = y} \frac{f_X(x)}{|g'(x)|}, \qquad \text{when the inverse branches are regular.}$$

12.2 Uniform and Triangular Densities

Uniform and triangular shapes offer a useful comparison. A density is valid when it is nonnegative and its total area is one. For a triangle with base from one to ten, the base length is nine. If the peak height is h, normalization requires

$$\frac{1}{2}(9)h = 1, \qquad \text{so } h = \frac{2}{9}.$$

Moving the triangular peak changes where probability mass is concentrated but not the height needed for normalization, as long as the base endpoints remain one and ten and the triangle still returns to zero at both endpoints. A peak near the left creates a right-skewed distribution with a long right tail. A peak near the right creates a left-skewed distribution. A peak at the midpoint 5.5 produces a symmetric triangular distribution.

For the symmetric triangular case, the mean and median both equal 5.5. For a skewed triangular distribution they generally differ. The mode is the location of the density peak, the median divides probability mass in half, and the mean is the balance point. These three measures answer different questions and need not coincide.

$$\text{Symmetric distribution about } c \implies \text{mean} = \text{median} = c \text{ when the mean exists;}$$
$$\text{for the symmetric triangle, } c = 5.5.$$

12.3 Median, Cumulative Probability, and the Discrete Convention

The median is a quantile. A continuous median m satisfies equal probability on either side when the density is continuous and has no flat ambiguity. More generally, a median satisfies that at least half the mass is at or below m and at least half is at or above m.

$$P(X \le m) \ge 0.5 \qquad \text{and} \qquad P(X \ge m) \ge 0.5.$$

Using the cumulative distribution function $F_X(x) = P(X \le x)$, the course convention for a discrete median is the smallest attainable value whose cumulative probability reaches or exceeds one half.

$$m = \inf\{x : F_X(x) \ge 0.5\}.$$

The family-size distribution assigns probabilities 0.15, 0.20, 0.35, and 0.30 to zero, one, two, and three children. The cumulative probabilities are 0.15, 0.35, 0.70, and 1. The first cumulative value reaching one half occurs at two, so the course median is two children.

$$F(1) = 0.35 < 0.5, \qquad \text{while} \qquad F(2) = 0.70 \ge 0.5; \qquad \text{therefore median} = 2.$$

12.4 Expected Value as a Probability-Weighted Balance Point

Expected value, also called the mean and written μ, is the probability-weighted average of possible outcomes. For a discrete variable it is a sum. For a continuous variable it is an integral against the density.

$$E[X] = \sum_x x\, p_X(x), \qquad \text{for discrete } X,$$

$$E[X] = \int_{-\infty}^{\infty} x\, f_X(x)\, dx, \qquad \text{for continuous } X.$$

The center-of-mass interpretation is useful. Imagine each possible value placed on a balance beam with a mass equal to its probability. The expected value is the balance point. Unlike the discrete median, the expected value is unique whenever it exists, but it need not be an attainable outcome. A family cannot have 1.8 children, yet 1.8 is a meaningful population average.

For the stated family distribution, the weighted mean is

$$E[X] = 0(0.15) + 1(0.20) + 2(0.35) + 3(0.30) = 1.80.$$

Expectation is linear even when variables are dependent. Constants move through the expectation operator in the natural way.

$$E[aX + b] = aE[X] + b, \qquad E[X_1 + \cdots + X_n] = E[X_1] + \cdots + E[X_n].$$

For one thousand families with the same expected family size, the expected total number of children is $1000 \times 1.8 = 1800$. If each child is assumed independently to be a girl with probability one half, the expected number of girls is 900. Linearity supplies the expected total; independence is not required for the addition of expectations, although the one-half conditional assumption is needed for the gender calculation.

$$E[\text{total children}] = 1000(1.8) = 1800; \qquad E[\text{girls}] = 0.5(1800) = 900.$$

This result illustrates a recurring theme: transformations of random variables induce transformations of summary measures. The mean responds linearly to shifts and changes of scale, while the next chapter shows that variance responds quadratically to scale.

13 | Variance, Standard Deviation, Dependence, and Risk

13.1 Why the Mean Is Not Enough

Two distributions can have the same expected value but very different risk. Consider lottery W, which pays -10 or $+10$ with equal probability, and lottery Y, which pays -100 or $+100$ with equal probability. Both means are zero, but Y has much greater dispersion and therefore exposes the decision maker to larger gains and losses.

$$E[W] = (-10)(0.5) + (10)(0.5) = 0, \qquad E[Y] = (-100)(0.5) + (100)(0.5) = 0.$$

Preference between these lotteries cannot be explained by expected value alone. A risk-averse person may prefer W, while a risk-seeking person may prefer Y. Variance and standard deviation describe dispersion, but they do not by themselves specify preferences. A complete decision model could apply expected utility, downside risk, value at risk, or another criterion.

$$\text{Expected utility} = \sum_x p(x)u(x), \qquad \text{where concave } u \text{ represents risk aversion.}$$

13.2 Variance and Standard Deviation

Variance is the expected squared deviation from the mean. If $\mu = E[X]$, subtracting μ centers each realization, squaring removes signs and emphasizes large deviations, and probability weighting recognizes that frequent deviations matter more than rare ones.

$$\text{Var}(X) = \sigma_X^2 = E\left[(X - \mu_X)^2\right].$$

For a discrete random variable, variance is calculated as a probability-weighted sum. For a continuous variable, it is an integral.

$$\text{Var}(X) = \sum_x (x - \mu_X)^2\, p_X(x), \qquad \text{Var}(X) = \int_{-\infty}^{\infty} (x - \mu_X)^2\, f_X(x)\, dx.$$

Variance has squared units. If X is measured in dollars, variance is measured in dollars squared. Standard deviation returns to the original units and is therefore easier to interpret.

$$SD(X) = \sigma_X = \sqrt{\text{Var}(X)}.$$

For W, both outcomes are ten units from the zero mean. The variance is one hundred and the standard deviation is ten.

$$\text{Var}(W) = (-10 - 0)^2(0.5) + (10 - 0)^2(0.5) = 100, \qquad SD(W) = \sqrt{100} = 10.$$

For equally likely Y values -100 and $+100$, the variance is ten thousand and the standard deviation is one hundred. This is consistent with $Y = 10W$.

$$\text{Var}(Y) = 10{,}000 \quad \text{and} \quad SD(Y) = 100.$$

13.3 The Computational Identity and Transformation Rules

Expanding the square gives an equivalent variance formula. It is often computationally convenient, although the centered formula is usually more numerically stable in software when the mean is very large relative to the spread.

$$\operatorname{Var}(X) = E[X^2] - (E[X])^2.$$

Shifting every outcome by a constant changes the mean but not the variance. Scaling every outcome by a multiplies variance by a^2 and standard deviation by $|a|$.

$$E[aX + b] = aE[X] + b, \qquad \operatorname{Var}(aX + b) = a^2 \operatorname{Var}(X), \qquad SD(aX + b) = |a|\, SD(X).$$

The absolute value matters for standard deviation because standard deviation cannot be negative. If $a = -2$, the distribution is reflected and stretched by two; its standard deviation doubles, and does not become negative.

Suppose the Y lottery is modified so that -100 occurs with probability 0.4 and $+100$ occurs with probability 0.6. The mean shifts to twenty because the positive outcome is more likely.

$$E[Y] = (-100)(0.4) + (100)(0.6) = 20.$$

The corresponding variance is nine thousand six hundred, and the standard deviation is approximately 97.98.

$$\operatorname{Var}(Y) = (-100 - 20)^2(0.4) + (100 - 20)^2(0.6) = 9{,}600, \qquad SD(Y) = \sqrt{9{,}600} \approx 97.98.$$

The decrease from variance ten thousand to nine thousand six hundred reflects the shifted mean and unequal masses. The $+100$ outcome is now closer to the mean and also more likely, reducing average squared deviation modestly.

13.4 Adding Random Variables: Independence and Covariance

Variance is not linear in the same way as expectation. For independent random variables, the variance of a weighted sum is the sum of weighted variances, with each coefficient squared.

$$\text{If } X_1 \text{ and } X_2 \text{ are independent,} \qquad \operatorname{Var}(c_1X_1 + c_2X_2) = c_1^2 \operatorname{Var}(X_1) + c_2^2 \operatorname{Var}(X_2).$$

For dependent variables, covariance captures how their deviations move together. Positive covariance means above-average values tend to occur together; negative covariance means one tends to be above average when the other is below average.

$$\operatorname{Cov}(X_1, X_2) = E[(X_1 - \mu_1)(X_2 - \mu_2)]\,,$$

$$\operatorname{Var}(c_1X_1 + c_2X_2) = c_1^2 \operatorname{Var}(X_1) + c_2^2 \operatorname{Var}(X_2) + 2c_1c_2 \operatorname{Cov}(X_1, X_2).$$

Independence implies zero covariance when second moments exist, but zero covariance does not generally imply independence. This distinction matters in regression, portfolio analysis, forecasting, and any system where multiple uncertain inputs move together.

The failure of the independent-sum formula is illustrated with $X_2 = 3X_1 + 5$. Because X_2 is constructed directly from X_1, the variables are perfectly dependent. First, the variance of X_2 is nine times the variance of X_1.

$$X_2 = 3X_1 + 5 \implies \mathrm{Var}(X_2) = 9\,\mathrm{Var}(X_1).$$

Adding them gives $4X_1 + 5$. Therefore the exact variance is sixteen times the variance of X_1, not ten times.

$$\mathrm{Var}(X_1 + X_2) = \mathrm{Var}(4X_1 + 5) = 16\,\mathrm{Var}(X_1).$$

The missing six units of variance come from twice the covariance. Since $\mathrm{Cov}(X_1, 3X_1{+}5) = 3\,\mathrm{Var}(X_1)$, the covariance term is six variance units.

$$2\,\mathrm{Cov}(X_1, X_2) = 2 \times 3\,\mathrm{Var}(X_1) = 6\,\mathrm{Var}(X_1).$$

Covariance depends on measurement units, so correlation is often used to express standardized dependence. Correlation divides covariance by the two standard deviations and therefore lies between -1 and $+1$. Values near $+1$ indicate strong positive linear co-movement, values near -1 indicate strong negative linear co-movement, and values near zero indicate little linear association.

$$\mathrm{Corr}(X_1, X_2) = \frac{\mathrm{Cov}(X_1, X_2)}{SD(X_1)\,SD(X_2)}, \qquad -1 \le \mathrm{Corr} \le 1.$$

For $X_2 = 3X_1 + 5$, assuming X_1 has positive variance, the correlation is $+1$ because X_2 is a perfectly increasing affine function of X_1. If the coefficient had been negative, the correlation would be -1. This makes the dependence failure in the variance calculation visible on a standardized scale.

The risk calculation also distinguishes population moments from estimates. The formulas above describe the true distributional mean and variance. With observed data, analysts estimate them using a sample mean and sample variance. The common unbiased sample variance divides the sum of squared deviations by $n - 1$ because the sample mean was estimated from the same data.

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i; \qquad s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2.$$

This is the key modeling lesson: whether risks diversify depends on dependence. Adding independent risks can smooth relative uncertainty, while adding highly positively correlated risks does not provide the same diversification benefit.

14 | The Normal Distribution, Standardization, and Z Probabilities

14.1 The Normal Density and Its Normalization

Unlike the uniform examples with finite lower and upper bounds, a normal random variable has support across the entire real line. Its density is symmetric, peaks at the mean μ, and decays rapidly as x moves away from μ. The parameter σ is the standard deviation, and σ^2 is the variance.

$$X \sim N(\mu, \sigma^2).$$

The normal probability density is

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right], \qquad -\infty < x < \infty.$$

The exponential term creates the bell shape. Squaring $x - \mu$ makes the curve symmetric around μ. The negative sign causes density to decrease as squared distance grows. The factor $1/(\sigma\sqrt{2\pi})$ is the normalizing constant that makes the total area equal one.

$$\int_{-\infty}^{\infty} f_X(x)\,dx = 1.$$

Changing μ shifts the entire curve without changing its shape. Increasing σ spreads the same total probability over a wider range, lowering the peak. Decreasing σ concentrates probability around the mean and raises the peak. For every normal distribution, symmetry implies that mean, median, and mode coincide at μ.

$$\text{For a normal distribution,} \qquad \text{mean} = \text{median} = \text{mode} = \mu.$$

14.2 The Empirical Rule and Tail Behavior

Normal distributions are unbounded, but most probability mass lies within a few standard deviations of the mean. The standard empirical rule gives approximately 68.27% within one standard deviation, 95.45% within two, and 99.73% within three.

$$P(\mu - \sigma \le X \le \mu + \sigma) \approx 0.6827,$$

$$P(\mu - 2\sigma \le X \le \mu + 2\sigma) \approx 0.9545,$$

$$P(\mu - 3\sigma \le X \le \mu + 3\sigma) \approx 0.9973.$$

These are the canonical 68–95–99.7 normal-distribution percentages, and this book uses these standard values throughout.

The probability outside plus or minus three standard deviations is approximately 0.0027 in total, or about 0.00135 in each tail. Rare does not mean impossible: the normal model still assigns positive probability to arbitrarily extreme finite observations.

$$P(|X - \mu| > 3\sigma) \approx 1 - 0.9973 = 0.0027.$$

14.3 Affine Transformations of Normal Variables

Normal distributions are closed under affine transformations. If X is normal and $Y = aX + b$, then Y is also normal. Its mean is transformed linearly and its variance is multiplied by a^2.

$$\text{If } X \sim N(\mu, \sigma^2), \text{ then } aX + b \sim N(a\mu + b,\ a^2\sigma^2).$$

The corresponding standard deviation is the absolute value of a times σ. A pure shift $X + b$ changes only the mean. A pure scale aX changes both the mean and the spread.

$$E[aX + b] = a\mu + b; \qquad SD(aX + b) = |a|\,\sigma.$$

Standardization converts any nondegenerate normal variable to the standard normal distribution. Subtracting μ shifts the mean to zero, and dividing by σ rescales the standard deviation to one.

$$Z = \frac{X - \mu}{\sigma}, \qquad Z \sim N(0, 1).$$

The standard normal cumulative distribution function, written Φ, returns the left-tail probability. A Z table is a printed lookup table for Φ.

$$\Phi(z) = P(Z \le z) = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{t^2}{2}\right) dt.$$

Symmetry connects positive and negative table values. The left-tail probability at $-z$ equals one minus the left-tail probability at $+z$.

$$\Phi(-z) = 1 - \Phi(z).$$

14.4 Worked Z-Table Calculations

Suppose X is normal with mean one hundred and standard deviation fifty. To find the probability that X is at most one hundred ten, standardize the cutoff.

$$z = \frac{110 - 100}{50} = 0.20.$$

The Z table gives $\Phi(0.20) \approx 0.57926$. Thus about 57.9% of this distribution lies at or below 110.

$$P(X \le 110) = \Phi(0.20) = 0.57926.$$

For an interval probability, standardize both endpoints and subtract cumulative left-tail areas. The interval from ninety to one hundred ten maps to -0.20 through $+0.20$.

$$P(90 \le X \le 110) = P(-0.20 \le Z \le 0.20) = \Phi(0.20) - \Phi(-0.20).$$

Using the table values 0.57926 and 0.42074, the desired central area is

$$0.57926 - 0.42074 = 0.15852.$$

The same method handles any interval. For a right-tail probability, subtract a left-tail table value from one. For a two-sided probability, standardize both bounds and subtract. It is important to always state whether the second normal parameter denotes variance or standard deviation; this book writes $N(\mu, \sigma^2)$ so that the standard deviation used in a Z score is σ.

$$P(X > c) = 1 - \Phi\left(\frac{c-\mu}{\sigma}\right), \qquad P(a \le X \le b) = \Phi\left(\frac{b-\mu}{\sigma}\right) - \Phi\left(\frac{a-\mu}{\sigma}\right).$$

These normal calculations prepare for confidence intervals and statistical inference, taken up in the chapters that follow. Standardization converts a problem with arbitrary units into a universal reference scale. Once the distribution of a standardized statistic is known, table values or software can translate distances from the mean into probabilities, tail areas, critical values, and confidence levels.

14.5 Chapter Summary

The chapter can be summarized as a pipeline: first define a numerical random variable and its support; second, specify a normalized PMF or PDF; third, distinguish theoretical probabilities from finite-sample frequencies; fourth, use transformations to connect variables and derive their means and variances; fifth, when a normal model is appropriate, standardize to Z and calculate areas with the standard normal distribution.

Model $\to$ distribution $\to$ sample $\to$ summary measures $\to$ standardized inference.

The AI examples fit the first half of this pipeline: model outputs are random variables, softmax supplies normalized conditional masses, and decoding samples from those masses. The statistical examples fit the second half: expectation summarizes center, variance summarizes spread, covariance captures dependence, and standardization makes normal probabilities comparable across measurement scales.

Rapid Review

- A random variable maps experimental outcomes to numbers; it may be discrete or continuous.
- A PMF sums to one. A PDF integrates to one, and probability is area under the density.
- For Uniform(a, b), density is $1/(b - a)$.
- An affine transformation $Y = aX + b$ shifts and rescales a distribution.
- The mean is a probability-weighted balance point; expectation is linear.
- Variance is expected squared distance from the mean; standard deviation is its square root.
- Scaling by a multiplies variance by a^2; shifting by b does not change variance.

- Variances add directly only under zero covariance; independence is a sufficient condition.
- A normal variable has bell-shaped density parameterized by μ and σ^2.
- Standardization $Z = (X - \mu)/\sigma$ converts normal probabilities to the $N(0, 1)$ reference scale.

Core Formula Sheet

- Uniform density: $f_X(x) = 1/(b - a)$, for $a \leq x \leq b$.
- Expectation: $E[X] = \sum x\, p_X(x)$, or $\int x f_X(x)\, dx$.
- Variance: $\mathrm{Var}(X) = E[(X - \mu)^2] = E[X^2] - \mu^2$.
- Affine transformation: $E[aX + b] = aE[X] + b$; $\mathrm{Var}(aX + b) = a^2\, \mathrm{Var}(X)$.
- Covariance sum rule: $\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\, \mathrm{Cov}(X, Y)$.
- Normal density: $f(x) = \left[1/(\sigma\sqrt{2\pi})\right] \exp\left[-(x - \mu)^2/(2\sigma^2)\right]$.
- Standardization: $Z = (X - \mu)/\sigma$.
- Interval probability: $P(a \leq X \leq b) = \Phi((b - \mu)/\sigma) - \Phi((a - \mu)/\sigma)$.

15 | From a Population Distribution to Statistical Estimation

The easiest way to understand this chapter is to see it as answering one fundamental question: *what can we learn about an unknown population when we cannot observe the entire population?* That question takes us from probability theory into statistics.

15.1 From a Known Distribution to an Unknown One

In the previous chapters, the starting point was a probability distribution. If we know the probability distribution of a random variable X, then we can calculate quantities such as its expected value, variance, and standard deviation.

For a continuous random variable, a probability density must satisfy two fundamental requirements: it must be nonnegative, and the total area underneath the density must equal one. An important distinction is that the density itself does not necessarily have to be below one — what must lie between zero and one is probability, obtained by calculating areas under the density.

Once the distribution is known, the expected value is conceptually straightforward: a probability-weighted average of possible outcomes. For a discrete random variable,

$$E[X] = \sum_x xP(x).$$

So expected value is simply a weighted average, where the probabilities provide the weights.

Variance uses the same basic idea but asks a different question. Instead of asking where the distribution is centered, variance asks how dispersed the outcomes are around that center. Thus,

$$\mathrm{Var}(X) = E[(X - \mu)^2].$$

We first calculate the difference between an observation and the population mean, square that difference, and then take its probability-weighted average. Squaring is important because deviations can be positive or negative; we care about their magnitude rather than their sign, and squaring also avoids some of the mathematical complications associated with absolute values.

Variance is therefore always nonnegative. It can equal zero only in the degenerate case where every observation is exactly the same. As variation among observations increases, variance increases.

The problem with variance is its unit. If X is measured in dollars, variance is measured in dollars squared. Therefore, we take its square root and obtain the standard deviation,

$$\sigma = \sqrt{\mathrm{Var}(X)}.$$

Standard deviation brings uncertainty back into the original units of the variable.

15.2 The Fundamental Practical Problem

All of these formulas appear simple when the population probability distribution $P(x)$ is known. But in real applications, **we usually do not know** $P(x)$. That is the central problem motivating the rest of this chapter.

Building or observing the complete probability distribution can be extraordinarily expensive or practically impossible. A population need not consist of people: it might consist of possible sales outcomes, words or sentences in a language, images, driving environments, customer behavior, or almost anything else that we want to model.

So instead of observing the entire population distribution, we collect a random sample:

$$X_1, X_2, \ldots, X_n.$$

This leads to one of the most important tricks in analytics. Rather than explicitly estimating the entire probability distribution $P(x)$, we replace probability weighting by equal weighting of randomly sampled observations. Each sampled observation receives weight

$$\frac{1}{n}.$$

Consequently, $E[X]$ is estimated by

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i.$$

This is the sample mean. The deeper message is more important than this particular formula: whenever an unknown population quantity can be represented as an expectation under $P(x)$, random sampling gives us a way of approximating that expectation without explicitly knowing the entire distribution. This is why ordinary sample averages connect to **Monte Carlo estimation** — sample means and sample variances can be viewed as simple applications of the much broader Monte Carlo principle.

15.3 Why Random Sampling Is Doing the Work

There is a crucial qualification: the observations must actually be collected in an appropriately random fashion. Suppose the sample consists of 100 observations. Giving each observation weight $1/100$ creates an empirical distribution over those observations. The hope is that this empirical distribution becomes increasingly representative of the underlying population as the sample becomes larger.

Consider an MBA salary dataset. The population contains more than 2,000 observations, and the population mean salary is approximately 111,208. A random sample of 100 observations produces a sample mean of approximately 111,568. Those numbers are not identical, and they should not be identical: sampling introduces randomness. If we drew another sample of 100 people, we would obtain another sample mean, and another sample would give yet another.

But something very important happens as the sample size increases: the sample mean tends to get closer to the population mean. This leads naturally to the **Law of Large Numbers**. In simplified notation,

$$\bar{X} \to \mu \qquad \text{as} \qquad n \to \infty.$$

The sample does not need to reproduce the population perfectly at any particular finite sample size. Instead, the procedure has the desirable property that increasing the amount of data moves us toward the population quantity.

15.4 The Important Complication: Estimating Variance

If the population mean μ were known, we could estimate variance by averaging $(X_i - \mu)^2$. But this creates an obvious problem: if we knew the true population mean, we would already know something about the population that we are trying to estimate. In practice, μ is unknown, so we replace it with the sample mean $\bar{X}$, giving squared deviations $(X_i - \bar{X})^2$.

But using $\bar{X}$ creates another consequence: these deviations are no longer completely free. Once the sample mean is calculated, the deviations around that sample mean must satisfy

$$\sum_{i=1}^{n}(X_i - \bar{X}) = 0.$$

If you know $n-1$ of those deviations, the final deviation is automatically determined because all deviations must add to zero. Therefore, although there are n observations, there are only $n-1$ independent deviations. We say that we have lost **one degree of freedom**. That is the intuition behind the familiar sample variance formula,

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2.$$

Why not divide by n? Because $\bar{X}$ was itself estimated using the same data. Dividing by n would systematically make the estimated variance too small. The $n-1$ adjustment compensates for this tendency. This idea becomes even more important later in regression, where the same dataset is used to estimate multiple parameters and consequently additional degrees of freedom are consumed. The general lesson: every time the data are used to estimate an unknown quantity that subsequently enters another calculation, degrees of freedom must be tracked carefully.

15.5 Accuracy Improves Surprisingly Slowly

The next major idea concerns how quickly statistical estimates improve as more data are collected. Suppose 100 observations give us some estimation error. If we want to cut that error in half, doubling the sample size is *not* sufficient — we need approximately four times as much data. This is the **square-root relationship**: statistical error generally shrinks according to

$$\frac{1}{\sqrt{n}}.$$

Therefore, if $n \to 4n$, then

$$\frac{1}{\sqrt{n}} \to \frac{1}{2\sqrt{n}}.$$

So quadrupling the sample size approximately halves the uncertainty. Likewise, if we want the error to become one-fourth as large, we need approximately sixteen times the sample size. More data help, but there are diminishing returns to additional data: moving from a poor estimate to a reasonably good estimate may be relatively inexpensive, while moving from a very good estimate to an extremely precise estimate can require enormous amounts of additional data. The same intuition applies to modern machine learning and AI, where reducing estimation error further and further can require very large increases in the amount of training data.

15.6 The Assumption Underneath Everything: Randomness and Independence

There is one condition underneath almost everything discussed so far: the sample must behave like a **random sample from the population**. If the data collection mechanism systematically favors certain observations, the empirical distribution created by assigning $1/n$ weight to observations may no longer properly represent the population.

Postal codes would generally be a poor mechanism for randomly sampling incomes because geography and income can be related. Similarly, conducting an election survey only in neighborhoods with particular political characteristics would obviously distort the results. The broader point is not about postal codes or elections; it is about the data-generating mechanism. Before applying statistical formulas, we should ask: how did these observations enter the dataset? Were observations selected independently? Was some part of the population systematically more likely to appear? Does observing one individual affect which individual will be observed next? Are there clusters or other dependence structures?

If the random-sampling assumption fails, simply increasing the sample size does not necessarily fix the problem. A huge biased sample can still produce an extremely precise estimate of the *wrong* quantity. Randomization, reshuffling, and resampling are practical ways of trying to preserve the statistical logic underlying estimation. In real company datasets, this becomes especially important because database records are rarely generated specifically for statistical inference.

15.7 The Conceptual Transition

We can now summarize the first major movement of this chapter. We began with the population, $P(X)$. If $P(X)$ were known, we could calculate μ, σ^2, σ directly. But $P(X)$ is usually unknown, so we take a random sample, $X_1, \ldots, X_n$. From that sample we calculate statistics such as $\bar{X}$ and S^2. The sample mean estimates the population mean. The sample variance estimates the population variance, with the $n-1$ correction because the sample mean has itself been estimated from the same data. As n increases, these estimates become increasingly informative about their population counterparts.

But now a new question arises. Suppose our sample tells us that the mean is 111,568. We know that this number is almost certainly not exactly equal to the true population mean. So how uncertain should we be about it? In other words, **how do we quantify the error around a point estimate?** Instead of reporting only $\bar{X}$, we want something like

$$\bar{X} \pm \text{margin of error.}$$

To determine that margin of error, we need to understand the probability distribution of the estimator $\bar{X}$ itself. That brings us to one of the central results of statistics: the **Central Limit Theorem**, the subject of the next chapter.

16 | Sampling Distributions and the Central Limit Theorem

The previous chapter ended with a new problem. We know how to estimate the population mean using the sample mean,

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i.$$

But once we calculate a sample mean, say 111,568, we know that this number is only an estimate. A different random sample would generally produce a different sample mean. So the key question becomes: **how much does the sample mean itself vary from sample to sample?** That question leads us to the concept of a **sampling distribution**.

16.1 The Sample Mean Is Itself a Random Variable

Suppose that we repeatedly draw random samples of size n from the same population. From the first sample, we calculate one sample mean; from the second, another; from the third, another. So we would obtain

$$\bar{X}_1, \bar{X}_2, \bar{X}_3, \ldots$$

These sample means vary because the underlying samples vary. Therefore, $\bar{X}$ is itself a random variable, and like every random variable, it has its own probability distribution. That distribution is called the **sampling distribution of the sample mean**.

This is conceptually different from the original population distribution. The population distribution describes the individual observations X. The sampling distribution describes the possible values of the statistic $\bar{X}$ that we would obtain across repeated samples. This distinction is one of the most important ideas in statistical inference.

16.2 Expected Value and Variance of the Sample Mean

What is $E[\bar{X}]$? Recall that $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$. Taking expectations,

$$E[\bar{X}] = E\left[\frac{1}{n}\sum_{i=1}^{n} X_i\right].$$

Because expectation is linear,

$$E[\bar{X}] = \frac{1}{n}\sum_{i=1}^{n} E[X_i].$$

Each X_i comes from the same population, so $E[X_i] = \mu$. Therefore,

$$E[\bar{X}] = \frac{1}{n}\sum_{i=1}^{n} \mu = \mu,$$

since there are n copies of μ. This is the mathematical reason that the sample mean is centered at the true population mean, and the key reason why $\bar{X}$ is a good estimator of μ: if we repeatedly took samples and averaged their sample means, that long-run average would equal the population mean.

The next question is even more important for confidence intervals: how variable is $\bar{X}$? We start again with $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$, so

$$\mathrm{Var}(\bar{X}) = \mathrm{Var}\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right).$$

Assuming the observations are independent,

$$\mathrm{Var}\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} \mathrm{Var}(X_i).$$

The $1/n$ outside the sum enters the variance squared, so

$$\mathrm{Var}(\bar{X}) = \frac{1}{n^2}\sum_{i=1}^{n} \mathrm{Var}(X_i).$$

Each observation has population variance σ^2, therefore

$$\mathrm{Var}(\bar{X}) = \frac{1}{n^2} n\sigma^2 = \frac{\sigma^2}{n}.$$

This result is fundamental. Taking the square root, the standard deviation of $\bar{X}$ is

$$\sqrt{\frac{\sigma^2}{n}} = \frac{\sigma}{\sqrt{n}}.$$

This quantity is the standard deviation of the sampling distribution of the sample mean. In practical statistics, we usually call it the **standard error of the mean**,

$$SE(\bar{X}) = \frac{\sigma}{\sqrt{n}}.$$

The sample-mean distribution is much narrower than the original population distribution because averaging reduces noise: an individual observation can vary substantially, but an average of many observations is more stable. This is exactly why sample means become more precise as sample size increases.

Notice that the standard error, $\sigma/\sqrt{n}$, follows the same square-root relationship discussed in the previous chapter. If we multiply the sample size by four, $n \to 4n$, then

$$SE = \frac{\sigma}{\sqrt{4n}} = \frac{1}{2}\frac{\sigma}{\sqrt{n}},$$

so the standard error is cut in half. If we multiply the sample size by sixteen, SE is divided by four. This is the mathematical reason behind the earlier statement that halving estimation error requires approximately quadrupling the sample size.

16.3 The Central Limit Theorem

We already know two things about the sampling distribution of $\bar{X}$: $E[\bar{X}] = \mu$ and $SD(\bar{X}) = \sigma/\sqrt{n}$. But we still need to know its **shape**. The Central Limit Theorem gives us the answer: as the sample size becomes sufficiently large, the distribution of $\bar{X}$ approaches a normal distribution,

$$\bar{X} \approx N\left(\mu, \frac{\sigma}{\sqrt{n}}\right).$$

The crucial point is that this happens even if the underlying population itself is not normally distributed. The population may be uniform, right-skewed, left-skewed, or have some other unusual shape. Nevertheless, as the sample size increases, the distribution of the sample mean increasingly resembles a bell-shaped normal distribution. That is the extraordinary power of the Central Limit Theorem.

This theorem gives us a standardized way to quantify uncertainty. We know that approximately 95 percent of the standard normal distribution's probability lies between -1.96 and $+1.96$. So after standardizing the sample mean,

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}},$$

we know that approximately

$$P\left(-1.96 \leq \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq 1.96\right) = 0.95.$$

This is the bridge from the Central Limit Theorem to confidence intervals.

16.4 Turning the Probability Statement Around

We do not actually want an interval for $\bar{X}$: we observe $\bar{X}$. What is unknown is μ. So we rearrange the inequality algebraically so that μ appears in the middle. This produces

$$P\left(\bar{X} - 1.96\frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + 1.96\frac{\sigma}{\sqrt{n}}\right) = 0.95.$$

This is the standard 95 percent confidence interval for a population mean when the population standard deviation is known. We therefore define

$$\text{Lower Bound} = \bar{X} - 1.96\frac{\sigma}{\sqrt{n}}, \qquad \text{Upper Bound} = \bar{X} + 1.96\frac{\sigma}{\sqrt{n}}.$$

The quantity $1.96\,\sigma/\sqrt{n}$ is called the **margin of error**, so the entire expression can be remembered as

$$\boxed{\text{Confidence interval} = \text{Point estimate} \pm \text{Margin of error}} \quad \text{or} \quad \boxed{\bar{X} \pm 1.96\frac{\sigma}{\sqrt{n}}}.$$

16.5 Point Estimator versus Interval Estimator

The sample mean itself, $\bar{X}$, is a **point estimator**: it gives one number. But one number alone tells us nothing about how precise that estimate is. The confidence interval is an **interval estimator**: it gives a range around the point estimate.

This distinction is very important. Two studies might both estimate the same mean, say $\bar{X} = 100$ in Study A and $\bar{X} = 100$ in Study B, yet have very different confidence intervals: $[99, 101]$ for Study A versus $[70, 130]$ for Study B. The point estimates are identical, but Study A provides far more precise information. That is why reporting uncertainty is so valuable.

16.6 What Does 95 Percent Confidence Mean?

Confidence level is the probability with which the interval procedure captures the unknown population parameter. For a 95 percent confidence interval, the central area of the relevant sampling distribution is 95 percent, and the normal distribution leaves 5% outside the interval. Because the standard confidence interval is symmetric, this 5 percent is divided equally: 2.5% in the left tail and 2.5% in the right tail. This means the cumulative probability up to the positive cutoff is 0.975. Looking this up in the Z-table gives $Z = 1.96$; likewise, the lower cutoff is -1.96.

The margin of error is $MOE = Z\,\sigma/\sqrt{n}$. Suppose the confidence level and population variability stay fixed. As n increases, $\sqrt{n}$ increases, so MOE decreases and the confidence interval becomes narrower — more data produce more precise estimates, although the improvement is only at the square-root rate.

There is another tradeoff. Suppose we move from a 95 percent confidence level to 99 percent. To capture more of the sampling distribution, we must move further into the tails, which requires a larger critical value. A larger critical value produces a larger margin of error, so **higher confidence produces a wider confidence interval**. There is always a tradeoff between confidence and precision: to be more confident that the procedure covers the true parameter, we must accept a wider interval unless we collect more data.

16.7 The Salary Example

Applying this procedure to the MBA salary data: a sample of size $n = 100$ produced a sample mean of roughly 111,568. The population standard deviation, available here for teaching purposes because the entire population dataset is available, is approximately 47,506. For 95 percent confidence, $Z = 1.96$. The margin of error is

$$1.96 \times \frac{47{,}506}{\sqrt{100}}.$$

Since $\sqrt{100} = 10$, the standard error is approximately 4,751, and the margin of error is roughly 9,312. So the confidence interval is approximately

$$111{,}568 \pm 9{,}312.$$

The important conceptual structure, more than the precise arithmetic, is

$$\boxed{\bar{X} \pm Z\frac{\sigma}{\sqrt{n}}}.$$

16.8 But There Is a Major Problem

The formula above contains σ, the **population standard deviation**. If we knew the entire population well enough to calculate σ, we would often have far more information than we normally have in real statistical problems. Usually, the population standard deviation is unknown, so we estimate it using the sample standard deviation, S. But now we are estimating two population quantities from the sample: the mean, μ, and the standard deviation, σ. That extra estimation introduces additional uncertainty. Therefore, we should not continue using the normal Z-distribution exactly as before; we need a distribution with heavier tails to acknowledge that extra uncertainty. That distribution is the **Student t-distribution**, the subject of the next chapter.

17 | Confidence Intervals, Z versus t, and Sample-Size Calculation

We ended the previous chapter with an important practical problem. The confidence interval for a population mean was derived as

$$\bar{X} \pm Z\frac{\sigma}{\sqrt{n}}.$$

This formula works when the population standard deviation σ is known. But in real applications, that is often unrealistic: usually we do not know the population mean, and we also do not know the population standard deviation. What changes when σ is unknown? The answer is that we estimate σ using the sample standard deviation, and because this introduces additional uncertainty, we replace the normal Z-distribution with the **Student t-distribution**.

17.1 Estimating the Unknown Population Standard Deviation

Recall the sample standard deviation,

$$S = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2}.$$

The $n-1$ appears because $\bar{X}$ has already been estimated from the same data. Once σ is unknown, we substitute S for it, but we cannot simply write $\bar{X} \pm Z\, S/\sqrt{n}$ and continue as though nothing happened, because S itself is random: it changes from sample to sample. Using the sample to estimate another unknown quantity means our margin of error needs to become larger, and consequently the resulting confidence interval becomes wider.

17.2 Enter the t-Distribution

To account for this extra uncertainty, we use the t-distribution. The confidence interval becomes

$$\boxed{\bar{X} \pm t\frac{S}{\sqrt{n}}} \qquad \text{instead of} \qquad \bar{X} \pm Z\frac{\sigma}{\sqrt{n}}.$$

This is one of the most important distinctions in this chapter:

$$\text{if } \sigma \text{ is known: } \boxed{\bar{X} \pm Z\frac{\sigma}{\sqrt{n}}}; \qquad \text{if } \sigma \text{ is unknown: } \boxed{\bar{X} \pm t\frac{S}{\sqrt{n}}}.$$

The normal distribution is more concentrated in the center, while the t-distribution is flatter and has more probability in its tails. Because the total area under any probability density must still equal one, shifting more area into the tails means there is less mass near the center. We want heavier tails because, when we estimate σ instead of knowing it exactly, we should acknowledge that extreme errors are somewhat more plausible. The heavier-tailed t-distribution does exactly that: it produces a larger critical value than the corresponding Z critical value, especially for small samples, and that larger critical value makes the confidence interval wider. The t-distribution is essentially a penalty for not knowing the true population standard deviation.

17.3 Degrees of Freedom Appear Again

The appropriate t-distribution depends on the number of **degrees of freedom**. For a one-sample mean,

$$df = n - 1.$$

This is the same degree-of-freedom logic discussed earlier for the sample variance: one parameter, the sample mean, has been estimated from the data, so one degree of freedom is lost. As the sample size increases, $df = n - 1$ becomes larger, and the t-distribution gradually approaches the normal distribution. In the limit, with infinitely many degrees of freedom, the t-distribution essentially becomes the normal distribution — the normal distribution is the limiting form of the t-distribution as degrees of freedom become very large. This makes intuitive sense: with a very large sample, the estimate S of σ becomes highly accurate, so there is less need for the extra t-distribution correction.

A subtle but important point concerns $n - 1$. When we calculate S, we use $n - 1$. But in the standard-error expression we still use $S/\sqrt{n}$, not $S/\sqrt{n-1}$. These two appearances of sample size come from different pieces of theory: the $n - 1$ inside S corrects the estimation of the population variance, while the $\sqrt{n}$ in the denominator of the standard error comes from the variance of the sample mean,

$$\mathrm{Var}(\bar{X}) = \frac{\sigma^2}{n}.$$

So even when σ is unknown and replaced by S, the standard-error structure remains $S/\sqrt{n}$. These are different corrections and should not be mixed.

17.4 Known Sigma versus Unknown Sigma, and Proportions

The decision rule can be summarized very simply.

Case 1: Population standard deviation known. Use Z:

$$\bar{X} \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}.$$

Case 2: Population standard deviation unknown. Estimate it using the sample standard deviation S, calculated using $n - 1$, and use t:

$$\bar{X} \pm t_{\alpha/2, n-1} \frac{S}{\sqrt{n}}.$$

A third case, briefly mentioned, is estimating a **population proportion**: the procedure uses the Z-score rather than the t-score. So the summary table has three cases: known population standard deviation uses Z; unknown population standard deviation uses t; and a population proportion uses Z. The detailed treatment of proportions is deferred to a later chapter.

17.5 Different Confidence Levels

The critical value depends on the confidence level. For a 95 percent confidence interval, we used $Z = 1.96$. For a 90 percent confidence interval, 90 percent of the area should be in the center, leaving 10% outside; with a symmetric interval, the two tails each contain 5%. To obtain the positive critical value, we look for cumulative probability 0.95, which gives $Z \approx 1.645$. The general lesson is not to memorize every possible Z-value, but to understand how the confidence level determines the tail probability.

This reinforces another tradeoff: for the same sample, a 90 percent confidence interval is narrower than a 95 percent interval, and a 95 percent interval is narrower than a 99 percent interval. Higher confidence requires capturing more of the probability distribution, which means moving farther into the tails, increasing the critical value and therefore the margin of error:

$$\text{higher confidence} \quad \Longrightarrow \quad \text{wider interval.}$$

There is no contradiction here — greater confidence comes at the price of lower precision.

The standard intervals used in this chapter are symmetric, $\bar{X} - MOE$ to $\bar{X} + MOE$, which corresponds to treating overestimation and underestimation equally. But in some applications we may care much more about one direction — for instance, underestimating demand may be much more costly than overestimating it. In such cases, one could construct a **one-sided confidence bound** or otherwise allocate tail probability asymmetrically. The symmetric interval is a modeling choice rather than an absolute necessity.

17.6 Reversing the Problem: How Much Data Do We Need?

Until now, the sample size n was given, and we asked how precise our estimate is. Now we reverse the question: before collecting data, suppose we decide how precise we want the estimate to be, and ask how many observations must we collect. This is simply reverse engineering the margin-of-error formula.

Consider the known-σ case. Margin of error is

$$MOE = Z\frac{\sigma}{\sqrt{n}}.$$

Suppose the maximum acceptable margin of error is E. Then set

$$E = Z\frac{\sigma}{\sqrt{n}}.$$

Solving for n: multiply both sides by $\sqrt{n}$ to get $E\sqrt{n} = Z\sigma$, so $\sqrt{n} = Z\sigma/E$. Squaring both sides,

$$\boxed{n = \left(\frac{Z\sigma}{E}\right)^2}.$$

This equation gives the minimum sample size required to reach the target margin of error.

17.7 The Salary Example, Revisited

The target margin of error is 10,000. For a 95 percent confidence interval, $Z = 1.96$. The population standard deviation is approximately 47,506.15. So,

$$n = \left(\frac{1.96 \times 47{,}506.15}{10{,}000}\right)^2.$$

The calculation gives approximately 86.69. But we cannot collect 86.69 observations, and if the requirement is *at least* the desired precision, we must always round upward. Therefore,

$$n = 87.$$

At least 87 observations are required to obtain a margin of error no larger than 10,000 under the assumptions of this example. Even if the result were 86.01, we would still need 87 observations — rounding downward would fail to guarantee the targeted precision.

17.8 The Deeper Lesson from the Sample-Size Formula

Consider again

$$n = \left(\frac{Z\sigma}{E}\right)^2.$$

This formula summarizes several important relationships. If the population is more variable, σ is larger, so we need more observations:

$$\sigma \uparrow \implies n \uparrow.$$

If we want higher confidence, the critical value Z becomes larger, so we need more observations:

$$\text{confidence} \uparrow \implies n \uparrow.$$

If we demand a smaller margin of error E, we need substantially more observations. Most importantly, E is in the denominator and is squared, so if we cut the desired margin of error in half, $E \to E/2$, the required sample size becomes $4n$. This is exactly the square-root relationship discussed earlier, now seen from the reverse direction.

17.9 Connecting the Whole Chapter Sequence

We can now see the complete logical progression across these chapters. We began with a population probability distribution: if we knew that distribution, we could calculate its expected value and variance directly. But in real applications, we usually do not know the distribution, so we take a random sample. Using that sample, we estimate population quantities: the population mean μ is estimated by $\bar{X}$, and the population variance σ^2 is estimated by $S^2 = \frac{1}{n-1}\sum(X_i - \bar{X})^2$. Random sampling is essential because it allows these sample statistics to meaningfully represent the population.

The sample mean itself is random: its expected value is $E[\bar{X}] = \mu$, and its variance is $\mathrm{Var}(\bar{X}) = \sigma^2/n$, so its standard error is $\sigma/\sqrt{n}$. The Central Limit Theorem tells us that the sampling distribution of $\bar{X}$ becomes approximately normal as sample size grows, which enables us to turn a point estimate into a confidence interval: $\bar{X} \pm Z\,\sigma/\sqrt{n}$ if σ is known, or $\bar{X} \pm t\,S/\sqrt{n}$ if σ is unknown. Finally, if we decide in advance how much estimation error we are willing to tolerate, we reverse the margin-of-error equation to determine how much data we need.

The Most Important Formulas to Remember

- Sample mean: $\bar{X} = \dfrac{1}{n}\sum_{i=1}^{n} X_i$.
- Sample variance: $S^2 = \dfrac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2$.
- Sampling distribution of the sample mean: $E[\bar{X}] = \mu,\ \ \mathrm{Var}(\bar{X}) = \dfrac{\sigma^2}{n}$.
- Standard error: $SE(\bar{X}) = \dfrac{\sigma}{\sqrt{n}}$.
- Known population standard deviation: $CI = \bar{X} \pm Z\dfrac{\sigma}{\sqrt{n}}$.
- Unknown population standard deviation: $CI = \bar{X} \pm t\dfrac{S}{\sqrt{n}}$.
- Sample size for a target margin of error E, when σ is known: $n = \left(\dfrac{Z\sigma}{E}\right)^2$.

This completes the chapter sequence on random variables, distributions, and statistical estimation. The next part turns to hypothesis testing, which builds directly on the sampling-distribution and confidence-interval logic developed here.

Part IV

Hypothesis Testing for Management Analytics (Module 4)

18 | The Logic of Hypothesis Testing

The main objective of this module is to understand what hypothesis testing is actually trying to accomplish before worrying about formulas.

This distinction is important because hypothesis testing can otherwise look like a mechanical sequence: calculate a test statistic, find a critical value, compare two numbers, and either reject or fail to reject. But the lecture emphasizes that there is a deeper logic underneath these calculations.

18.1 Start with a business or scientific conjecture

Suppose a company operates a website and observes a relatively low purchase or click-through rate.

There could be many explanations. Perhaps prices are too high. Perhaps customers find competing products elsewhere. Perhaps the website is difficult to navigate. Perhaps the placement of information or even something such as the company logo affects behavior.

The company therefore develops hypotheses about possible causes.

For example:

Giving customers a 10 percent price discount will increase the click-through rate.

That is initially only a conjecture.

The next question is:

How can data provide convincing evidence that the conjecture is correct?

One approach is to conduct an experiment.

Customers can be randomly divided into two groups. A control group sees the normal price. A treatment group sees the discounted price. We then compare outcomes such as click-through rates or purchase rates.

The same basic logic appears in randomized controlled medical trials. One group receives a treatment while another receives a placebo, and researchers ask whether the difference in outcomes is sufficiently large to attribute it to the treatment rather than random variation.

In technology companies, essentially the same procedure is often called A/B testing.

So hypothesis testing provides a formal mechanism for deciding whether an observed difference represents genuine evidence or could reasonably have occurred simply because of randomness.

18.2 Null versus alternative hypothesis

The next idea is probably the most important conceptual point in the lecture.

There are two competing statements.

The null hypothesis, written H_0, represents the benchmark or default position.

The alternative hypothesis, written H_A, represents the claim we are trying to establish.

The lecture gives a useful practical rule:

Put what you want to demonstrate into the alternative hypothesis. Put its opposite into the null hypothesis.

Suppose your research claim is:

The new design increases the click-through rate.

Then that statement belongs in H_A.

The null hypothesis represents the opposite position.

This structure is intentional. Hypothesis testing is constructed so that we begin by provisionally assuming the null hypothesis is true and then ask whether the observed evidence is sufficiently inconsistent with that assumption.

18.3 The proof-by-contradiction interpretation

The lecture explains hypothesis testing using an analogy with proof by contradiction.

Suppose we want to establish H_A.

Rather than starting by assuming H_A is true, we temporarily assume its opposite:

Assume H_0 is true.

Then collect data.

Now ask:

If H_0 were really true, how plausible would the data we just observed be?

If the observed data would be extremely unlikely under H_0, then we have produced a contradiction between our assumption and our evidence.

The reasoning becomes:

Assume H_0.

Observe the data.

Determine how likely such data would be if H_0 were true.

If the data are extremely unlikely under H_0, the evidence is inconsistent with H_0.

Therefore, reject H_0.

This provides support for H_A.

This is the conceptual foundation behind both the critical-value method and the p-value method.

18.4 Why "fail to reject" matters

There is an important asymmetry in hypothesis testing.

There are normally two conclusions:

Reject H_0

or

Fail to reject H_0.

We generally should not say:

Accept H_0.

Why?

Because failing to find sufficient evidence against a proposition does not demonstrate that the proposition itself is true.

The lecture uses the legal analogy.

A defendant begins from the position of not guilty. The prosecutor presents evidence. If the evidence becomes sufficiently compelling, the initial position is rejected.

But suppose the prosecutor does not provide enough evidence.

The conclusion is not necessarily:

We have proved the defendant innocent.

Rather:

We do not have sufficient evidence to establish guilt.

Statistical hypothesis testing works similarly.

If we fail to reject H_0, we are saying:

The available data do not provide sufficiently strong evidence against H_0.

We are not saying:

H_0 has been proved true.

The lecture explicitly emphasizes this distinction.

18.5 Formulating hypotheses correctly

The lecture then works through several examples.

Suppose we want to demonstrate that teenagers' average weekly cellphone usage is different from 16 hours.

The alternative is

H_A: $\mu \neq 16$.

Therefore,

H_0: $\mu = 16$.

This is a two-sided test.

Why two-sided?

Because evidence against H_0 could come from either direction.

The true average might be substantially below 16 or substantially above 16.

Both possibilities contradict H_0.

Now consider another example.

Suppose we want to demonstrate that the proportion of lost bags at an airport > 3 percent.

Our claim is:

H_A: $p > 0.03$.

The null therefore becomes:

H_0: $p \leq 0.03$.

This is a one-sided test because only sufficiently large values provide evidence supporting our particular claim.

Consider a third example.

Suppose we want to demonstrate that the average age of a population < 50.

Then:

H_A: $\mu < 50$.

H_0: $\mu \geq 50$.

Notice something that occurs repeatedly:

Equality appears in the null hypothesis.

The lecture initially postpones the complete explanation for this because it becomes clearer once critical values and Type I error are introduced.

But this feature is not merely a notational convention.

It is connected to how we construct the probability distribution under the null hypothesis and identify the worst-case boundary of the null hypothesis.

We will return to this point in Chapter 2.

18.6 Comparing two populations

The lecture also briefly introduces a two-population example.

Suppose we want to establish that second-year baseball players hit more home runs on average than first-year players.

Let:

μ_1 = average home runs for first-year players.

μ_2 = average home runs for second-year players.

Our substantive claim is:

H_A: $\mu_2 > \mu_1$.

Therefore:

H_0: $\mu_2 \leq \mu_1$.

An important feature here is that we do not need to specify the actual values of either mean.

We don't have to say that one population has an average of 10 and another has an average of 12.

We care about their relative ordering.

The lecture emphasizes that correctly identifying your substantive position is critical: the alternative hypothesis should express what you are trying to establish, while the null represents the opposing position.

18.7 The Hyatt fuel-efficiency example

A particularly useful example concerns Hyatt Hotels.

Suppose the company establishes a fuel-efficiency target of 20 miles per gallon for its fleet.

You are the manager, and you want to demonstrate to senior management that the fleet is exceeding the target.

Higher MPG means greater fuel efficiency because the vehicle travels more miles for each gallon of fuel.

Your substantive claim is therefore:

H_A: $\mu > 20$.

Consequently:

H_0: $\mu \leq 20$.

The sample contains 55 observations, with a sample mean of 22 MPG and a sample standard deviation of 5.

At first glance, 22 appears to support our claim.

But this is where statistical reasoning becomes necessary.

We cannot simply say:

Twenty-two > twenty, therefore the fleet exceeds the target.

Why not?

Because the sample mean is random.

Even if the true population mean were exactly 20, we could obtain a sample mean of 22 simply through sampling variation.

Therefore, the real question is not:

Is $22 > 20$?

The relevant question is:

Is 22 sufficiently > 20 that it would be unusually difficult to obtain such a result if the true population mean were 20 or lower?

That distinction is the transition from descriptive statistics to hypothesis testing.

18.8 Probability, surprise, and information

The lecture then introduces a very useful interpretation.

Suppose H_0 is true.

We observe some sample outcome.

How surprised should we be?

If an outcome has a high probability under H_0, observing it tells us relatively little.

But if an outcome has an extremely small probability under H_0, observing it is surprising.

And surprising observations contain more evidence against H_0.

So there is an inverse relationship:

high probability → low surprise → weak evidence

and

low probability → high surprise → strong evidence.

The lecture connects this intuition to the broader idea of information and notes that information can be represented using the negative logarithm of likelihood, although that formula is not required for the course.

This perspective is extremely helpful for understanding hypothesis testing.

A statistically significant result is essentially saying:

If H_0 were true, what we just observed would be sufficiently surprising that we are unwilling to continue treating H_0 as a reasonable explanation.

18.9 Significance level

But now we need to define how surprising is surprising enough.

This is where α enters.

Alpha is the significance level.

Common values are:

0.05,

0.01,

and sometimes 0.10 depending on the application.

Suppose $\alpha = 0.05$.

Conceptually, we are saying:

We will reject H_0 only when the evidence falls into a sufficiently unlikely region under H_0—one whose probability is controlled at 5 percent.

This threshold separates ordinary random fluctuations from outcomes we regard as statistically significant.

The boundary separating these regions is the critical value.

On one side of the critical value is the rejection region.

If the observed test statistic falls sufficiently far into that region, we reject H_0.

If it does not, we fail to reject H_0.

18.10 Type I error

There is still a problem.

Even extremely unlikely events occasionally occur.

Suppose H_0 really is true.

Nevertheless, because sampling is random, we could happen to obtain an extreme sample and reject H_0.

That mistake is called a Type I error.

So:

Type I error = rejecting H_0 when H_0 is actually true.

And the probability of Type I error is controlled by α.

Thus, if $\alpha = 0.05$, we design the test so that the probability of incorrectly rejecting a true null hypothesis is no $>$ approximately 5 percent under the relevant null boundary.

This does not mean that every individual rejection has a literal 5 percent probability of being wrong.

Instead, α describes the long-run error rate of the testing procedure when H_0 is true.

The lecture later illustrates this by imagining repeatedly drawing samples. If the null is true and we repeated a 5-percent-level procedure many times, only about 5 percent of samples should fall into the rejection region.

18.11 Type II error and the fundamental trade-off

There is another possible mistake.

Suppose H_A is actually true.

But our data are not sufficiently extreme to reject H_0.

Then we fail to reject H_0 even though H_0 is false.

That is a Type II error.

So:

Type I error: reject a true H_0.

Type II error: fail to reject a false H_0.

The probability of Type I error is α.

The probability of Type II error is beta.

And these errors generally trade off against one another.

Making it easier to reject H_0 reduces Type II errors but increases Type I errors.

Making it harder to reject H_0 reduces Type I errors but increases Type II errors.

This motivates an important interpretation of classical hypothesis testing presented in the lecture:

Control Type I error at an acceptable level such as 5 percent, and subject to that constraint, try to make Type II error as small as possible.

The lecture explicitly describes this as a constrained optimization problem: minimize Type II error subject to Type I error being bounded by α.

This connects hypothesis testing directly back to the optimization perspective introduced earlier in Module 1.

The main message to retain from this Chapter is:

Hypothesis testing is fundamentally an evidence-versus-default framework.

We begin with H_0 as the default position.

We ask what kinds of observations would be expected if H_0 were true.

We collect data.

If the observed data are sufficiently improbable under H_0, we regard them as strong contradictory evidence and reject H_0.

But because random events can always produce unusual samples, we explicitly control the probability of making a Type I error using the significance level α.

Next chapter will continue from exactly this point and explain the most technical and important part of the lecture: why equality belongs in H_0, how the worst-case null value is chosen, how the Central Limit Theorem leads to the test statistic, why the t-distribution appears when σ is unknown, and how the critical-value method is actually carried out numerically.

19 | From Hypotheses to the Critical-Value Test

In previous chapter, we developed the logic of hypothesis testing. We start with a null hypothesis, assume it is true, collect data, and ask whether what we observed is sufficiently surprising under that assumption.

Now we move to the operational question:

How do we turn that idea of "surprising enough" into an actual numerical test?

This is where the lecture introduces the critical value, the rejection region, the Central Limit Theorem, and eventually the t statistic.

19.1 The alternative hypothesis determines where to look

Suppose our claim is

H_A: $\mu < 50$.

Then the null hypothesis is

H_0: $\mu \geq 50$.

If we want evidence against H_0, which direction should we look?

We should look toward small sample means.

If the population mean really is 50 or greater, then observing a sample mean substantially below 50 would be surprising.

Therefore, the rejection region should appear on the left side of the sampling distribution.

Now reverse the claim.

Suppose

H_A: $\mu > 50$.

Then

H_0: $\mu \leq 50$.

Now unusually large sample means contradict H_0.

So the rejection region moves to the right side.

Finally, consider

H_A: $\mu \neq 50$.

Then

H_0: $\mu = 50$.

Now both extremely low and extremely high sample means contradict H_0.

So there are two rejection regions, one in each tail. The lecture explicitly describes these three possibilities: left-tailed, right-tailed, and two-tailed tests.

This gives us an important practical rule:

The alternative hypothesis tells you where the rejection region goes.

Less than means left tail.

Greater than means right tail.

Not equal means both tails.

19.2 Why equality belongs in the null hypothesis

Now we can return to a question postponed in previous Chapter.

Why do we write something like

H_0: $\mu \geq 40,000$,

instead of simply

H_0: $\mu > 40,000$?

The deeper reason concerns how we control the probability of Type I error.

Suppose a tire manufacturer claims its average tire life is at least 40,000 miles.

You are concerned that this claim may be false.

So your position is:

H_A: $\mu < 40,000$.

And therefore:

H_0: $\mu \geq 40,000$.

The lecture emphasizes that the phrase "you are concerned about this claim" tells us which side we want to challenge. The goal is to gather evidence that the manufacturer's claimed minimum lifetime may not be satisfied.

But notice something.

Under H_0, μ could be:

40,000,

41,000,

45,000,

50,000,

or even larger.

So which value of μ should we use when constructing the test?

The answer is:

Use the boundary value, 40,000.

Why?

Because this produces the worst-case probability of Type I error.

Imagine our rejection region consists of very low sample means.

If the actual population mean were 50,000, obtaining a sample mean low enough to suggest $< 40,000$ would be extremely unlikely.

If the actual population mean were 45,000, it would still be unlikely.

But as the true mean moves downward toward 40,000, obtaining such a low sample mean becomes progressively more likely.

Therefore, within the entire null-hypothesis region,

$\mu \geq 40,000$,

the value that creates the largest Type I error probability is the boundary:

$\mu = 40,000$.

The lecture describes this explicitly as a worst-case calculation: move μ through the null region toward its smallest permitted value, and the Type I error probability increases until the boundary is reached.

This explains why equality is so important.

The equality point provides the benchmark distribution we use to calibrate the test.

So when you see

H_0: $\mu \geq \mu_0$,

the actual critical-value calculation is typically performed at

$\mu = \mu_0$.

19.3 Constructing the critical value

Now suppose $\alpha = 0.05$.

Remember what α represents.

We want to design the rejection region such that, under the worst-case null hypothesis,

the probability of falling into the rejection region is no more than 5 percent.

For the tire example, we therefore conceptually solve:

Probability that sample mean is sufficiently small, given that $\mu = 40,000$,

$= 0.05$.

There will be some threshold separating ordinary sample means from sufficiently extreme sample means.

That threshold is the critical value.

Everything beyond the critical value in the relevant direction belongs to the rejection region.

So the critical value is engineered before observing the sample result, based on:

the null hypothesis,

α,

the sampling distribution,

and the sample size.

This is important.

We do not look at the data first and then move the critical value until we obtain the result we want.

The testing rule must be defined independently of the particular realization of the sample.

19.4 Why the sample mean has less variability than individual observations

To construct this rejection region, we need to know the probability distribution of the sample mean.

This is where the Central Limit Theorem becomes central.

Suppose the observations themselves have standard deviation σ.

The sample mean does not have standard deviation σ.

Instead,

$$SD(\bar{X}) = \frac{\sigma}{\sqrt{n}}$$

That quantity is called the standard error of the mean.

The intuition in the lecture is useful.

When we average many observations, unusually high observations and unusually low observations tend to offset each other.

Therefore, sample averages cluster much more tightly around the population mean than individual observations do.

As the sample size increases, this clustering becomes stronger.

Mathematically:

$$SE = \frac{\sigma}{\sqrt{n}}$$

This square-root-of-n term is critical.

The lecture explicitly warns not to forget it: the variability of $\bar{X}$ is smaller than the variability of an individual observation because averaging produces a smoothing effect.

This also explains why larger samples provide stronger statistical evidence.

When n increases, the standard error decreases.

Therefore, it becomes easier to distinguish a genuine departure from H_0 from ordinary sampling noise.

19.5 Standardizing the sample mean

Suppose we know population standard deviation σ.

Then we could form a standardized statistic:

$$Z = \frac{\bar{X} - \mu_0}{\frac{\sigma}{\sqrt{n}}}$$

Under the null boundary and appropriate Central Limit Theorem conditions, this follows approximately a standard normal distribution.

This converts everything into a common standardized scale.

Instead of saying:

“The sample mean is 2,000 miles below the manufacturer’s claim,”

we can say:

“The sample mean is a certain number of standard errors below the null-hypothesis benchmark.”

That second statement is much more informative.

A difference of 2,000 units could be enormous if the standard error is 500.

But it could be tiny if the standard error is 10,000.

So statistical significance depends not simply on the raw difference

$\bar{X} - \mu_0$,

but on the difference relative to sampling uncertainty.

19.6 What happens when σ is unknown?

In real applications, we often do not know the population standard deviation σ.

That is exactly what happens in the tire example.

We have only the sample.

So we replace σ with the sample standard deviation, S.

But once we make that substitution, the standardized statistic is no longer treated as standard normal.

Instead, we use the Student t distribution.

The resulting test statistic is

$$T = \frac{\bar{X} - \mu_0}{\frac{S}{\sqrt{n}}}$$

And the degrees of freedom are

n -1.

The lecture explicitly makes this transition: when population σ is unknown, replace it with the sample standard deviation, and use the t distribution with n -1 degrees of freedom.

This distinction is one of the most important mechanical rules from the lecture.

If σ is known, use the normal or Z framework.

If σ is unknown and is replaced by S, use the t framework.

19.7 Why the t distribution is wider

Conceptually, using S creates additional uncertainty.

When σ is known, the denominator of our statistic contains a fixed population quantity.

When σ is unknown, we estimate it from the same sample.

That introduces additional randomness.

The t distribution accounts for this uncertainty by having heavier tails than the normal distribution, especially with small sample sizes.

As sample size becomes larger, S becomes a more reliable estimate of σ.

Consequently, the t distribution approaches the normal distribution.

So t is not an arbitrary replacement.

It is the probability distribution that incorporates uncertainty caused by estimating σ.

19.8 Worked tire example

Now let us go through the lecture's main example.

The manufacturer claims that average tire lifetime is at least 40,000 miles.

We are concerned that the actual lifetime may be lower.

Therefore:

H_0: $\mu \geq 40,000$.

H_A: $\mu < 40,000$.

Suppose $\alpha = 0.05$.

A random sample of 12 tires is collected.

Because n = 12,

degrees of freedom equal

12 -1,

which = 11.

Because this is a left-tailed test, we need a left-tail t critical value containing 5 percent probability.

The appropriate one-tailed critical magnitude for 11 degrees of freedom is about

1.796.

Because we are looking in the left tail, the critical value becomes approximately

-1.796.

Therefore, our decision rule is:

If the calculated t statistic < -1.796, reject H_0.

Otherwise, fail to reject H_0.

The lecture spends some time distinguishing tables that report two-tailed versus one-tailed probabilities. For a two-tailed 5-percent test, α is split into 2.5 percent in each tail. For a one-tailed 5-percent test, the entire 5 percent belongs in one tail.

This table distinction matters because using the wrong column produces the wrong critical value.

19.9 Calculate the test statistic

From the tire sample, the lecture obtains approximately:

$\bar{X}$ = 37.28 thousand miles.

S = 2.73 thousand miles.

n = 12.

The null benchmark is 40 thousand miles.

Therefore:

$$T = \frac{37.28 - 40}{\frac{2.73}{\sqrt{12}}}$$

The resulting statistic is approximately

-3.44.

Now compare:

test statistic approximately -3.44,

critical value approximately -1.796.

Because

-3.44 is farther into the left tail than -1.796,

the observation falls inside the rejection region.

Therefore:

Reject H_0.

The sample provides statistically significant evidence that the true average tire lifetime is below the manufacturer's claimed 40,000-mile benchmark.

The lecture describes this result as a sufficiently surprising observation under the null hypothesis and therefore as evidence inconsistent with the manufacturer's claim.

19.10 What does -3.44 actually mean?

This is worth interpreting carefully.

A t statistic of -3.44 does not mean the average tire life is 3.44 units below 40,000.

Instead, it says:

The observed sample mean lies roughly 3.44 estimated standard errors below the null benchmark.

That is a very substantial standardized departure.

The whole purpose of standardization is to express the observed difference relative to the amount of sampling variation we would normally expect.

That is why the test statistic—not simply $\bar{X} - \mu_0$—is compared with the critical value.

19.11 Another example: Hyatt fuel efficiency

The lecture then returns to the Hyatt Hotels example.

Recall the data:

n = 55.

$\bar{X}$ = 22 MPG.

S = 5 MPG.

The company wants to establish that average fuel efficiency exceeds 20 MPG.

So:

H_0: $\mu \leq 20$.

H_A: $\mu > 20$.

This is a right-tailed test.

The t statistic is

$$T = \frac{22 - 20}{\frac{5}{\sqrt{55}}}$$

The lecture obtains approximately

2.966.

For about 54 degrees of freedom, the table does not have an exact row of 54, so the lecture uses the nearby row of 50 and obtains a one-tailed 5-percent critical value of approximately

1.676.

Now compare:

$2.966 > 1.676$.

Therefore, the test statistic lies in the right-tail rejection region.

So we reject H_0.

There is statistically significant evidence that Hyatt's average fleet fuel efficiency exceeds 20 MPG.

19.12 Understanding the 5 percent Type I error operationally

The lecture closes this portion with a particularly useful interpretation.

Suppose the Hyatt null hypothesis were actually true.

Imagine repeating the entire experiment 100 times.

Each time, take a new random sample of 55 vehicles.

Each time, compute the test statistic.

Because we deliberately constructed the critical value using $\alpha = 0.05$, only roughly 5 out of those 100 experiments should fall into the rejection region when the null hypothesis is actually true.

Those cases represent Type I errors.

The other approximately 95 experiments would not incorrectly reject H_0.

So when we observe a result such as T = 2.966, which lies beyond the critical threshold, our argument is essentially:

This kind of result belongs to a region that should occur no more than about 5 percent of the time under the worst-case null hypothesis.

Therefore, we regard it as sufficiently surprising to reject H_0.

The lecture emphasizes that mistakes are still possible. Hypothesis testing does not eliminate uncertainty. It controls the frequency of one particular kind of mistake.

The core algorithm to remember

You can summarize the entire critical-value method as follows.

First, identify what you want to show.

Put that statement in H_A.

Put the opposing statement, including the equality boundary, into H_0.

Second, determine whether the alternative gives you a left-tailed, right-tailed, or two-tailed test.

Third, identify α.

Fourth, use the null-hypothesis boundary μ_0.

Fifth, calculate the appropriate critical value.

If σ is unknown, use the t distribution with n -1 degrees of freedom.

Sixth, calculate

$$T = \frac{\bar{X} - \mu_0}{\frac{S}{\sqrt{n}}}$$

Finally, compare the test statistic to the critical value.

If the statistic falls inside the rejection region:

Reject H_0.

Otherwise:

Fail to reject H_0.

And always remember:

Failing to reject H_0 does not mean proving H_0 true.

The most important conceptual lesson in this Chapter is that the critical value is designed before looking at the observed result so that the worst-case probability of incorrectly rejecting H_0 is controlled by α.

Then the sample tells us where we actually landed.

The test statistic answers:

How many estimated standard errors away from the null benchmark is my observation?

The critical value answers:

How far away would I need to be before I regard the observation as too surprising to comfortably maintain H_0?

If the test statistic crosses that boundary, we reject H_0.

Chapter 3 will continue with the remaining material from the lecture, including the interpretation of these tests, the relationship between one-tailed and two-tailed critical values, the later examples and questions, and the lecture's closing takeaways.

20 | Interpreting the Test Correctly and the Lecture's Final Takeaways

This final Chapter focuses less on introducing new formulas and more on how to think correctly about the procedure you have just constructed.

One important clarification from the lecture is that, although the lecture initially announces both the critical-value and p-value techniques, this particular session develops the critical-value method in detail. The lecture does not provide a full worked p-value calculation before the lecture ends. Instead, it closes after reinforcing α, Type I error, and the Hyatt example, with Type II error and two-sample tests deferred to the Wednesday class.

So I will stay faithful to what this lecture actually covers.

20.1 The first thing to get right is your position

A recurring theme throughout the lecture is that many mistakes in hypothesis testing occur before any calculation begins.

They come from formulating H_0 and H_A incorrectly.

Consider again the tire-manufacturer example.

The company says:

Our average tire life is at least 40,000 miles.

But the question also says that you are concerned about this claim.

That second sentence is crucial.

You are not trying to establish the manufacturer's claim.

You are effectively asking whether the data provide evidence that the company's claim is too optimistic.

Therefore, your position is:

H_A: $\mu < 40,000$.

The manufacturer's position becomes:

H_0: $\mu \geq 40,000$.

Why?

Because the methodology is organized around rejecting H_0.

The lecture emphasizes that in research you can choose which substantive position you want to investigate, but once you choose it, the hypothesis formulation must be consistent with what you are trying to reject.

This is an excellent exam strategy.

Before writing any formulas, ask yourself:

What exactly am I trying to provide evidence for?

That sentence usually becomes H_A.

20.2 Use the data as a sanity check on your formulation

The lecture gives another useful practical idea.

Suppose you formulate the hypotheses and then observe a sample result that strongly points in the opposite direction from what you intended to investigate.

That may be a signal that you misunderstood the question.

For example, suppose you intended to investigate whether tire life was below 40,000 miles, but somehow formulated the alternative in the other direction.

Then you may discover that the observed sample mean lies completely opposite to your rejection region.

That should prompt you to return to the verbal question.

So hypothesis testing is not only algebra.

There is a conceptual consistency check among three things:

the verbal research question,

the alternative hypothesis,

and the direction of the rejection region.

These three must agree.

20.3 A useful triangle to remember

You can think about the entire setup using three connected pieces.

First:

Your claim determines H_A.

Second:

H_A determines the direction of the rejection region.

Third:

The rejection region determines which critical value you need.

For example:

If

H_A: $\mu < \mu_0$,

then you need a left-tailed test.

If

H_A: $\mu > \mu_0$,

then you need a right-tailed test.

If

H_A: $\mu \neq \mu_0$,

then you need a two-tailed test.

The lecture explicitly summarizes that the rejection region can lie on the left, on the right, or in both tails depending on how H_0 and H_A are formulated.

So you should almost be able to look at H_A and immediately draw the rejection region.

20.4 One-tailed versus two-tailed tests

This distinction becomes particularly important when using a t table.

Suppose $\alpha = 0.05$.

For a one-tailed test, the entire 5 percent probability goes into one tail.

For a two-tailed test, the 5 percent is divided:

2.5 percent in the left tail,

and

2.5 percent in the right tail.

That means the critical value for a two-tailed 5-percent test is farther away from zero than the corresponding one-tailed critical value.

This makes intuitive sense.

In a two-tailed test you are allowing rejection in either direction.

Therefore, each individual rejection region receives only half the total Type I error budget.

The lecture discusses a source of confusion here because different t tables are presented differently. Some tables directly show one-tail probabilities; others are organized around two-tail probabilities. The lecture notes that for the exam a single-tail table would be provided to avoid this ambiguity.

The underlying mathematics does not change.

Only the way the table labels its columns changes.

20.5 The critical value is not calculated from the sample outcome

This point deserves repetition.

There are two conceptually different stages.

Stage one: design the test

Before seeing the sample outcome, determine:

H_0,

H_A,

α,

the appropriate reference distribution,

degrees of freedom,

and the critical value.

This defines the decision rule.

Stage two: observe the data

Now collect the sample and calculate:

$\bar{X}$,

S,

and the test statistic.

Then ask where the test statistic lands relative to the previously constructed rejection region.

The lecture explicitly notes that the standardization and critical-value construction are being done before observing the data.

That separation is fundamental.

Otherwise, you could manipulate the decision rule after seeing the outcome, which would destroy the stated Type I error guarantee.

20.6 Why standardization is so powerful

Suppose one company measures tire life in miles.

Another measures customer spending in dollars.

Another examines fuel efficiency in miles per gallon.

Another studies waiting time in minutes.

The raw quantities have completely different units.

But once we standardize using

$$T = \frac{\bar{X} - \mu_0}{\frac{S}{\sqrt{n}}}$$

we convert all these problems to the same common scale.

The test statistic tells us:

How many estimated standard errors away from the null benchmark is the observed sample mean?

That is why one t distribution can be used across many different applications.

A t statistic of 3 does not mean three miles, three dollars, or three minutes.

It means approximately:

three standard errors away from the null benchmark.

That makes results comparable across very different problems.

20.7 Revisit the tire example carefully

For the tire problem, the observed sample mean is roughly

37.28 thousand miles,

while the null benchmark is

40 thousand miles.

The sample standard deviation is approximately

2.73 thousand miles,

and there are 12 observations.

Because the standard error is

$\frac{S}{\sqrt{12}}$,

the observed difference becomes a test statistic somewhere around

-3.44.

The lecture then compares that number with the appropriate left-tail critical value.

Because the statistic lies well inside the rejection region, H_0 is rejected.

Conceptually, the conclusion is not simply:

37.28 is smaller than 40.

Instead:

37.28 is sufficiently far below 40 relative to the amount of sampling uncertainty that such an outcome would be difficult to reconcile with the null hypothesis.

That is what makes the conclusion statistical rather than merely arithmetic.

20.8 The Hyatt example shows the mirror image

For Hyatt, the hypotheses were reversed in direction.

H_0:

$\mu \leq 20$ MPG.

H_A:

$\mu > 20$ MPG.

The sample produced:

$\bar{X} = 22$,

S = 5,

n = 55.

The calculated t statistic was about

2.966.

The critical value was approximately

1.676.

Because this is a right-tailed test,

$2.966 > 1.676$

puts the observation inside the rejection region.

Therefore:

Reject H_0.

The evidence supports the position that the average fleet efficiency exceeds 20 MPG.

Notice that the tire test rejected because the test statistic was very negative.

The Hyatt test rejected because the test statistic was very positive.

There is no contradiction.

The direction comes entirely from H_A.

20.9 Alpha is a property of the testing procedure

The final discussion in the lecture provides one of the most valuable interpretations of α.

Suppose Hyatt's null hypothesis really is true.

Now imagine repeating the entire process 100 times.

Each experiment uses a new random sample of 55 observations.

For each sample, calculate the same t statistic and apply exactly the same critical-value rule.

If $\alpha = 0.05$, the procedure is constructed so that only about 5 percent of these experiments should incorrectly fall into the rejection region when the null is true.

The lecture describes this as roughly 5 out of 100 repeated samples.

This interpretation is extremely important.

Alpha does not mean:

There is a 5 percent probability that H_0 is true.

It also does not mean:

After I reject H_0, there is a 5 percent probability that my conclusion is wrong.

Instead, it refers to the long-run behavior of the decision rule:

If H_0 were true and we repeatedly used this testing procedure, the probability of rejecting H_0 is controlled at α.

That is the frequentist interpretation.

20.10 Why an individual rejection can still be wrong

Suppose $\alpha = 0.05$ and you reject H_0.

Could you still have made a mistake?

Yes.

That is precisely what Type I error means.

Rare events still occur.

Something with probability 5 percent is unusual, but certainly not impossible.

Therefore statistical significance does not mean certainty.

Instead it means:

The observation belongs to a region that we deliberately classified as sufficiently unlikely under H_0 that we are willing to reject H_0 when it occurs.

That language is much more accurate.

The lecture explicitly emphasizes this point: even after rejection, a Type I error remains possible, but the testing procedure is constructed so that this probability is bounded by the chosen α level.

20.11 Statistical significance is fundamentally relative to noise

One major insight running through the lecture is that an observed difference alone tells us very little.

Suppose the null mean is 20 and the sample mean is 22.

Is a difference of 2 statistically significant?

You cannot answer without additional information.

You need:

the variability,

and

the sample size.

If S is extremely large and n is small, a difference of 2 could easily arise from randomness.

If S is modest and n is large, the same difference could be highly significant.

The test statistic incorporates all three components:

the observed difference,

the variability,

and the sample size.

This is why the denominator is the standard error:

$\frac{S}{\sqrt{n}}$.

That $\sqrt{n}$ explains something very important.

As sample size increases, the standard error gets smaller.

So the same raw difference produces a larger absolute test statistic.

That means larger studies are generally capable of detecting smaller departures from H_0.

20.12 Significance and substantive importance are not the same thing

The lecture does not explicitly develop this distinction as a separate topic, so I will phrase this only as an implication of the mechanics already presented.

Because the test statistic becomes larger as sample size grows, a very large sample can potentially make a relatively small raw difference cross a statistical threshold.

Therefore, when interpreting a real business result, it is useful to distinguish:

statistical significance

from

the magnitude of the observed difference.

The lecture itself focuses on statistical significance and does not develop effect-size analysis, so that topic belongs beyond the scope of this particular session.

20.13 The role of Type II error in this lecture

Earlier in the lecture, Type II error was introduced conceptually.

Recall:

Type I error means rejecting H_0 when H_0 is true.

Type II error means failing to reject H_0 when H_A is actually true.

The lecture also described a trade-off.

If we make rejection easier, Type II error tends to decrease but Type I error tends to increase.

If we make rejection harder, Type I error decreases but Type II error tends to increase.

The lecture characterizes hypothesis testing as controlling Type I error at α and, subject to this constraint, trying to reduce Type II error.

However, an important boundary for this particular lecture is:

The numerical calculation of Type II error is not completed in this session.

At the end of the class, the instructor explicitly says that Type II error and two-sample tests will be continued on Wednesday.

So for this lecture, students primarily need to understand the conceptual meaning of Type II error rather than calculate beta numerically.

20.14 What about the p-value method?

At the beginning of the lecture, both critical values and p-values are announced as methods that will be discussed.

The lecture also motivates the idea underlying a p-value: ask how unlikely or surprising the observed data are under H_0.

But in the lecture provided here, the detailed numerical work concentrates on the critical-value technique.

So the operational decision rule developed in this lecture is:

calculate the test statistic,

calculate a critical value,

and determine whether the statistic lies inside the rejection region.

A full numerical p-value procedure is not developed before the class ends.

Conceptually, however, the same logic is already visible:

small probability under H_0 means greater surprise and stronger evidence against H_0.

The lecture repeatedly connects low likelihood with high informational content and greater inconsistency with the null.

20.15 The complete conceptual flow of the lecture

It is useful to reconstruct the whole lecture as one continuous argument.

You begin with a substantive question.

For example:

Does a new website design improve conversion?

Does a medication improve outcomes?

Does a fleet exceed 20 MPG?

Does a manufacturer's product really last at least 40,000 miles?

Then identify what you want to establish.

That becomes H_A.

Construct the opposite benchmark as H_0.

Assume H_0 is true.

Recognize that sample outcomes are random.

Therefore, even observations favoring H_A can occasionally occur under H_0.

Ask:

How extreme would the sample result need to be before I regard it as too unlikely to comfortably attribute to random variation?

Choose α.

Use the worst-case null boundary.

Construct the sampling distribution.

If population σ is unknown, estimate it with S and use the t distribution.

Define the critical value so that the rejection region has probability α under the worst-case null.

Then collect the sample.

Calculate

$$T = \frac{\bar{X} - \mu_0}{\frac{S}{\sqrt{n}}}$$

Finally determine whether the statistic falls into the rejection region.

If yes:

Reject H_0.

If no:

Fail to reject H_0.

And remember that failure to reject does not prove H_0.

That is essentially the entire architecture developed in this lecture.

20.16 The most important exam-ready rules

If you were reducing the lecture to a mental checklist, the most important rules are these.

First, identify the claim you actually want to support.

Second, put that claim in H_A.

Third, equality belongs to H_0 and provides the boundary used to calibrate the test.

Fourth, use H_A to determine the tail.

Less than means left.

Greater than means right.

Not equal means two tails.

Fifth, α controls the probability of Type I error.

Sixth, when σ is unknown, use the sample standard deviation and the t distribution with n -1 degrees of freedom.

Seventh, remember the $\sqrt{n}$:

$$SE = \frac{S}{\sqrt{n}}$$

Eighth, calculate the standardized test statistic.

Ninth, compare it with the critical value in the correct direction.

And finally, report:

reject H_0,

or

fail to reject H_0.

Do not automatically write "accept H_0."

Final synthesis

The deepest message of the lecture is not really the t formula.

It is the logic of decision-making under uncertainty.

Observed data will almost never line up perfectly with a theoretical benchmark because samples are random.

Therefore, hypothesis testing asks us to distinguish two explanations for an observed difference.

One explanation is:

This difference reflects ordinary sampling randomness.

The competing explanation is:

This difference is too large, relative to expected sampling variation, to comfortably attribute to randomness under H_0.

The significance level tells us how demanding we want to be before choosing the second interpretation.

The critical value operationalizes that standard.

The test statistic tells us how extreme our particular observation is.

And the comparison between those two quantities produces the decision.

That is the central framework developed across this entire lecture.

This completes the summary of the uploaded lecture. The lecture ends by noting that the next session will continue with Type II error and two-sample hypothesis tests, rather than developing those topics fully here.

21 | Introduction to Two-Sample Hypothesis Testing, Motivation, Independence Assumption, and Two-Tailed Tests

21.1 Introduction to Two-Sample Hypothesis Testing

In the previous lessons, we studied hypothesis testing with **one sample**.

The main idea was:

- We start with a default assumption called the **null hypothesis**.
- We collect data from a population.
- We determine whether the observed data provide enough evidence to reject that default assumption.

In this lesson, we extend this idea to the case of **two samples**.

The main question becomes:

How can we determine whether two populations are different from each other?

The lecture introduces this idea using a business example.

21.2 Motivating Example: Comparing Two Customer Populations

Imagine a company provides a service to customers.

The company wants to understand whether customers from two different regions behave differently.

The two regions are:

- Chicago
- San Francisco

The company measures service utilization, such as the number of minutes customers use the service.

The research question is:

Do customers from Chicago and San Francisco have different average service utilization?

In statistical language, we want to compare two population means.

Let μ_C represent the true average utilization of Chicago customers. Let μ_{SF} represent the true average utilization of San Francisco customers. The question is whether μ_C and μ_{SF} are different.

21.3 The Most Important Assumption: Independence

Before performing a two-sample hypothesis test, we need an important assumption.

The two samples must be:

Independent.

This means:

The observations collected from one population should not influence the observations collected from the other population.

In this example:

Chicago customers and San Francisco customers should represent separate groups.

A Chicago customer's behavior should not affect the data collected from a San Francisco customer.

21.4 Why Independence Matters

The professor emphasizes that independence is not a minor technical detail.

If independence is violated, the statistical conclusion can be wrong.

The reason is that the formulas used in two-sample testing rely on the assumption that the two samples provide separate information.

When the samples are related, the uncertainty calculation changes.

This affects:

- The standard error.
- The test statistic.
- The final decision.

Therefore, before applying a two-sample test, we must ask:

Are these two samples truly independent?

21.5 Example of Violating Independence

The professor provides an example.

Suppose we collect data from 66 customers in Chicago.

We measure their service utilization.

Then, instead of randomly collecting new customers from San Francisco, suppose we find their relatives or close friends living in San Francisco.

Now we collect utilization data from those people.

At first glance, we have two groups:

- Chicago customers.
- San Francisco customers.

However, these observations are connected.

Why?

Because people who are friends or relatives may have similar behaviors.

For example:

- They may communicate with each other.
- They may have similar lifestyles.
- They may have similar preferences.

Therefore, their service usage may be correlated.

The two groups are no longer independent.

21.6 More General Examples of Dependence

The professor explains that this issue can occur in many settings.

For example:

Suppose we want to study salaries in Chicago.

If we collect salary information from people who are close friends or relatives, the salaries may be related.

Family background, education, career choices, and social networks may create similarities.

Therefore, the observations are not completely independent.

A better approach would be collecting observations randomly from unrelated individuals.

Random sampling helps create independence.

21.7 Formulating the Hypotheses

Now we return to the original Chicago versus San Francisco example.

The research question is:

"Is there any difference between these two populations?"

The alternative hypothesis is:

$$H_A : \mu_C \neq \mu_{SF}$$

This means:

The two populations have different average service utilization.

The null hypothesis is:

$$H_0 : \mu_C = \mu_{SF}$$

This represents the default assumption:

The two populations behave the same.

The purpose of hypothesis testing is to determine whether the data provide enough evidence against this equality assumption.

21.8 Why This Is a Two-Tailed Test

This test is called a:

Two-tailed test.

The reason is that we are not interested in only one direction.

We do not specifically ask:

"Is Chicago larger than San Francisco?"

Or:

"Is Chicago smaller than San Francisco?"

Instead, we ask:

"Are they different?"

A difference in either direction is meaningful.

There are two possibilities:

21.8.1 Case 1:

$$\bar{X}_C > \bar{X}_{SF}$$

Chicago has a larger sample average.

21.8.2 Case 2:

$$\bar{X}_C < \bar{X}_{SF}$$

Chicago has a smaller sample average.

Either outcome provides evidence against:

$$H_0 : \mu_C = \mu_{SF}$$

Therefore, evidence can appear in either tail of the distribution.

21.9 Comparison with One-Tailed Tests

Previously, we studied one-tailed tests.

In a one-tailed test, we are interested in only one direction.

For example:

$$\mu_C > \mu_{SF}$$

Would be a right-tailed test.

We would only look for evidence on the positive side.

Alternatively:

$$\mu_C < \mu_{SF}$$

Would be a left-tailed test.

However, in the Chicago versus San Francisco example, we do not have a preferred direction.

We only care whether there is a difference.

Therefore, we use a two-tailed test.

21.10 The Critical Value Approach

The professor introduces the critical value approach.

The main question is:

How large must the difference be before we consider it statistically significant?

A small difference between sample averages may simply occur because of random sampling variation.

A very large difference may indicate that the populations are genuinely different.

The critical value provides the threshold.

It tells us:

"If the observed difference exceeds this value, we reject the null hypothesis."

21.11 Statistical Testing as Searching for Rare Events

The logic of hypothesis testing is based on rare events.

The procedure is:

1. Assume the null hypothesis is true.
2. Determine how likely the observed data would be under that assumption.
3. If the observed data would be extremely unlikely, reject the null hypothesis.

In other words:

A rare observation creates evidence against the assumption that produced it.

21.12 Summary of Chapter

The major ideas from this section are:

1. Two-sample hypothesis testing compares two populations.
2. The key example compares Chicago and San Francisco customers.
3. The null hypothesis assumes equal population means.
4. The alternative hypothesis tests whether the means are different.
5. This is a two-tailed test because differences in either direction matter.
6. The independence assumption is essential.
7. Violating independence can produce incorrect conclusions.
8. The critical value approach determines how large an observed difference must be before we reject the null hypothesis.

22 | Critical Value Approach, Significance Level, Rare Events Logic, and Computing the Test Threshold

22.1 Moving from the Research Question to a Statistical Decision

In the previous section, we discussed the motivation for comparing two populations. The main question was:

Do customers from Chicago and San Francisco behave differently in terms of service utilization?

The statistical objective is to determine whether there is enough evidence to reject the default assumption.

The hypotheses are:

Null hypothesis, H_0:

The two populations have the same mean.

$$H_0 : \mu_C = \mu_{SF}$$

Where:

- μ_C represents the true average utilization of Chicago customers.
- μ_{SF} represents the true average utilization of San Francisco customers.

The alternative hypothesis is:

$$H_A : \mu_C \neq \mu_{SF}$$

This means we are looking for evidence that the two populations are different.

Because we are not specifically interested in whether Chicago is larger or smaller than San Francisco, this is called a **two-tailed test**.

22.2 Understanding the Two-Tailed Test

In a two-tailed test, evidence against the null hypothesis can come from two directions.

The first possibility:

$$\bar{X}_C > \bar{X}_{SF}$$

This means the Chicago sample average is significantly larger.

The second possibility:

$$\bar{X}_C < \bar{X}_{SF}$$

This means the Chicago sample average is significantly smaller.

Either situation can provide evidence against:

$$H_0 : \mu_C = \mu_{SF}$$

Therefore, rejection regions exist on both sides of the distribution.

This is why the test is called **two-tailed**.

22.3 The Logic Behind Statistical Testing: Searching for Rare Events

The critical value approach is based on one important idea:

If the null hypothesis is true, how surprising is the observed data?

The procedure starts by assuming:

$$H_0 \text{ is true}$$

Then we ask:

"If the two populations really have the same mean, how likely is it that we observe such a large difference between the two samples?"

The probability we calculate is:

$$P(\bar{X}_C > \bar{X}_{SF} \text{ or } \bar{X}_C < \bar{X}_{SF})$$

Under the assumption that:

$$\mu_C = \mu_{SF}$$

This probability measures how unusual our observation would be if the null hypothesis were correct.

22.4 The Meaning of Statistical Significance

Suppose the probability of observing our data under the null hypothesis is very small.

For example:

$$P = 0.01$$

This means:

"If the null hypothesis were true, we would only see this type of data 1% of the time."

Such an event is considered rare.

Because it is rare, we become suspicious of the null hypothesis.

The reasoning is:

1. Assume the null hypothesis is true.
2. Calculate how likely the observed data is.
3. If the probability is extremely small, the assumption is probably wrong.
4. Reject the null hypothesis.

This is the fundamental logic behind hypothesis testing.

22.5 The Role of the Significance Level Alpha

The threshold for deciding whether something is rare is called the **significance level**.

It is represented by:

$$\alpha$$

In business applications, the most common choice is:

$$\alpha = 0.05$$

This means:

"We consider an event sufficiently surprising if it would happen less than 5% of the time under the null hypothesis."

Another interpretation:

A probability of 5 out of 100 observations, or 1 out of 20 observations, is considered rare enough.

22.6 Why Do We Use 0.05?

The professor explained that the choice of 0.05 is not a universal law.

Different fields use different standards.

For example, in physics experiments such as the CERN particle discovery experiments, the standard is much more restrictive.

They use approximately:

$$5 \text{ sigma}$$

As the evidence threshold.

A 5-sigma result corresponds to an extremely small probability.

The probability is approximately:

$$0.000002$$

Or roughly:

3 chances out of 10 billion.

This means physicists require extremely strong evidence before announcing a new discovery.

However, this standard is too strict for most business applications.

In business analytics, management science, and social sciences, researchers commonly use:

$$\alpha = 0.05$$

Because the goal is usually decision-making rather than discovering fundamental physical laws.

22.7 Understanding Sigma and Statistical Distance

The professor used the normal distribution to explain significance levels.

A normal distribution has:

- A center representing the expected value.
- Spread represented by standard deviation.

A result becomes more unusual as it moves farther away from the center.

For a significance level of 0.05 in a two-tailed test:

- The rejection area is split into two parts.

- Each tail contains:

$$\frac{\alpha}{2} = 0.025$$

Therefore:

- 2.5% of probability is in the left tail.
- 2.5% of probability is in the right tail.

The critical values determine where these rare regions begin.

22.8 Transforming the Problem into a Difference

Instead of comparing two separate sample averages, we combine them into one difference:

$$\bar{X}_C - \bar{X}_{SF}$$

This simplifies the analysis.

Now the question becomes:

"Is the difference between these two sample averages large enough to be considered unusual?"

The rejection rule becomes:

Reject H_0 if:

$$\bar{X}_C - \bar{X}_{SF} < -CV$$

Or

$$\bar{X}_C - \bar{X}_{SF} > CV$$

Where:

(CV) is the critical value threshold.

In other words:

- A very negative difference leads to rejection.
- A very positive difference leads to rejection.
- A difference close to zero does not provide enough evidence.

22.9 Connecting the Critical Value to Alpha

The critical value is selected so that:

$$P(\text{difference falls in rejection region}) = \alpha$$

For a two-tailed test:

$$P(\text{left tail}) = \frac{\alpha}{2}$$

And:

$$P(\text{right tail}) = \frac{\alpha}{2}$$

Adding them:

$$\frac{\alpha}{2} + \frac{\alpha}{2} = \alpha$$

Therefore, the total probability of rejecting a true null hypothesis is exactly the chosen significance level.

22.10 Moving Toward the t-Distribution

To calculate the critical value, we need to know the distribution of:

$$\bar{X}_C - \bar{X}_{SF}$$

Under the null hypothesis:

$$\mu_C = \mu_{SF}$$

Therefore, the expected difference is:

$$0$$

The Central Limit Theorem tells us that the difference between sample means follows approximately a normal distribution.

However, there is a problem:

The true population standard deviations are unknown.

Therefore, instead of using the normal distribution, we estimate variability using sample standard deviations.

Once we replace the unknown population standard deviation with sample estimates, the distribution becomes a:

t-distribution.

The t-distribution requires another parameter:

degrees of freedom.

The next section explains how degrees of freedom are calculated for two samples and how the final test statistic is constructed.

23 | Two-Sample t-Test Derivation, Degrees of Freedom, Standard Error, and Test Statistic Calculation

23.1 From the Difference Between Two Samples to a Statistical Test

In the previous chapter, we transformed the original question:

"Are Chicago and San Francisco customers different?"

Into a question about the difference between two sample averages:

$$\bar{X}_C - \bar{X}_{SF}$$

The key idea is:

If the null hypothesis is true:

$$H_0 : \mu_C = \mu_{SF}$$

Then the expected difference between the two populations is:

$$D_0 = \mu_C - \mu_{SF} = 0$$

The term D_0 represents the difference assumed by the null hypothesis.

In this particular problem:

$$D_0 = 0$$

Because we are testing whether the two populations have exactly the same average.

23.2 Central Limit Theorem and Distribution of the Difference

The Central Limit Theorem allows us to determine the distribution of the difference between two sample averages.

Under the null hypothesis:

$$\mu_C = \mu_{SF}$$

The difference:

$$\bar{X}_C - \bar{X}_{SF}$$

Follows approximately a normal distribution.

The center of this distribution is:

$$D_0 = 0$$

Therefore:

$$\bar{X}_C - \bar{X}_{SF}$$

Is centered around zero.

The intuition is:

- If the two populations truly have the same mean, the difference between their sample averages should usually be close to zero.
- Large positive or negative differences would be unusual.
- Very unusual differences provide evidence against the null hypothesis.

23.3 Why Do We Use the t-Distribution?

The next issue is variability.

The standard deviation of the difference between two sample means depends on the true population standard deviations.

The formula would require:

$$\sigma_C$$

And

$$\sigma_{SF}$$

However, in practice, these values are unknown.

Instead, we estimate them using sample standard deviations:

$$S_C$$

And:

$$S_{SF}$$

Whenever we replace unknown population parameters with estimates from data, the distribution changes.

Therefore:

Instead of using the normal distribution, we use the:

t-distribution.

The t-distribution is similar to the normal distribution but has heavier tails because of additional uncertainty caused by estimating variability.

23.4 Degrees of Freedom in a Two-Sample Test

The t-distribution requires a parameter called:

degrees of freedom.

For a single sample, we learned:

$$Df = n - 1$$

Because one degree of freedom is lost when calculating the sample mean.

For two independent samples, the exact formula is more complicated.

The professor explained that the general formula accounts for:

1. Different sample sizes.
2. Different sample variances.

The exact formula gives a weighted combination of information from both samples.

However, for this course, we use an approximation.

The approximation is:

$$Df = (n_1 - 1) + (n_2 - 1)$$

Where:

- n_1 is the first sample size.
- n_2 is the second sample size.

The intuition is:

Each sample loses one degree of freedom because each sample uses one observation of information to estimate its own sample mean.

23.5 Applying Degrees of Freedom to the Example

The example has:

Chicago sample:

$$N_1 = 66$$

San Francisco sample:

$$N_2 = 53$$

Therefore:

$$Df = (66 - 1) + (53 - 1)$$

Which gives:

$$Df = 65 + 52$$

$$Df = 117$$

The degrees of freedom determine which t-distribution table value we should use.

23.6 Standard Error of the Difference Between Two Means

The next step is calculating the uncertainty associated with the difference.

The standard deviation of:

$$\bar{X}_C - \bar{X}_{SF}$$

Is called the standard error.

The formula is:

$$SE = \sqrt{\frac{S_C^2}{n_C} + \frac{S_{SF}^2}{n_{SF}}}$$

Where:

- S_C is the sample standard deviation of Chicago.
- S_{SF} is the sample standard deviation of San Francisco.
- n_C and n_{SF} are the sample sizes.

23.7 Why Do We Add the Two Variances?

A very important concept discussed in class is the variance of a difference.

Many people incorrectly think:

"If we subtract two quantities, the uncertainty should cancel."

This is not true.

The variance rule says:

If two random variables are independent:

$$Var(X - Y) = Var(X) + Var(Y)$$

The minus sign disappears because variance squares the coefficient.

For example:

$$(-1)^2 = 1$$

Therefore:

$$Var(\bar{X}_C - \bar{X}_{SF}) = Var(\bar{X}_C) + Var(\bar{X}_{SF})$$

The uncertainties add together.

23.8 The Critical Role of Independence

The previous variance formula depends on a very important assumption:

$$\bar{X}_C$$

And

$$\bar{X}_{SF}$$

Must be independent.

If the two samples are not independent, then the variance decomposition is incorrect.

This means:

- The standard error calculation becomes wrong.
- The test statistic becomes wrong.
- The final statistical conclusion may be wrong.

This independence issue becomes extremely important later in the lecture when the professor discusses the MBA salary example.

23.9 Calculating the Critical Values

For a two-tailed test with:

$$\alpha = 0.05$$

We divide the significance level into two parts:

$$\frac{\alpha}{2} = 0.025$$

Therefore, we need two critical values:

$$-t_{\alpha/2}$$

And:

$$+t_{\alpha/2}$$

These create two rejection regions.

The rejection rule is:

Reject H_0 if:

$$T < -t_{\alpha/2}$$

Or:

$$T > t_{\alpha/2}$$

Otherwise, we fail to reject the null hypothesis.

23.10 The Test Statistic

The standardized test statistic is:

$$T = \frac{(\bar{X}_C - \bar{X}_{SF}) - D_0}{\sqrt{\frac{S_C^2}{n_C} + \frac{S_{SF}^2}{n_{SF}}}}$$

For the Chicago versus San Francisco example:

$$D_0 = 0$$

Because the null hypothesis assumes equal means.

Therefore:

$$T = \frac{\bar{X}_C - \bar{X}_{SF}}{\sqrt{\frac{S_C^2}{n_C} + \frac{S_{SF}^2}{n_{SF}}}}$$

The numerator represents the observed difference.

The denominator represents the expected variation of that difference.

The test statistic measures:

"How many standard errors away from zero is our observed difference?"

23.11 Summary of the Two-Sample Independent Test Procedure

The complete procedure is:

23.11.1 Step 1: Define hypotheses

Example:

$$H_0 : \mu_C = \mu_{SF}$$

$$H_A : \mu_C \neq \mu_{SF}$$

23.11.2 Step 2: Calculate degrees of freedom

Using:

$$Df = (n_1 - 1) + (n_2 - 1)$$

23.11.3 Step 3: Find critical values

For a two-tailed test:

$$\pm t_{\alpha/2}$$

23.11.4 Step 4: Calculate the test statistic

$$T = \frac{(\bar{X}_1 - \bar{X}_2) - D_0}{SE}$$

23.11.5 Step 5: Make the decision

If:

$$T$$

Falls outside the critical region:

Reject H_0.

If:

$$-T_{\alpha/2} < T < T_{\alpha/2}$$

Fail to reject H_0.

23.12 Key Takeaways

The central message of this section is:

A two-sample hypothesis test is essentially a comparison between:

1. **The observed difference between two sample means**, and
2. **The amount of variation we would naturally expect if the two populations were actually identical.**

The independence assumption is the foundation of the calculation. If independence is violated, the entire statistical conclusion may become unreliable.

24 | Complete Two-Sample Independent Test Example, Interpretation, and Right-Tailed Tests

24.1 Applying the Two-Sample t-Test: Chicago versus San Francisco

In this example, the company wants to understand whether customers from two regions behave differently.

The two populations are:

- Chicago customers
- San Francisco customers

The variable being measured is service utilization, measured in minutes.

The collected data are:

24.1.1 Chicago sample

Sample size:

$$N_C = 66$$

Sample mean:

$$\bar{X}_C = 433.9$$

Sample standard deviation:

$$S_C = 73$$

24.1.2 San Francisco sample

Sample size:

$$N_{SF} = 53$$

Sample mean:

$$\bar{X}_{SF} = 424.0$$

Sample standard deviation:

$$S_{SF} = 82$$

The research question is:

"Is there a significant difference between Chicago and San Francisco customers?"

Therefore, the hypotheses are:

$$H_0 : \mu_C = \mu_{SF}$$

And:

$$H_A : \mu_C \neq \mu_{SF}$$

This is a two-tailed test.

24.2 Step 1: Calculate Degrees of Freedom

The course uses the approximation:

$$Df = (n_C - 1) + (n_{SF} - 1)$$

Substituting the sample sizes:

$$Df = (66 - 1) + (53 - 1)$$

$$Df = 65 + 52$$

$$Df = 117$$

Therefore, we use the t-distribution with approximately:

$$Df = 117$$

24.3 Step 2: Determine the Critical Value

The significance level is:

$$\alpha = 0.05$$

Because this is a two-tailed test:

$$\alpha/2 = 0.025$$

We search the t-table using:

- Degrees of freedom approximately 120.
- Probability level 0.025.

The critical value is approximately:

$$T_{\alpha/2} = 1.98$$

Therefore, the rejection regions are:

$$T < -1.98$$

Or:

$$T > 1.98$$

The interpretation is:

The observed difference must be more than about 1.98 standard errors away from zero before we consider it statistically significant.

24.4 Step 3: Calculate the Test Statistic

The formula is:

$$T = \frac{\bar{X}_C - \bar{X}_{SF}}{\sqrt{\frac{S_C^2}{n_C} + \frac{S_{SF}^2}{n_{SF}}}}$$

First, calculate the difference between sample means:

$$\bar{X}_C - \bar{X}_{SF}$$

$$= 433.9 - 424.0$$

$$= 9.9$$

The Chicago sample average is higher by approximately 9.9 minutes.

Now calculate the standard error:

$$SE = \sqrt{\frac{73^2}{66} + \frac{82^2}{53}}$$

The denominator represents the expected variation in the difference between the two sample means.

After calculation:

$$SE \approx 14.4$$

Therefore:

$$T = \frac{9.9}{14.4}$$

Which gives:

$$T = 0.687$$

24.5 Step 4: Statistical Decision

The decision rule is:

Reject H_0 if:

$$T < -1.98$$

Or:

$$T > 1.98$$

The calculated value is:

$$T = 0.687$$

This value lies between:

$$-1.98$$

And:

$$1.98$$

Therefore, it does not fall into the rejection region.

The conclusion is:

We fail to reject the null hypothesis.

24.6 Interpretation of the Result

Failing to reject the null hypothesis does not mean that the two populations are exactly the same.

It means:

"The observed difference is not large enough compared with the natural variability in the data to conclude that the populations are different."

In this example:

Chicago customers appear to use the service about 9.9 minutes more than San Francisco customers.

However, because the variability among customers is large, this difference is not statistically significant.

The observed difference could easily occur due to random sampling variation.

24.7 Important Statistical Lesson

The professor emphasizes that hypothesis testing is not about proving that something is true.

Instead, it is about evaluating whether the data provide enough evidence against a default assumption.

The logic is:

1. Assume the null hypothesis is true.
2. Determine how unusual the observed data would be.
3. If the observation is extremely unlikely, reject the null hypothesis.
4. Otherwise, maintain the null hypothesis as a reasonable explanation.

24.8 Example of a Non-Zero Difference Hypothesis

The professor also discussed an important extension.

Sometimes the null hypothesis does not assume equality.

For example:

Suppose we want to test whether Chicago customers use the service at least 10 minutes more than San Francisco customers.

Then the hypothesis becomes:

$$\mu_C - \mu_{SF} = 10$$

The expected difference under the null hypothesis is:

$$D_0 = 10$$

The test statistic becomes:

$$T = \frac{(\bar{X}_C - \bar{X}_{SF}) - D_0}{SE}$$

The purpose of subtracting D_0 is to center the distribution around zero.

The general idea is:

The null hypothesis always defines the expected difference.

We compare the observed difference against that expected difference.

24.9 Moving from Two-Tailed Tests to One-Tailed Tests

The professor then introduced another example:

“Does an MBA program increase salary potential?”

Here, the research question is directional.

The goal is not simply to find any difference.

The goal is specifically:

$$\mu_{After} - \mu_{Before} > 0$$

The hypotheses are:

Null hypothesis:

$$H_0 : \mu_A - \mu_B \leq 0$$

Alternative hypothesis:

$$H_A : \mu_A - \mu_B > 0$$

This is called a:

right-tailed test.

24.10 Why Is It Called a Right-Tailed Test?

The alternative hypothesis is looking for positive differences.

The evidence we want to observe is on the right side of the distribution.

Therefore:

- Negative differences do not help us reject the null hypothesis.
- Only sufficiently large positive differences provide evidence.

The rejection rule becomes:

$$T > t_\alpha$$

There is only one critical value.

Unlike the two-tailed test:

Two-tailed test splits alpha:

$$\alpha/2 + \alpha/2$$

Right-tailed test uses all alpha on one side:

$$\alpha$$

24.11 Right-Tailed Test Example

The sample sizes are:

Before MBA:

$$N_B = 36$$

After MBA:

$$N_A = 36$$

The degrees of freedom are:

$$Df = (36 - 1) + (36 - 1)$$

$$Df = 70$$

For:

$$\alpha = 0.05$$

And:

$$Df = 70$$

The critical value is approximately:

$$T_\alpha = 1.667$$

The rejection rule is:

Reject H_0 if:

$$T > 1.667$$

24.12 The Initial MBA Test Result

The calculated test statistic is:

$$T = 1.34$$

Since:

$$1.34 < 1.667$$

The test statistic does not enter the rejection region.

Therefore:

We fail to reject the null hypothesis.

At this stage, it appears that the MBA program does not have statistically significant evidence of increasing salaries.

However, the professor then points out a critical problem:

The independence assumption is violated.

This changes the entire analysis.

24.13 The Major Lesson Before Moving Forward

The same data can produce different conclusions depending on whether the statistical assumptions are satisfied.

The independent two-sample test assumes:

Sample 1 and Sample 2 are independent

But in the MBA example:

The same individuals are measured:

- Salary before MBA.
- Salary after MBA.

These observations are naturally connected.

Therefore, the two samples are dependent.

Using an independent two-sample test in this situation can lead to incorrect conclusions.

The next chapter explains how to handle dependent samples using the **matching technique** and why it provides a much stronger analysis.

25 | Dependent Samples, Matching Technique, Average Treatment Effect, and Final Conclusions

25.1 Why the Independent Two-Sample Test Fails for the MBA Example

In the previous chapter, we analyzed an MBA salary example.

The question was:

Does completing an MBA program increase salary potential?

The initial approach treated the data as two independent samples:

- Salary before MBA.
- Salary after MBA.

The independent two-sample t-test produced:

$$T = 1.34$$

And the critical value was:

$$1.667$$

Because:

$$1.34 < 1.667$$

We failed to reject the null hypothesis.

However, the professor emphasized that this conclusion is not reliable because an important assumption is violated.

The assumption is:

The two samples must be independent.

25.2 Why Are the Samples Dependent?

The reason is that the same individuals appear in both samples.

For example:

Before MBA:

Alex earns \$195,000.

After MBA:

Alex earns \$226,000.

These two observations are not unrelated.

They come from the same person.

Alex's:

- Experience,
- Ability,
- Industry background,
- Previous education,
- Professional network,
- Personal characteristics,

Affect both salary measurements.

Therefore:

The before-MBA salary and after-MBA salary are highly correlated.

They are not independent observations.

25.3 Why Does Dependence Matter?

The independent two-sample test assumes:

$$Cov(X_1, X_2) = 0$$

Meaning:

The observations in one sample do not provide information about the other sample.

But in the MBA example:

$$Cov(X_{Before}, X_{After}) \neq 0$$

Because the same person generates both observations.

This affects the variance calculation.

The independent sample formula assumes:

$$Var(X_1 - X_2) = Var(X_1) + Var(X_2)$$

But when observations are correlated, the correct formula includes a covariance term:

$$Var(X_1 - X_2) = Var(X_1) + Var(X_2) - 2Cov(X_1, X_2)$$

The covariance term changes the uncertainty calculation.

Therefore, using the independent sample formula produces an incorrect standard error.

Once the standard error is wrong, the test statistic is wrong.

25.4 The Solution: Matching Technique

The professor introduces the idea of:

Matched-pair testing

Or:

Matched two-sample testing.

The main idea is:

Instead of comparing two separate samples, compare each individual with themselves.

For each person:

$$Difference = Salary_{After} - Salary_{Before}$$

This creates a new sample:

$$D_1, D_2, \ldots, D_n$$

Where each observation represents the individual improvement after the MBA program.

25.5 Why Does Matching Work?

Consider Alex again.

Before MBA:

$$Salary_{Before} = 195,000$$

After MBA:

$$Salary_{After} = 226,000$$

The difference is:

$$226,000 - 195,000 = 31,000$$

This difference removes characteristics that do not change.

For example:

- Alex's intelligence.
- Alex's years of experience.
- Alex's previous career achievements.

These factors existed before and after the MBA.

When we subtract:

$$After - Before$$

These constant factors cancel out.

The remaining difference captures the effect associated with obtaining the MBA degree.

25.6 The Matching Philosophy

The professor explains that matching is a powerful idea in statistical inference.

The purpose is:

Remove unwanted variation caused by individual differences.

Instead of asking:

"Are MBA graduates different from non-MBA graduates?"

We ask:

"For the same person, how much did salary change after receiving the MBA?"

This eliminates many hidden factors.

The resulting analysis focuses on the treatment effect.

25.7 Average Treatment Effect

The difference created by matching is called the treatment effect.

The average treatment effect is:

$$ATE = \frac{1}{n}\sum_{i=1}^{n}(After_i - Before_i)$$

In this example:

The average matched difference is:

$$19,729$$

This means:

On average, the MBA program appears to increase salary potential by approximately:

$$\$19,729$$

The remaining question is:

Is this increase statistically significant?

25.8 New Hypotheses After Matching

After matching, we no longer have two samples.

We have one sample:

$$D_i = After_i - Before_i$$

The hypotheses become:

Null hypothesis:

$$H_0 : \mu_D \leq 0$$

Alternative hypothesis:

$$H_A : \mu_D > 0$$

Where:

$$\mu_D$$

Is the average treatment effect.

The question becomes:

"Is the average salary increase after MBA significantly greater than zero?"

25.9 Why Does the Degree of Freedom Change?

Before matching:

We had two samples:

- Before MBA.
- After MBA.

Therefore, we used:

$$(n_1 - 1) + (n_2 - 1)$$

Degrees of freedom.

After matching:

We have only one sample:

$$D_1, D_2, \ldots, D_n$$

Therefore, we use:

$$Df = n - 1$$

The sample size is:

$$N = 36$$

Therefore:

$$Df = 36 - 1$$

$$Df = 35$$

25.10 Critical Value for the Matched Test

This is a right-tailed test.

Therefore, we use:

$$T_\alpha$$

With:

$$Df = 35$$

And:

$$\alpha = 0.05$$

From the t-table:

$$T_{0.05,35} = 1.690$$

The rejection rule becomes:

Reject H_0 if:

$$T > 1.690$$

25.11 Calculating the Matched Test Statistic

The test statistic is:

$$T = \frac{\bar{D} - 0}{S_D/\sqrt{n}}$$

Where:

- $\bar{D}$ is the average matched difference.
- S_D is the standard deviation of the matched differences.
- (n) is the number of individuals.

The average difference is:

$$\bar{D} = 19,729$$

The standard deviation of differences is:

$$S_D = 7,706$$

The sample size is:

$$N = 36$$

Therefore:

$$T = \frac{19,729}{7706/\sqrt{36}}$$

Since:

$$\sqrt{36} = 6$$

The denominator becomes:

$$7706/6$$

The resulting test statistic is:

$$T = 15.36$$

25.12 Final Statistical Conclusion

The critical value is:

$$1.690$$

The calculated statistic is:

$$15.36$$

Since:

$$15.36 > 1.690$$

The test statistic falls far inside the rejection region.

Therefore:

$$\text{Reject } H_0$$

The conclusion is:

There is very strong statistical evidence that the MBA program increases salary potential.

25.13 The Importance of Choosing the Correct Method

The professor highlights a very important lesson:

The same dataset produced two completely different conclusions.

25.13.1 Incorrect independent two-sample test:

- Ignored correlation.
- Produced:

$$T = 1.34$$

Conclusion:

Fail to reject the null hypothesis.

25.13.2 Correct matched-pair test:

- Used the relationship between observations.
- Removed individual-specific effects.
- Produced:

$$T = 15.36$$

Conclusion:

Reject the null hypothesis strongly.

25.14 Overall Summary of the Lecture

The lecture covered three major ideas:

25.14.1 Independent Two-Sample Testing

Used when:

- Two groups are unrelated.
- Observations come from independent populations.

Example:

Chicago customers versus San Francisco customers.

The procedure:

1. Define hypotheses.
2. Calculate degrees of freedom.
3. Find critical values.
4. Calculate test statistic.
5. Compare statistic with rejection region.

25.14.2 Importance of Independence

Independence is not a technical detail.

It determines whether:

- The variance calculation is correct.
- The standard error is correct.
- The conclusion is valid.

Violating independence can completely change the result.

25.14.3 Matched-Pair Testing

Used when:

- The same individuals are measured twice.
- Data naturally come in pairs.

Examples:

- Before and after treatment.
- Before and after education.
- Before and after policy intervention.

The method:

1. Compute individual differences.
2. Treat differences as one sample.
3. Perform a one-sample t-test.

25.15 Final Takeaway

The main message of this lesson is:

Statistical testing is not only about applying formulas. The first and most important step is understanding the structure of the data.

If two samples are independent, use an independent two-sample test.

If observations are naturally paired, remove the dependence through matching and analyze the differences.

Choosing the wrong model can lead to completely wrong managerial conclusions.

Part V

Simulation, Monte Carlo Methods, and Optimization with Simulation (Module 5)

26 | Monte Carlo Simulation and the Inverse Transformation Method

26.1 The Central Question of Simulation

Simulation is essentially about generating random samples. Suppose we know a probability distribution; the question is how a computer can actually generate observations from that distribution. This sounds simple when the distribution is something familiar, such as a normal distribution, but the same question appears in far more complicated settings. A large language model, for example, can be viewed as sampling from a complicated probability distribution over possible tokens: given exactly the same prompt twice, the model does not necessarily produce the same answer, because sampling is involved. Random-number generation therefore matters far beyond traditional simulation.

This module develops three methods for generating random observations: **inverse transformation**, **acceptance–rejection**, and **Metropolis**. The important progression is that each method removes an important limitation of the previous one, so rather than memorizing three unrelated algorithms, it is useful to think of them as an evolutionary sequence.

26.2 Uniform Random Numbers as the Building Block

The starting point for every method is the uniform distribution between zero and one, written $U \sim \mathrm{Uniform}(0, 1)$. Computers can readily generate pseudo-random numbers that behave approximately like independent draws from this distribution — repeated draws such as $0.17,\ 0.83,\ 0.42,\ 0.006,\ 0.71, \ldots$ with every region of equal length between zero and one equally likely. These are called **pseudo-random numbers**, since the randomness is produced computationally rather than being ideal mathematical randomness.

The natural question is then: suppose what we actually want is not a uniform random variable, but a normal random variable, a triangular random variable, or a discrete variable taking specific values with specified probabilities. How do we turn a uniform random number into the distribution we actually want? That question leads directly to inverse transformation.

26.3 Stretching and Squeezing: An Intuition for Transformation

A useful intuition is to think of the interval from zero to one as a piece of elastic material: some regions can be **stretched** and others **squeezed**. Doing this correctly transforms a uniform distribution into another probability distribution.

Consider the simplest example, $W = 4U$. Because U ranges between 0 and 1, W ranges between 0 and 4 — the horizontal range has been stretched by a factor of four. But probabilities must still sum to one, so if the original uniform density has height 1 over an interval of length 1, the new density must have height $\frac{1}{4}$ over an interval of length 4. In short: **stretch horizontally $\to$ shrink vertically**, which preserves total probability. This idea generalizes: instead of stretching every part of the interval by the same factor, we can stretch different regions by different amounts, and that nonlinear stretching is exactly what allows us to create complicated target distributions.

26.4 From Density to CDF to Inverse CDF

Suppose the target distribution has density g. From that density we obtain its cumulative distribution function (CDF), denoted G, which answers: what is the probability that the random variable is less than or equal to a particular value x? For a normal distribution, for instance, the CDF starts near zero far to the left, equals 0.5 at the mean, and approaches one far to the right. The CDF therefore maps values of X into numbers between zero and one.

Our simulation problem requires exactly the opposite direction: we already possess a number between zero and one and want to turn it into X. Reversing the mapping gives the **inverse CDF**, G^{-1}, which is why the technique is called the **inverse transformation method**.

26.5 The Inverse Transformation Algorithm

The procedure has three steps:

1. Determine the CDF of the target distribution.
2. Invert that CDF.
3. Generate $U \sim \text{Uniform}(0, 1)$ and compute the corresponding value using the inverse CDF.

Symbolically, the central formula is

$$X = F^{-1}(U).$$

For interpretation purposes this is more important than the symbol itself: generate a percentile at random, and then ask what value of X corresponds to that percentile. That is inverse transformation.

26.6 A Discrete Example

Suppose X can take values $0, 1, 2, 3, 4$ with probabilities 0.2, 0.4, 0.2, 0.1, 0.1. The cumulative probabilities are therefore 0.2, 0.6, 0.8, 0.9, 1. Turning this into intervals on the uniform $(0, 1)$ scale:

- If $U \in (0, 0.2)$, output 0.
- If $U \in (0.2, 0.6)$, output 1.
- If $U \in (0.6, 0.8)$, output 2.
- If $U \in (0.8, 0.9)$, output 3.
- If $U \in (0.9, 1)$, output 4.

The interval corresponding to the value 1 has length $0.6 - 0.2 = 0.4$, so a uniform random number lands in it with probability 0.4 — exactly the desired probability of $X = 1$. Similarly, the interval corresponding to 3 has length 0.1, so 3 appears with probability 0.1. This is the essence of the method.

26.7 A Second Discrete Example

Consider a random variable taking values $10, 15, 20, 25$ with probabilities 0.1, 0.7, 0.1, 0.1. The inverse transformation mapping is:

- $U \in (0, 0.1) \Rightarrow 10$
- $U \in (0.1, 0.8) \Rightarrow 15$
- $U \in (0.8, 0.9) \Rightarrow 20$
- $U \in (0.9, 1) \Rightarrow 25$

The value 15 receives such a large interval because it should occur seventy percent of the time, and the interval from 0.1 to 0.8 occupies seventy percent of the uniform $(0, 1)$ range. This is **squeezing an entire region of uniform numbers into the same output value** — an interpretation worth remembering because it provides intuition rather than just an algorithm.

26.8 Continuous Distributions: The Normal Example

Now consider generating a normal random variable with mean 100 and standard deviation 20. Generate $U \sim \text{Uniform}(0, 1)$.

- If $U = 0.5$: the 50th percentile of a normal distribution is its mean, so $X = 100$.
- If $U = 0.7$: the standard normal value with cumulative probability approximately 0.7 is $Z \approx 0.52$–0.53, so $X \approx 100 + 0.52(20) \approx 110$.
- If $U = 0.99$: the corresponding $Z \approx 2.33$, so $X \approx 100 + 2.33(20) \approx 146.6$.

The exact numerical value is less important than the procedure: we are asking *which normal value has cumulative probability equal to* U, using the cumulative normal table (or Φ^{-1}). As U approaches one, the transformed value grows increasingly large; as U approaches zero, it moves further into the negative tail. This is the stretching phenomenon again: the middle of the uniform distribution is stretched relatively little, while the extreme regions near 0 and 1 are stretched enormously.

26.9 The Triangular Distribution

Consider a triangular distribution with minimum 20, mode 23, and maximum 26. Because half of the triangular distribution lies below 23, $U = 0.5$ maps to $X = 23$. But suppose $U = 0.25$: we now need the value X for which the area under the triangular density from 20 to X equals 0.25, which becomes a geometry problem.

An important observation is that X is **not** simply halfway between 20 and 23. The density is increasing as X moves from 20 toward 23, so if we stopped at the midpoint 21.5, both the base and the height of the resulting small triangle would be reduced by one-half, so its area is reduced by $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$. The cumulative probability at the midpoint is therefore only one quarter of the probability contained in the entire left half of the distribution. This illustrates a general principle: **percentiles are determined by probability area, not by geometric distance along the x-axis.**

Key Takeaways

- Simulation begins with uniform $(0, 1)$ pseudo-random numbers.
- The CDF converts observations into cumulative probabilities; the inverse CDF does the opposite, converting cumulative probabilities into observations: $X = F^{-1}(U)$.
- Inverse transformation can be understood as stretching and squeezing the uniform distribution until it has the shape of the target distribution.

26.10 The Limitation of Inverse Transformation

To use inverse transformation we need enough information about the target distribution to obtain its CDF and then invert it. In many realistic problems this is difficult, because the target density generally needs to be **normalized** — its total probability must equal one. In a gigantic state space involving sentences, images, or other high-dimensional objects, we might only know that "object A looks twice as plausible as object B." That is a **score**, not a probability, and converting all such scores into a normalized distribution can be computationally prohibitive. Without a normalized distribution, constructing and inverting the CDF becomes impractical. This motivates the second method developed in this module: **acceptance–rejection sampling**, whose conceptual advantage is that we can often work with **unnormalized scores** rather than requiring the complete normalized probability distribution.

27 | Acceptance–Rejection Sampling

27.1 Motivation: Working with Unnormalized Scores

Inverse transformation is conceptually elegant, but it generally requires a properly normalized probability distribution from which to construct and invert a CDF. In complicated real-world situations normalization can be extremely difficult: with an enormous number of possible sentences or images, we might have a function assigning a score to each outcome without any convenient way to compute the total score over every possible outcome. Acceptance–rejection is motivated as a method that operates directly on scores, without requiring that normalization constant to be calculated.

27.2 Scores versus Probabilities

The distinction between a **score** and a **probability** is central. Suppose four outcomes have scores $5, 10, 5, 5$. These do not sum to one, so they are not probabilities; dividing each by their sum (25) gives the normalized probabilities 0.20, 0.40, 0.20, 0.20. Acceptance–rejection asks whether that normalization step can be avoided: if all we need is to generate observations with frequencies proportional to the scores, perhaps we can work with the scores directly. In very large spaces, summing over every possible outcome could be computationally enormous, so acceptance–rejection instead uses **relative** scores.

27.3 The Acceptance Ratio and the Mode

Acceptance–rejection requires knowing the **mode** — the outcome with the highest score — which serves as a benchmark. For any candidate outcome x,

$$\text{acceptance probability} = \frac{\text{score}(x)}{\text{score}(\text{mode})}.$$

Because the mode has the maximum score, this ratio is always between zero and one, making it a valid acceptance probability. Crucially, the maximum score itself does not have to be normalized — it could be 0.4, or 10, or any other scale; only the *relative* scores matter.

Since the mode's acceptance probability is score/score $= 1$, **the mode is always accepted**, which makes intuitive sense: it is supposed to be the most frequently occurring outcome. A candidate with score 0.2 relative to a mode of 0.4 has acceptance probability 0.5; a candidate with score 0.1 has acceptance probability 0.25. The rejection mechanism changes the frequencies of the initially generated observations until their relative frequencies match the target scores.

27.4 The Two-Stage Sampling Process

Acceptance–rejection is a **two-stage generation process**. In the first stage we generate a candidate, which is not necessarily the final output. In the second stage we decide whether to accept that candidate: if accepted, it becomes part of the sample; if rejected, we discard it and generate another candidate. Verbally: generate something, inspect it, and if it satisfies the acceptance criterion let it pass — otherwise reject it and regenerate. That is the fundamental architecture of acceptance–rejection sampling.

27.5 Why Rejection Reproduces the Target Distribution

Suppose a first-stage generator produces outcomes $1, 2, 3, 4$ uniformly, and the target scores are 0.20, 0.15, 0.25, 0.40 respectively, so value 4 is the mode. The acceptance probabilities are

$$\frac{0.20}{0.40} = 0.50, \qquad \frac{0.15}{0.40} = 0.375, \qquad \frac{0.25}{0.40} = 0.625, \qquad \frac{0.40}{0.40} = 1.$$

These are *acceptance* probabilities, not the probabilities of the final distribution — they need not sum to one. Imagine the first-stage generator produces 100 observations of each value. Applying the acceptance rates leaves approximately 50, 37.5, 62.5, and 100 accepted observations respectively, out of 250 total. Dividing by 250 gives approximately 0.20, 0.15, 0.25, 0.40 — exactly the target probabilities. So although the initial generator is uniform, the rejection mechanism reshapes the distribution into the target one.

27.6 Why Normalization Disappears

Suppose instead of probabilities we had raw scores $20, 15, 25, 40$. The acceptance probabilities are still 0.50, 0.375, 0.625, 1 — identical to before. If every score were multiplied by one million, nothing would change, because the common scaling factor cancels in every ratio. This is why unnormalized scores are sufficient: models can score objects, sentences, or other outcomes even though those scores do not naturally sum to one.

27.7 Sampling from the Triangular Score Function

Return to the triangular distribution with minimum 20, mode 23, maximum 26. Where inverse transformation required calculating areas under the triangle, acceptance–rejection offers another way: assign the triangular function height 1 at its peak. The resulting triangle has base 6 and height 1, so its area is $\frac{1}{2}(6)(1) = 3$ — **not** normalized (a proper density would need area one), but acceptance–rejection does not care; the triangle can still be used directly as a score function.

First generate a candidate uniformly between 20 and 26. Suppose the candidate is 23 (the mode): its score is 1, so the acceptance probability is $1/1 = 1$, i.e. always accept. Suppose the candidate is 21.5, halfway between 20 and 23: its height under the triangle is 0.5, so the acceptance probability is $0.5/1 = 0.5$ — accept with probability 50%. Similarly, a candidate of 24.5, halfway between 23 and 26 on the descending side, also has score 0.5 and is accepted with probability 50%. Notice how much simpler this is than solving the inverse-CDF geometry problem: we only need the height of the score function at the candidate relative to its maximum height, not total areas.

27.8 Implementing "Accept with Probability p"

Given an acceptance probability for a candidate, how do we actually implement "accept with probability p"? Generate a second, independent uniform random number $V \sim \text{Uniform}(0, 1)$ and apply the rule: accept if $V \leq p$, otherwise reject. For an acceptance probability of 0.375, accept if $V \leq 0.375$; for 0.625, accept if $V \leq 0.625$. For the mode, whose acceptance probability equals one, every possible $V \in (0, 1)$ satisfies $V \leq 1$, so the mode is always accepted. This gives a general implementation rule for any acceptance probability.

27.9 The Complete Acceptance–Rejection Algorithm

For a discrete example, the process is:

1. Generate a candidate uniformly from the possible outcomes.
2. Determine the candidate's acceptance probability, score(candidate)/score(mode).
3. Generate an independent $V \sim \text{Uniform}(0, 1)$.
4. If V falls inside the acceptance region, accept the candidate; otherwise reject it and generate another candidate.
5. Repeat until enough accepted observations have been obtained.

A rejected observation is simply discarded, not converted into something else; the algorithm then tries again. For example, if candidate 2 has acceptance probability 0.375 and the second uniform draw is $0.70 > 0.375$, candidate 2 is rejected and a new candidate is generated. If the next candidate is 4, with acceptance probability 1, it is accepted automatically. This repeats until the desired sample size is reached.

27.10 Exam Guidance

For an acceptance–rejection question involving discrete outcomes, it is generally not necessary to simulate thousands of observations. The critical answer is the **acceptance table** — for example, value 1 has acceptance probability 0.50, value 2 has 0.375, value 3 has 0.625, and value 4 has 1 — together with a clear explanation of the candidate-generation step. Presenting the acceptance rates and the procedure is enough for exam purposes; actually programming the sampling algorithm is more relevant to a computational assignment.

27.11 A Continuous Example: Sampling Up to an Unknown Normalization Constant

A particularly important continuous example: suppose the target density has the form

$$p(x) = C\,f(x), \qquad -2 \le x \le 2,$$

where f is the standard normal density and C is an unknown normalization constant. At first this looks problematic, since C is unknown. But acceptance–rejection does not require knowing it, because when we form a ratio of scores, C cancels. The mode of the standard normal distribution is zero, so zero has the maximum score and is accepted with probability one; other candidates are accepted relative to the standard normal density evaluated at zero. This is an excellent illustration of why acceptance–rejection is useful when the target density is known only **up to a normalization constant**.

27.12 Generating Uniform Candidates on an Arbitrary Interval

To sample candidates uniformly between -2 and 2 from a base $U \sim \text{Uniform}(0,1)$, use

$$X = 4U - 2.$$

Multiplying U by 4 changes the range from $(0,1)$ to $(0,4)$; subtracting 2 shifts that interval to $(-2,2)$. More generally, to obtain a uniform random variable on (a,b), use

$$X = a + (b-a)U.$$

The guiding principle is conceptual rather than a formula to memorize: **stretch first, then shift.**

27.13 Comparing Inverse Transformation with Acceptance–Rejection

Inverse transformation is direct: generate U and immediately transform it into X. If the inverse CDF is easy to calculate, the method is elegant and efficient, but it requires the CDF and its inverse, which can be difficult to obtain. Acceptance–rejection avoids that requirement, needing only relative scores and the maximum score — but at a price: many candidates may be rejected, wasting computational effort. If the target distribution is very concentrated relative to the proposal distribution, most generated candidates might be rejected. There is always a tradeoff between the two methods.

27.14 The Remaining Weakness of Acceptance–Rejection

Acceptance–rejection removes the need to calculate the normalization constant, but it still requires knowing something important: **the mode and its score**, since the maximum score is the denominator of the acceptance ratio. If the state space is incredibly complicated — for example, if finding the globally most likely sentence, image, or configuration is itself extremely difficult — then acceptance–rejection still has a limitation. Finding the mode of a distribution can itself become very challenging in real-world problems, which motivates **Metropolis** as a more generic technique that removes this requirement.

Key Takeaways

- Acceptance–rejection generates candidates easily and then selectively filters them according to their relative scores: acceptance probability $= \text{score}(x)/\text{score}(\text{mode})$.
- The mode is accepted with probability one; lower-score outcomes are increasingly filtered out.
- Advantage over inverse transformation: no normalization constant or inverse CDF is required.
- Remaining disadvantage: the mode (maximum score) must still be known, and rejection can waste samples.

28 | Markov Chains and the Metropolis Algorithm

28.1 Where Metropolis Fits

There are three major random-number-generation techniques in this module. The first is the **transformation technique**: start with a uniform random variable, which computers generate easily, and transform it into a random variable with the desired target distribution. The second is **acceptance–rejection sampling**: generate random candidates and use another random mechanism to decide whether each proposal should be accepted or rejected, selectively reshaping the proposal distribution into the target distribution. The third technique, and the principal topic of this chapter, is the **Metropolis algorithm**, which belongs to the broader family of methods called **Markov Chain Monte Carlo (MCMC)**. Other MCMC methods exist, such as Gibbs sampling, but this module focuses on Metropolis.

The important conceptual difference is that Metropolis does not attempt to generate an independent observation from the target distribution in one step. Instead, it constructs a **Markov chain** whose long-run behavior corresponds to the target distribution.

28.2 What Is a Markov Chain?

Suppose the target random variable can take ten values, $1, 2, \dots, 10$, thought of as ten possible states, and suppose we are currently at state 3. The defining property of a Markov chain is that the probability distribution of the **next state depends only on the current state**, not on the complete history of the process: it does not matter whether we arrived at state 3 from state 2, from state 4, or were already there in the previous iteration. All that matters is that we are currently at state 3. A useful analogy is a person wandering between locations who does not remember how they arrived at their present location; their next movement is determined only by where they currently are. This is the **Markov property**.

28.3 What Are We Trying to Accomplish?

Suppose the desired probabilities of states 1 through 10 are unequal — for example, $P(1) = 0.11$, $P(2) = 0.12$, $P(3) = 0.09$, and so forth. If we simply generated integers uniformly between 1 and 10, every state would appear approximately 10% of the time, which is not our target. Instead, we want to construct a random walk such that, after running it for a sufficiently long time, the fraction of time spent in each state approaches its target probability: if $P(1) = 0.11$, we want approximately 11% of observations to be state 1, and so on. The objective is not to obtain the correct probability immediately, but to construct transition rules whose **long-run visitation frequencies** reproduce the target distribution. This is the central idea of MCMC.

28.4 Designing the Random Movement

Consider an interior state such as state 3. A simple local movement rule allows the chain to move left to state 2, stay at state 3, or move right to state 4. If these three proposed movements were treated equally (each with probability $\frac{1}{3}$) and simply accepted, the resulting long-run distribution would tend toward something close to uniform. We therefore need a mechanism that encourages movement toward states with higher target scores and discourages movement toward states with lower scores — the **Metropolis acceptance probability**.

28.5 The Metropolis Acceptance Ratio

Suppose we are currently at state i and propose moving to state j. The acceptance probability is based on the ratio P_j/P_i. But this ratio might exceed one, and since probabilities cannot exceed one, we clip the ratio at one:

$$\alpha(i \to j) = \min\left(1,\ \frac{P_j}{P_i}\right).$$

This is perhaps the single most important formula of the chapter. For a proposed move from state 3 to state 4, the acceptance probability is $\min(1, P_4/P_3)$, and the overall probability of actually moving from 3 to 4 is

$$\frac{1}{3} \times \min\left(1, \frac{P_4}{P_3}\right),$$

because two things must happen: we must first propose the rightward move (probability $\frac{1}{3}$), and then, conditional on proposing it, we must accept it. The multiplication rule gives the total transition probability. The same reasoning gives the probability of moving from 3 to 2 as $\frac{1}{3} \times \min(1, P_2/P_3)$.

28.6 Why the Ratio Works

If the proposed state has a higher score than the current state, $P_j/P_i > 1$, and after clipping the acceptance probability becomes one: moves toward a higher-score state are accepted automatically. If the proposed state has a lower score, the ratio is below one and the move is accepted only probabilistically — for example, a ratio of 0.8 means we accept the move with probability 0.8. This creates a systematic tendency toward high-score states without completely preventing movement toward lower-score states.

That last point is important: Metropolis does *not* simply move uphill all the time. If it always moved toward the highest-probability state, the chain would eventually become trapped there. Instead, Metropolis still permits downward moves, merely making them less likely, so the process continues exploring the entire state space while spending proportionally more time in high-probability regions — intuitively, directing more "traffic" toward popular states.

28.7 The Probability of Staying

Once the probabilities of moving left and moving right are calculated, the easiest way to calculate the probability of remaining at the current state is to use the fact that probabilities must sum to one:

$$P(\text{stay at } 3) = 1 - P(\text{move left}) - P(\text{move right}).$$

This staying probability includes several possibilities: we might explicitly propose staying, or we might propose moving left or right but reject the proposal. All of those outcomes leave the Markov chain at the current state. For exam purposes, the easiest method is simply to calculate all outgoing transition probabilities and assign the remaining probability to staying in the current state.

28.8 Boundary States

Interior states such as 3 are straightforward because they have neighbors on both sides, but state 1 and state 10 are different: state 1 cannot move left to 0, and state 10 cannot move right to 11. At state 1, only two proposals are possible — stay at 1, or move to 2 — so the proposal probability becomes $\frac{1}{2}$ rather than $\frac{1}{3}$. The transition probability from 1 to 2 is consequently

$$\frac{1}{2} \times \min\left(1, \frac{P_2}{P_1}\right),$$

and the probability of remaining at 1 is one minus that quantity. State 10 is handled symmetrically.

28.9 Why Normalization Constants Cancel

So far $P_1, P_2, P_3, \ldots$ have been described as probabilities, but in real applications they do **not** actually have to be normalized probabilities. Suppose instead a model gives scores $F(1) = 100$, $F(2) = 200$, $F(3) = 250, \ldots$ that do not need to sum to one. If the true probability has the form $P(x) = F(x)/Z$ for an unknown normalization constant Z, then

$$\frac{P_j}{P_i} = \frac{F(j)/Z}{F(i)/Z} = \frac{F(j)}{F(i)}.$$

The Z terms cancel, so we never need to calculate the normalization constant. This is an extraordinarily important property: Metropolis only uses *ratios* of scores.

28.10 Why Normalization Can Be Difficult

For a small distribution with ten possible states, normalization is easy: sum all ten scores and divide. But an enormous state space — millions, billions, or vastly more possible configurations, as arise with modern machine-learning models — can make computing the normalization constant computationally impossible. A neural network can produce scores for different outcomes even when explicitly enumerating and normalizing over the entire output space is infeasible. Metropolis can work directly with relative scores because the unknown common normalization factor cancels in the ratio. This is one of the major conceptual advantages of MCMC over acceptance–rejection, which typically requires additional global information about the target and proposal distributions: if we can evaluate a score proportional to the target density, we can potentially construct a Metropolis sampler.

28.11 Burn-In and Convergence

Suppose the chain is initialized at state 1. Initially, observations may depend strongly on that arbitrary starting location, but as the chain continues moving it gradually loses information about where it began, eventually approaching the target stationary distribution. The early observations are therefore often discarded — this is commonly called the **burn-in period**. Allowing only local movements (left, right, or stay) may require many iterations before the process settles into equilibrium behavior; allowing longer-distance proposals can speed up movement through the state space, although it makes the mechanism more complicated. There is a tradeoff: a simple proposal mechanism is easier to implement but may mix slowly, while a richer proposal mechanism explores more quickly but requires more computational effort. This module emphasizes the simpler neighboring-state construction.

28.12 Does the Initial State Matter?

In the long run, assuming the chain has the appropriate properties, the initial state should not determine the stationary behavior — the chain could be started at state 1, 5, or 10, and after enough transitions it effectively "forgets" its starting location. What ultimately matters is the transition mechanism, not the arbitrary initialization; this is another way to understand burn-in, since early observations contain information about initialization while later observations increasingly reflect the stationary behavior of the chain.

Key Takeaways

- Metropolis converts relative scores into a random walk whose long-run visitation frequencies reproduce the target distribution.
- Propose random movements through the state space, then accept them with probability $\min(1, \text{proposed-state score}/\text{current-state score})$.
- Higher-score states attract more traffic; lower-score states are still visited, but less frequently.
- Because Metropolis works with ratios, unknown normalization constants cancel, making it exceptionally useful for complicated probability distributions.

29 | Implementing Metropolis and Extending It to Continuous Distributions

29.1 Two Stages: Construction and Generation

It is useful to separate the Metropolis procedure into two stages. In the **preparation stage** we calculate the state-transition probabilities. In the **generation stage** we repeatedly generate uniform random numbers and use those transition probabilities to decide where the Markov chain moves. For a ten-state example, the first and last states have two possible transitions while the eight interior states have three each, giving a transition table with 28 entries. Manually producing a large transition table is not practical — with 1,000 states, the probabilities would naturally be generated through a loop in code. For exam purposes, the emphasis is on knowing *how* the transition probabilities are constructed, rather than writing the corresponding code.

29.2 From Transition Probabilities to a Random Move

Suppose we are currently in state 1, where the only possibilities are to stay at 1 or move to 2, each with transition probability $\frac{1}{2}$. To implement this, generate $U \sim \text{Uniform}(0, 1)$ and partition the interval from 0 to 1 according to the transition probabilities: if U falls in the first half, move from 1 to 2; if it falls in the second half, stay at 1. For example, if $U = 0.07$, it falls in the first region and the chain moves from state 1 to state 2; if the next draw is $U = 0.56$, it falls in the second region and the chain stays at 1. The transition probabilities determine **how wide each region should be**; the uniform random number determines **which transition actually occurs** — the same general simulation principle used throughout this module: map intervals of a uniform random variable into particular outcomes.

29.3 Interior States Have Three Regions

For an interior state such as state 2, there are three possible outcomes: move left to 1, stay at 2, or move right to 3. Suppose these transition probabilities, computed from the Metropolis formulas, are approximately 0.25 and 0.305 (with the remainder assigned to staying). We construct cumulative intervals: the first from 0 to 0.25, the second from 0.25 to about 0.555, and the remainder for the third outcome. Generating $U = 0.09$ falls in the first region, $U = 0.40$ falls in the second, and $U = 0.82$ falls in the third. A single uniform random number therefore determines the next state once the transition probabilities have been translated into cumulative regions.

29.4 Popular States and Long-Run Frequencies

If a state has a relatively high target probability, the transition mechanism will create relatively large probability mass associated with staying there or returning there; as a result, whenever the chain reaches such a "popular" state, it tends to spend more time there, and repeated simulation produces more observations of that state. A less probable state attracts less traffic. After many iterations, these differences in visitation frequencies create the target distribution. This is the underlying mechanism of Metropolis: we never directly instruct the simulation to "generate state 2 exactly 12% of the time." Instead we construct local transition probabilities, and the Markov chain automatically produces the desired global frequencies after sufficiently many iterations.

29.5 The Complete Discrete Metropolis Algorithm

Suppose we have states 1 through K with target scores $f(1), \ldots, f(K)$.

1. Start at any state i.
2. For an interior state, identify the three possible proposals $i-1$, i, and $i+1$.
3. Calculate the actual transition probabilities using the Metropolis score ratios, e.g. the probability of moving right is $\frac{1}{3}\min(1, f(i+1)/f(i))$, and similarly for moving left.
4. Calculate the stay probability as the residual: one minus the other outgoing probabilities.
5. Generate $U \sim \text{Uniform}(0,1)$, divide the interval $(0,1)$ into regions according to the transition probabilities, and observe where U falls to determine the next state.
6. Let the next state become the current state, and repeat many times (on the order of thousands of iterations).

Eventually the Markov chain reaches its stable behavior, and the empirical frequencies with which it visits the states approximate the target distribution.

29.6 Moving to the Continuous Case

Suppose the target distribution is continuous. In the discrete example, moving left or right was straightforward: if we were at state 3, the neighbors were 2 and 4. But suppose X is continuous and we are currently at $X = 21$ — is the next state 21.1? 21.01? 21.001? There is no unique "next" real number, so the discrete left-right-neighbor construction cannot be used literally. The solution is to generate a **continuous random proposal**.

29.7 The Triangular Distribution Example

Consider the same triangular target used earlier: lower endpoint 20, mode 23, upper endpoint 26 (the distribution need not generally be symmetric, though this particular example has the mode in the middle). Suppose the current state is $X = 21$. How should we propose a new state?

Instead of "move one state left" or "move one state right," generate $U \sim \text{Uniform}(-1, 1)$, representing a random movement: if U is positive we move in one direction, if negative the other. The proposed state is

$$X_{\text{proposal}} = X + U.$$

If the current state is 21 and $U \approx -0.104$, then $X_{\text{proposal}} \approx 21 - 0.104 = 20.896$. This is only a **proposal**: the key vocabulary is *current state*, *proposal*, *acceptance probability*, and then *accepted or rejected next state.*

29.8 The Acceptance Rule Is Essentially Unchanged

Once the proposal is generated, calculate the score at the proposed point, $f(X_{\text{proposal}})$, and at the current point, $f(X)$, then compute

$$\alpha = \min\left(1, \frac{f(X_{\text{proposal}})}{f(X)}\right).$$

This is essentially the same formula used in the discrete problem; the only difference is how the proposed state is generated. In the discrete case the proposal is left or right; in the continuous case a continuous random perturbation determines how far in either direction we propose to move.

29.9 Constructing an Unnormalized Score Function for the Triangle

We do not need the normalized triangular probability density — only a function with the appropriate relative shape. From 20 to 23 the density is increasing, so we can use the score $f(x) = x - 20$ for $20 \leq x \leq 23$. After 23 the triangle decreases to zero at 26, so we use $f(x) = 26 - x$ for $23 \leq x \leq 26$. At $x = 23$ both expressions give 3, so the two pieces meet, and the resulting piecewise function has the required triangular shape without needing to be normalized. If the entire score function were multiplied by 100, the acceptance ratio $100f(X_{\text{proposal}})/100f(X)$ would be unchanged, since the 100 cancels — again, Metropolis requires relative scores, not necessarily normalized probabilities.

29.10 A Numerical Acceptance Example

Suppose the current state is $X = 21$ and the proposed state is $X_{\text{proposal}} \approx 20.895$; both lie on the increasing part of the triangle. The score at the current state is $21 - 20 = 1$; the score at the proposed state is approximately $20.895 - 20 = 0.895$. The ratio is $0.895/1 \approx 0.895$, so $\alpha \approx 0.895$: the proposed move is accepted with probability around 89.5%. It is not accepted with probability one because the proposal moved slightly away from the mode, so its score is slightly lower.

Now consider the opposite case: at 21, propose moving to 21.5. Since this moves upward toward the mode at 23, the score at 21.5 exceeds the score at 21, the ratio exceeds one, and after clipping $\alpha = 1$ — the move is accepted automatically. This illustrates the fundamental Metropolis behavior: moves toward higher-score regions are accepted, while moves toward lower-score regions are accepted with probability less than one, but downward moves remain possible. This prevents the algorithm from degenerating into an optimization procedure that climbs to the mode and stays there; the objective is **sampling**, not optimization.

29.11 Implementing the Accept/Reject Decision

After obtaining α, generate a second independent uniform random number $V \sim \text{Uniform}(0, 1)$. If $V \leq \alpha$, accept the proposal and set $X_{\text{next}} = X_{\text{proposal}}$; if $V > \alpha$, reject it and set $X_{\text{next}} = X$. Repeat the entire procedure — generate another proposal perturbation, calculate a new ratio, accept or reject — many times. The resulting sequence of X values forms the Markov chain.

29.12 Boundary Handling in the Continuous Case

The continuous target exists only between 20 and 26. Near 20, a negative perturbation could propose a value below 20, which lies outside the support of the target distribution (similarly for a proposal above 26). One way to view the algorithm is that points outside the target domain have zero target score and therefore zero acceptance probability, so an invalid proposal is simply not accepted; the chain remains at the current legitimate value and proceeds to the next iteration. Proposals can also be explicitly restricted or clipped near the boundaries as an implementation detail.

29.13 Discrete and Continuous Metropolis Are the Same Idea

At first glance the discrete and continuous versions look different, but conceptually they are almost identical. For the discrete case: current state, propose a neighboring state, compare proposed and current scores, accept according to the ratio, otherwise stay. For the continuous case: current value, generate a random perturbation, create a proposed value, compare proposed and current scores, accept according to the ratio, otherwise stay. The critical formula in both cases remains $\alpha = \min(1, \text{proposed score}/\text{current score})$; the main difference is only the **proposal mechanism** — a naturally defined “neighbor” in discrete space versus a randomly generated perturbation in continuous space.

29.14 Two Kinds of Randomness

There are two different kinds of randomness inside the continuous Metropolis algorithm. The first randomness creates the **proposal** (e.g. $U \sim \text{Uniform}(-1, 1)$), answering “where should I consider moving?” The second randomness determines **acceptance** (e.g. $V \sim \text{Uniform}(0, 1)$), answering “given that proposal, should I actually move there?” This proposal-versus-acceptance distinction is one of the most important ways to understand MCMC: the proposal mechanism explores, and the acceptance mechanism reshapes that exploration so that the resulting long-run sample has the target distribution.

29.15 The Complete Continuous Metropolis Algorithm

A compact, exam-ready statement of the algorithm:

1. Start with an initial state X within the support of the target distribution.
2. Generate a random proposal perturbation U (e.g. uniform on $(-1, 1)$) and set $X_{\text{proposal}} = X + U$.
3. Evaluate the target score at the proposed and current states, and compute $\alpha = \min(1, f(X_{\text{proposal}})/f(X))$.
4. Generate $V \sim \text{Uniform}(0, 1)$; if $V \leq \alpha$, accept the proposed state, otherwise remain at the current state.
5. Repeat many times, discarding the initial burn-in observations if necessary.

The remaining states form a sample approximately distributed according to the target distribution.

Key Takeaways

- Metropolis has two layers: generate a candidate move, then decide whether to accept it using a target-score ratio.
- For discrete variables, candidate moves are neighboring states; for continuous variables, candidate moves are generated by adding continuous random noise.
- In both cases, repeated accepted and rejected movements create a Markov chain whose long-run frequencies reproduce the target distribution.
- We can generate complicated random variables without explicitly calculating their normalized probability distribution — we only need a way to compare the relative target score of two states.

30 | Optimization with Simulation: The Hotel Overbooking Problem

30.1 What Does "Optimization with Simulation" Mean?

Simulation does not have to be the end of an analysis; it can be used inside an optimization problem. Optimization with simulation moves repeatedly between two stages: a **simulation** stage, in which we generate random outcomes, and an **optimization** stage, in which we evaluate a decision based on those random outcomes and compare alternative decisions. The overall logic is: choose a decision, simulate its uncertain consequences, estimate its expected performance, change the decision, simulate again, compare, and continue until the best decision is identified. This connects the optimization material developed earlier in the course with the simulation techniques of this module.

30.2 The Hotel Overbooking Example

Let's consider a hotel that has 100 rooms. Suppose that each reservation generates revenue of \$150. If a customer actually arrives and occupies a room, the hotel incurs a preparation/cleaning cost of \$30. Customers have a 5% probability of not showing up, so the probability that a customer *does* show up is 95%. The customer has already paid, and a late no-show does not receive a refund. This creates an opportunity: if some customers fail to arrive, the hotel could accept more than 100 reservations while still having no more than 100 people physically arrive — the economic logic behind **overbooking**.

30.3 Why Overbooking Can Increase Profit

If the hotel accepts exactly 100 reservations, roughly 5% of customers will fail to arrive, leaving empty rooms even though every available room was technically sold (e.g. only 95 customers arrive, leaving 5 rooms unused, even though the hotel still received the booking revenue). Because some customers will probably not arrive, accepting 101, 102, or 103 reservations can reduce unused physical capacity and initially increase expected profit. But the hotel cannot keep increasing reservations indefinitely: eventually too many customers will arrive, and when more than 100 customers show up, some cannot be accommodated, creating an **overbooking penalty** of \$200 per displaced customer. Expected profit as a function of the booking limit should therefore initially increase but eventually decline once the expected penalties dominate the benefit of additional reservations. The optimization problem is to identify that turning point.

30.4 The Decision Variable

Let N denote the number of reservations the hotel accepts. Since the hotel has 100 rooms, the interesting values of N begin at 100: we can evaluate $N = 100, 101, 102, 103, \ldots$. For each value of N we simulate the number of customers who actually show up, calculate the corresponding profit, and after many simulations estimate the expected profit associated with that booking limit. Comparing expected profit across booking limits, we expect it to reach a maximum and then begin to decline; that value of N is the optimal reservation limit under the model's assumptions.

30.5 The Random Variable to Simulate

Suppose the hotel accepts 101 reservations. The uncertain quantity is the **number of customers who show up**, X, taking values $0, 1, 2, \ldots, 101$. Since each reservation holder independently shows up with probability 0.95, X follows a binomial distribution: the number of successes among 101 independent trials. Any of the three simulation techniques developed in this module — transformation, acceptance–rejection, or Metropolis — can in principle be used to simulate X. Using the Metropolis framework as an illustration: if the current state is 3, the neighboring states are 2, 3, and 4, and calculating the transition probability from 3 to 4 requires the ratio $P(X{=}4)/P(X{=}3)$.

30.6 The Binomial Probabilities and Why Their Ratio Simplifies

The probability that exactly 4 of 101 customers show up is

$$P_4 = \binom{101}{4}(0.95)^4(0.05)^{97},$$

and the probability that exactly 3 show up is

$$P_3 = \binom{101}{3}(0.95)^3(0.05)^{98}.$$

The Metropolis acceptance ratio for moving from state 3 to state 4 uses P_4/P_3, and much of this complicated expression cancels: the large factorial terms and large powers of 0.95 and 0.05 mostly cancel, simplifying the ratio to approximately

$$\frac{P_4}{P_3} \approx \frac{98}{4} \times \frac{0.95}{0.05}.$$

The exact simplification matters less than the conceptual lesson: even when individual probabilities look extremely complicated, the **ratio of neighboring probabilities can be simple**. This is exactly why Metropolis is attractive — we often do not need to calculate every probability in full, only enough information to compare neighboring states.

30.7 Choosing a Simulation Technique

The hotel problem does not force the use of any single method; the transformation technique, acceptance–rejection, and Metropolis are all valid choices, and the easiest one to explain and implement is acceptable. A textbook solution might use the transformation technique for the binomial distribution, but Metropolis can be easier to formulate because it avoids explicitly constructing the cumulative distribution function. The general principle — producing a valid simulator for the random number of show-ups — matters more than matching one particular computational method.

30.8 Simulation Is Only Half the Problem

Successfully simulating the number of show-ups is not sufficient: the random-number generator gives X, but the business question concerns **profit**. A second component is required: a profit function mapping the simulated number of show-ups into money. Developing only the random-number simulator represents only part of the answer; it is equally necessary to show how to calculate revenue, cost, and profit in each possible state.

30.9 The Profit Function

Let the hotel accept N reservations and let X be the number of customers who actually arrive.

Case 1: $X \leq 100$. The hotel can accommodate everybody. Revenue is $150N$ (all N reservations were sold and paid for, including no-shows), and the operating/cleaning cost is $30X$ (only actual arrivals generate this cost). Hence

$$\text{Profit} = 150N - 30X, \qquad X \leq 100.$$

Case 2: $X > 100$. The hotel has only 100 rooms, so it can serve only 100 customers. Revenue remains $150N$, and the room preparation cost is limited to 100 occupied rooms, $30(100)$. The number of customers who cannot be accommodated is $X - 100$, each generating a penalty of \$200, giving a penalty cost of $200(X - 100)$. Hence

$$\text{Profit} = 150N - 30(100) - 200(X - 100), \qquad X > 100.$$

This piecewise structure is fundamental because the economics change exactly when physical capacity is exceeded: below capacity, an additional arrival only creates the \$30 operating cost, while above capacity the hotel cannot serve another arrival and instead incurs the \$200 overbooking penalty.

30.10 One Simulation Replication

Suppose $N = 101$. First simulate the number of show-ups X. If the simulator produces $X = 97$ (below 100), use the first branch:

$$\text{Profit} = 150(101) - 30(97).$$

That gives one simulated profit observation. Repeating the random-number generation, suppose the next draw produces $X = 101$; capacity is now exceeded by one customer, so the second branch applies:

$$\text{Profit} = 150(101) - 30(100) - 200(1).$$

That gives another simulated profit observation, and the process repeats many times.

30.11 Estimating Expected Profit

Suppose we generate 10,000 simulated values of X, calculate the corresponding profit for each, and average those 10,000 simulated profits. That sample average estimates the **expected profit** associated with booking limit N:

$$E[\widehat{\text{Profit} \mid N}] = \frac{1}{10{,}000} \sum_{k=1}^{10{,}000} \text{Profit}_k.$$

For a computational project, students should actually implement this and can also calculate quantities such as standard errors and confidence intervals.

30.12 Searching Over the Decision Variable

After estimating expected profit for $N = 101$, change N to 102 and repeat the entire simulation; the distribution of X changes because there are now 102 potential customers. Generate show-ups, calculate profits, and average to obtain the estimated expected profit associated with 102 reservations. Then try 103, 104, and so forth. Eventually the estimated expected-profit curve should reach a maximum: the process is one of repeatedly increasing N and looking for the point where expected profit stops increasing and starts declining. That turning point is the solution to the optimization-with-simulation problem.

30.13 Why Profit Eventually Declines

Initially, overbooking is profitable because the hotel has predictable no-shows: selling one additional reservation generates an additional \$150 of revenue, and most of the time there is still enough physical capacity because some customers do not appear. But as N increases, the probability that more than 100 customers arrive also increases, and once that happens frequently enough, the expected \$200 penalties begin to dominate the benefit of accepting additional reservations. There is a balance between the benefit of selling additional capacity and the expected cost of excessive arrivals, which simulation evaluates when the randomness makes direct reasoning cumbersome.

30.14 Simulation versus Optimization

This example makes an important conceptual distinction. The **simulation problem** asks: for a given N, what is the distribution of profit, or more specifically, what is the expected profit associated with this N? The **optimization problem** asks: which N gives the highest expected profit? Optimization with simulation is essentially using simulation to evaluate the objective function, and then optimizing over the decision variable — a pattern that appears in many real-world problems where the objective function may be impossible or inconvenient to calculate analytically, but can be estimated numerically if the system can be simulated for any proposed decision.

30.15 What Is Required for the Final Exam

For a final-exam problem of this kind, two things are expected. First, **develop the simulator for the uncertain quantity** — in this example, explaining how to simulate the number of show-ups. Second, **show how the resulting state determines revenue, cost, or profit** — in this example, giving the piecewise profit expression. Those two pieces constitute the expected final-exam solution: you are not expected to run thousands of simulations, numerically search over all booking limits, or write code during the exam. The emphasis is on the correct simulation logic and the correct economic performance calculation.

30.16 What Is Required for the Group Assignment

The group assignment goes further: students must actually implement the simulator, using Python or another suitable computational environment, to generate many random show-up values, calculate profit for each observation, average the simulated profits, estimate expected profit, and potentially calculate a standard error and confidence interval. This is repeated for multiple reservation limits ($N = 101, 102, 103, \ldots$), and the reservation limit producing the highest estimated expected profit is identified. This optimization and numerical implementation goes beyond what is required on the final exam.

30.17 A Compact Exam Template

When faced with a simulation-based business problem, it can help to memorize the structure rather than individual numbers, by asking four questions:

1. **What is the decision?** Here: the number of reservations N.
2. **What is random?** Here: the number of customers who show up, X.
3. **How do I simulate X?** Use transformation, acceptance–rejection, or Metropolis, as appropriate.
4. **Given X, how do I calculate the performance measure?** Here: profit as a function of N and X.

For the exam, those steps are usually the core answer. For a full computational project, a fifth question is added: **how do I compare decisions?** — run many simulations for every candidate N, estimate expected profit, and choose the largest.

30.18 The Broader Lesson

The hotel example is more than an overbooking problem; it illustrates a general framework for decision analytics under uncertainty. There is a **controllable decision**, an **uncertain outcome**, and an **economic consequence** produced jointly by the decision and the random outcome. Simulation handles the uncertainty; optimization handles the decision. Together they provide a way to solve complex problems in which expected performance cannot easily be calculated directly. The same structure appears in inventory problems, staffing problems, capacity planning, revenue management, transportation, financial risk, and many other settings.

Final Summary of Module 5

This module placed Metropolis alongside transformation and acceptance–rejection as the third major random-number-generation method. Metropolis constructs a Markov chain whose long-run visitation frequencies reproduce a desired target distribution, using the acceptance rule

$$\alpha = \min\left(1, \frac{\text{proposed-state score}}{\text{current-state score}}\right),$$

which is powerful because common normalization constants cancel. For discrete problems, state-transition probabilities are constructed explicitly and uniform random numbers determine actual transitions; for continuous problems, a random perturbation creates a proposed state and the same acceptance-ratio idea applies. Finally, the hotel overbooking problem demonstrates how simulation becomes part of optimization: for each booking limit, simulate show-ups, convert show-ups into profit, estimate expected profit, and compare alternative booking limits.

Simulation tells us what might happen under a decision; optimization with simulation tells us which decision we should choose.